

Lecture 1: Calculus Review & 3D Coordinate Systems

Background & Motivation

Welcome to Calculus III. Everything you learned in Calc I and II—limits, derivatives, integrals, Taylor series—was about functions of a single variable. This semester, we generalize all of it to functions of two, three, or more variables. Today we start with a rapid review of the tools you'll need, then introduce the 3D coordinate systems that let us describe points in space. By the end of the semester, you'll be integrating over surfaces and applying theorems that unify all of calculus.

1. Core Concepts

- Limit: $\lim_{x \rightarrow a} f(x) = L$ means **as x approaches a , $f(x)$ gets arbitrarily close to L**
- Derivative: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$
- L'Hôpital's Rule: For $\frac{0}{0}$ or $\frac{\infty}{\infty}$: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

2. Essential Rules

Derivatives and Integrals

Derivatives:

$$\begin{aligned} (uv)' &= u'v + uv' \\ \left(\frac{u}{v}\right)' &= \frac{u'v - uv'}{v^2} \\ \frac{d}{dx}[f(g(x))] &= f'(g(x))g'(x) \\ \frac{d}{dx}[x^n] &= nx^{n-1} \\ \frac{d}{dx}[\ln(x)] &= 1/x \\ \frac{d}{dx}[e^x] &= e^x \end{aligned}$$

Integrals:

$$\begin{aligned} \int_a^b f(x) dx &= F(b) - F(a) \\ \int u dv &= uv - \int v du \\ \int x^n dx &= \frac{x^{n+1}}{n+1} + C \\ \int \frac{1}{x} dx &= \ln|x| + C \\ \int e^x dx &= e^x + C \\ \int \sin(x) dx &= -\cos(x) + C \end{aligned}$$

3. Taylor Series

- General form around $x = a$:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

- Common series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

4. 3D Coordinate Systems

Rectangular Coordinates

- Point in 3D: (x, y, z) means x **units right**, y **units forward**, z **units up**

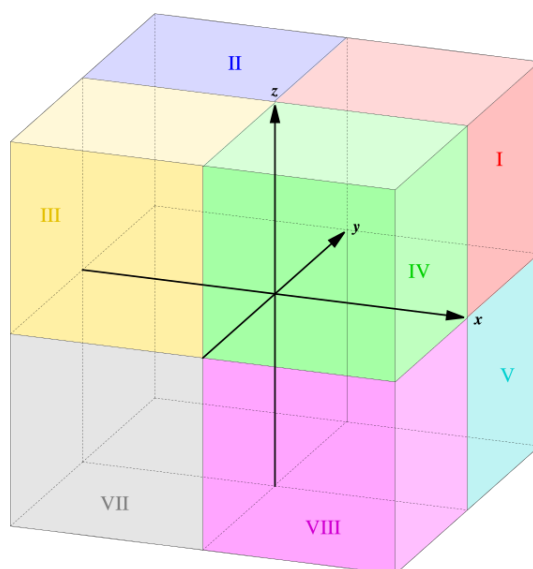
Distance and Midpoint in 3D

Distance between (x_1, y_1, z_1) and (x_2, y_2, z_2) :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Midpoint:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$



The eight octants of 3D space.

- Coordinate planes: xy -plane: $z = 0$ xz -plane: $y = 0$ yz -plane: $x = 0$

Alternative Coordinate Systems

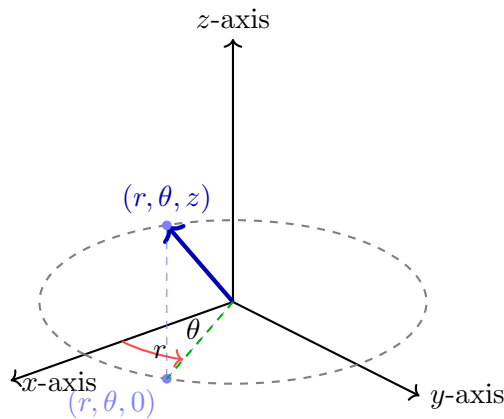
Polar and Cylindrical

Polar (r, θ) :

- r : distance from origin
- θ : angle from x -axis
- To rectangular:
 $x = r \cos \theta, \quad y = r \sin \theta$
- From rectangular:
 $r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}(y/x)$

Cylindrical (r, θ, z) :

- r : distance from z -axis
- θ : angle from x -axis
- z : height
- To rectangular:
 $x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$



A point in cylindrical coordinates (r, θ, z) : radius r from the z -axis, angle θ in the xy -plane, and vertical height z .

Spherical Coordinates (ρ, θ, ϕ)

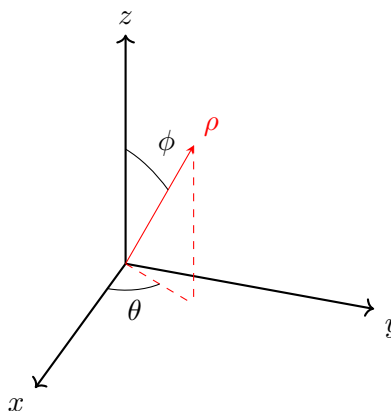
- ρ : distance from origin
- θ : angle from x -axis in xy -plane
- ϕ : angle from positive z -axis

– To rectangular:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

– From rectangular:

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1}(y/x), \quad \phi = \cos^{-1}(z/\rho)$$



A point in spherical coordinates (ρ, θ, ϕ) : distance ρ from the origin, azimuthal angle θ in the xy -plane, and polar angle ϕ measured from the $+z$ -axis.

Practice Problems

Problem 1. Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$.

Answer: $\frac{1}{6}$

Problem 2. Find first three terms of Taylor series for $\ln(1+x)$ at $a=0$.

Answer: $x - \frac{x^2}{2} + \frac{x^3}{3}$

Problem 3. Convert cylindrical point $(3, \pi/4, 2)$ to rectangular coordinates.

Answer: $(3\sqrt{2}/2, 3\sqrt{2}/2, 2)$

Problem 4. Find distance between points $(1, 1, 1)$ and $(4, 5, 2)$.

Answer: $\sqrt{26}$

Problem 5. Convert rectangular point $(0, 2, 2)$ to spherical coordinates.

Answer: $(\rho = 2\sqrt{2}, \theta = \pi/2, \phi = \pi/4)$

Lecture 2: Vectors in $\mathbb{R}^3$

Background & Motivation

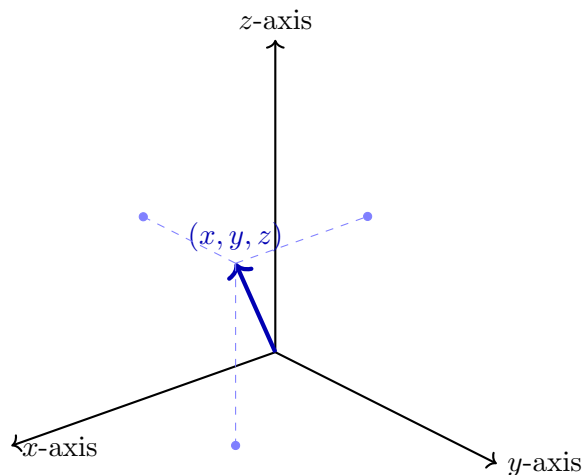
Last time we set up 3D coordinates. Now we need a language to describe direction and magnitude in space—that's what vectors give us. Vectors are the building blocks of everything in this course: lines, planes, curves, derivatives, integrals, and fields all use vector notation.

1. Vectors — Definitions

Definition 1: Vector

A **vector** is a quantity carrying both **magnitude** and **direction** — visually, an arrow. Two arrows are considered the *same* vector whenever they have the same length and direction, no matter where they are drawn, so a vector is really an equivalence class of parallel arrows. Placing its tail at the origin, the tip lands at a specific point (v_1, v_2, v_3) , and those coordinates are the **components** of $\mathbf{v}$.

– Written $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$; angle brackets distinguish vectors from points (a, b, c)



A vector in $\mathbb{R}^3$ represented by its components along the x , y , and z axes. Dashed lines mark the projections onto the three coordinate planes.

Definition 2: Magnitude (Norm)

The **magnitude** (or **norm**) $\|\mathbf{v}\|$ is the length of the vector — the Euclidean distance from its tail to its tip. When $\mathbf{v}$ is drawn with its tail at the origin, $\|\mathbf{v}\|$ is literally the distance from the origin to

the terminal point, and it is unchanged by rotating or translating the coordinate system. In short, magnitude strips away direction and records only *how much* of the vector there is. It is computed by the Pythagorean theorem applied to the components:

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Definition 3: Unit Vector

A vector with magnitude **1**. To **normalize**:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

Standard unit vectors: $\hat{\mathbf{i}} = \langle 1, 0, 0 \rangle$, $\hat{\mathbf{j}} = \langle 0, 1, 0 \rangle$, $\hat{\mathbf{k}} = \langle 0, 0, 1 \rangle$

Definition 4: Direction Cosines

For $\mathbf{v} = \langle a, b, c \rangle$, the angles α, β, γ with the x -, y -, z -axes:

$$\cos \alpha = \frac{a}{\|\mathbf{v}\|}, \quad \cos \beta = \frac{b}{\|\mathbf{v}\|}, \quad \cos \gamma = \frac{c}{\|\mathbf{v}\|}$$

Key identity: $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

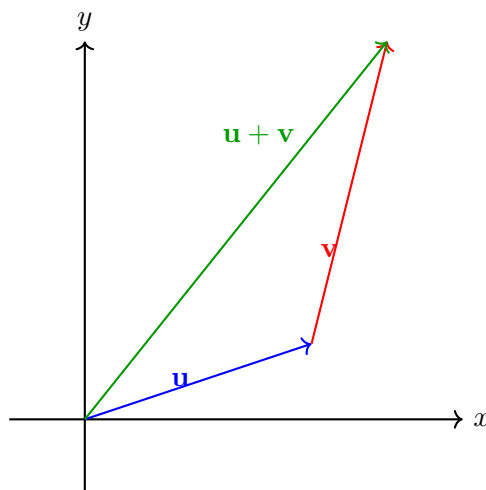
Vector Arithmetic

If $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$:

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$$

$$c\mathbf{u} = \langle cu_1, cu_2, cu_3 \rangle$$

– Addition is “tip-to-tail”; scalar multiplication scales length



Tip-to-tail method of vector addition: placing $\mathbf{v}$ at the tip of $\mathbf{u}$ results in $\mathbf{u} + \mathbf{v}$.

2. The Dot Product

Definition 5: Dot Product

Component formula:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3$$

Geometric formula:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

– The dot product is a **scalar** (number, not a vector)

Dot Product Facts

- $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$
- Angle between vectors: $\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right)$

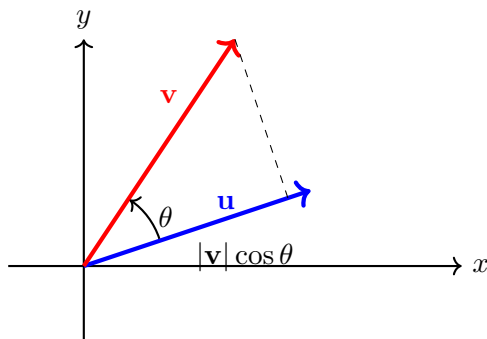
Definition 6: Vector Projection

The **projection** of $\mathbf{u}$ onto $\mathbf{v}$ is the “shadow” that $\mathbf{u}$ casts along the line of $\mathbf{v}$ — the part of $\mathbf{u}$ that points in the direction of $\mathbf{v}$. It is a vector parallel to $\mathbf{v}$ whose length is the *scalar component* $\text{comp}_{\mathbf{v}} \mathbf{u} = \|\mathbf{u}\| \cos \theta$.

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$$

Scalar component:

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}$$



The dot product interpreted as the projection of $\mathbf{v}$ onto $\mathbf{u}$, scaled by $\cos \theta$.

Example 1: Crane Cable Forces

Two crane cables exert forces $\mathbf{F}_1 = \langle 120, 0, 80 \rangle$ N and $\mathbf{F}_2 = \langle -50, 75, 60 \rangle$ N.

- Find the total force $\mathbf{F}_1 + \mathbf{F}_2$.
- Find a unit vector in the direction of the total force.
- What force $\mathbf{F}_3$ would perfectly counterbalance $\mathbf{F}_1 + \mathbf{F}_2$?

1. Total force:

$$\mathbf{F}_1 + \mathbf{F}_2 = \langle 70, 75, 140 \rangle \text{ N}$$

2. Magnitude:

$$\|\mathbf{F}_1 + \mathbf{F}_2\| = \sqrt{70^2 + 75^2 + 140^2} = \sqrt{4900 + 5625 + 19600} = \sqrt{30125} \approx 173.6 \text{ N}$$

3. Unit vector:

$$\hat{\mathbf{F}} = \frac{1}{173.6} \langle 70, 75, 140 \rangle \approx \langle 0.403, 0.432, 0.807 \rangle$$

4. Counterbalance: $\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2) = \langle -70, -75, -140 \rangle \text{ N}$

Example 2: Dot Product and Angle

Let $\mathbf{a} = \langle 3, 0, 4 \rangle$ and $\mathbf{b} = \langle 1, 2, -1 \rangle$. Find $\mathbf{a} \cdot \mathbf{b}$ and the angle between them.

1. Dot product:

$$\mathbf{a} \cdot \mathbf{b} = (3)(1) + (0)(2) + (4)(-1) = 3 + 0 - 4 = -1$$

2. Magnitudes:

$$\|\mathbf{a}\| = \sqrt{9 + 0 + 16} = 5, \quad \|\mathbf{b}\| = \sqrt{1 + 4 + 1} = \sqrt{6}$$

3. Angle:

$$\cos \theta = \frac{-1}{5\sqrt{6}} \implies \theta = \arccos\left(\frac{-1}{5\sqrt{6}}\right) \approx 94.7^\circ$$

Since $\theta > 90^\circ$, the vectors point in “generally opposite” directions.

Example 3: Projection

Project $\mathbf{a} = \langle 3, 4, 5 \rangle$ onto $\mathbf{n} = \langle 1, 1, 1 \rangle$.

1. Compute pieces:

$$\mathbf{a} \cdot \mathbf{n} = 3 + 4 + 5 = 12, \quad \|\mathbf{n}\|^2 = 3$$

2. Projection:

$$\text{proj}_{\mathbf{n}} \mathbf{a} = \frac{12}{3} \langle 1, 1, 1 \rangle = \langle 4, 4, 4 \rangle$$

3. Perpendicular component: $\mathbf{a} - \text{proj}_{\mathbf{n}} \mathbf{a} = \langle 3, 4, 5 \rangle - \langle 4, 4, 4 \rangle = \langle -1, 0, 1 \rangle$

3. The Cross Product

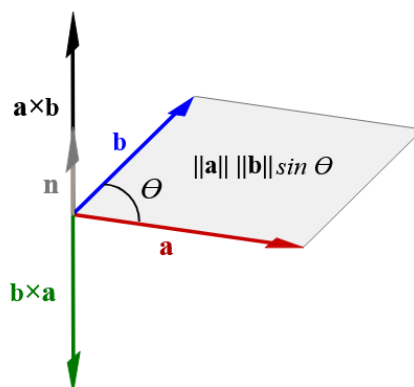
Definition 7: Cross Product

Determinant formula:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \langle u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1 \rangle$$

Properties:

- $\mathbf{u} \times \mathbf{v}$ is **perpendicular to both $\mathbf{u}$ and $\mathbf{v}$**
- $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$ – area of parallelogram
- Direction via the **right-hand rule**
- $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$ (anti-commutative)



Cross product: $\mathbf{a} \times \mathbf{b}$ is perpendicular to both, with magnitude = $\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$ (parallelogram area).

Cross Product Facts

- $\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}$
- $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$
- $\mathbf{u} \times \mathbf{u} = \mathbf{0}$ always

Example 4: Cross Product Computation

Compute $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 3, 0, 4 \rangle$ and $\mathbf{b} = \langle 1, 2, -1 \rangle$.

1. Set up the determinant:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 0 & 4 \\ 1 & 2 & -1 \end{vmatrix}$$

2. Expand:

$$= \hat{\mathbf{i}}(0 \cdot (-1) - 4 \cdot 2) - \hat{\mathbf{j}}(3 \cdot (-1) - 4 \cdot 1) + \hat{\mathbf{k}}(3 \cdot 2 - 0 \cdot 1) = \langle -8, 7, 6 \rangle$$

3. Verify orthogonality: $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = (3)(-8) + (0)(7) + (4)(6) = -24 + 0 + 24 = 0 \quad \checkmark$

4. Scalar Triple Product

Scalar Triple Product

$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ – gives the **volume of the parallelepiped** spanned by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

$$\text{Volume} = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

Example 5: Scalar Triple Product — Volume

Find the volume of the parallelepiped spanned by $\mathbf{a} = \langle 1, 0, 2 \rangle$, $\mathbf{b} = \langle 3, 1, 0 \rangle$, $\mathbf{c} = \langle 0, 2, 1 \rangle$.

1. Cross product $\mathbf{b} \times \mathbf{c}$:

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = \langle 1, -3, 6 \rangle$$

2. Triple product:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (1)(1) + (0)(-3) + (2)(6) = 13$$

3. Result: Volume = $|13| = \boxed{13}$ cubic units.

Show all work for full credit.

Practice Problems

Problem 1. Let $\mathbf{v} = \langle 3, 4, 12 \rangle$. Find $\|\mathbf{v}\|$, the unit vector $\hat{\mathbf{v}}$, and the direction cosines.

Answer: $\|\mathbf{v}\| = 13$, $\hat{\mathbf{v}} = \langle 3/13, 4/13, 12/13 \rangle$

Direction cosines: $\cos \alpha = 3/13$, $\cos \beta = 4/13$, $\cos \gamma = 12/13$

Problem 2. Compute $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 4, 0, -2 \rangle$.

Answer: $\mathbf{u} \cdot \mathbf{v} = 2$, $\mathbf{u} \times \mathbf{v} = \langle 2, 16, 4 \rangle$

Problem 3. Find the angle between $\mathbf{u} = \langle 1, 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 1 \rangle$.

Answer: $\theta = \pi/3 = 60^\circ$

Problem 4. Find $\text{proj}_{\mathbf{n}} \mathbf{a}$ where $\mathbf{a} = \langle 3, 4, 5 \rangle$ and $\mathbf{n} = \langle 1, 1, 1 \rangle$. Also find the perpendicular component.

Answer: $\text{proj}_{\mathbf{n}} \mathbf{a} = \langle 4, 4, 4 \rangle$, perpendicular component = $\langle -1, 0, 1 \rangle$

Problem 5. Three points $P_1 = (1, 2, 3)$, $P_2 = (4, -1, 2)$, $P_3 = (-2, 3, 5)$ define a plane. Use the cross product to find a normal vector.

Answer: $\mathbf{n} = \langle -5, -3, -6 \rangle$

Problem 6. Find the volume of the parallelepiped spanned by $\mathbf{a} = \langle 2, 1, 0 \rangle$, $\mathbf{b} = \langle 1, -1, 3 \rangle$, $\mathbf{c} = \langle 0, 2, 1 \rangle$.

Answer: $\text{Volume} = 13$

Lecture 3: Lines and Planes in Space

Background & Motivation

With vectors in hand, we can now describe geometric objects in 3D. A line needs a point and a direction; a plane needs a point and a normal. These are the simplest objects in space, and the techniques here—parametric equations, dot products for angles, cross products for normals—come back throughout the course.

1. Lines in $\mathbb{R}^3$

Definition 1: Parametric Equation of a Line

A line through $P_0 = (x_0, y_0, z_0)$ with direction vector $\mathbf{d} = \langle a, b, c \rangle$:

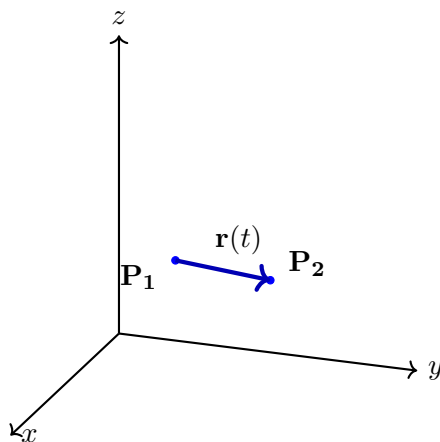
Vector form:

$$\mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{d} = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

Component form:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

– Each value of $t \in (-\infty, \infty)$ gives a point on the line



The line segment connecting points $\mathbf{P}_1$ and $\mathbf{P}_2$ in 3D, represented by $\mathbf{r}(t) = (1 - t)\mathbf{P}_1 + t\mathbf{P}_2$ for $t \in [0, 1]$.

Definition 2: Line Segment

The segment from P_1 to P_2 :

$$\mathbf{r}(t) = (1 - t) \mathbf{r}_1 + t \mathbf{r}_2, \quad t \in [0, 1]$$

– $t = 0$ gives P_1 ; $t = 1$ gives P_2

Example 1: Parametric Line

Write the parametric equation of the line through $P_0 = (1, -2, 3)$ in direction $\mathbf{d} = \langle 4, -1, 2 \rangle$.

1. Apply the formula $\mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{d}$:

$$\mathbf{r}(t) = \langle 1 + 4t, -2 - t, 3 + 2t \rangle$$

2. Component form: $x = 1 + 4t, \quad y = -2 - t, \quad z = 3 + 2t$

2. Planes in $\mathbb{R}^3$

Definition 3: Equation of a Plane

A plane through $P_0 = (x_0, y_0, z_0)$ with normal vector $\mathbf{n} = \langle A, B, C \rangle$:

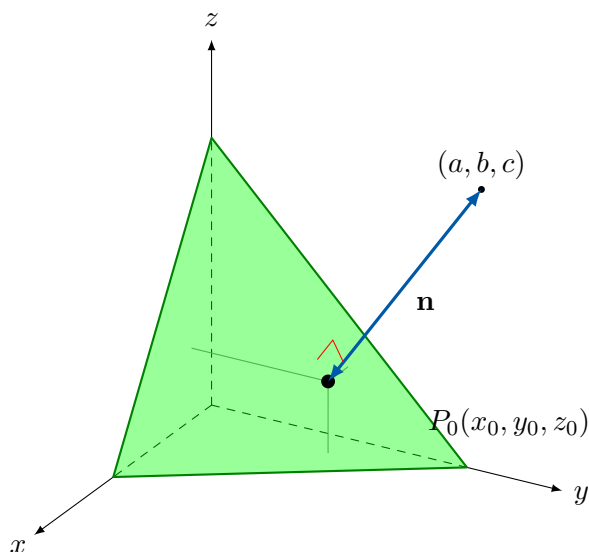
Point-normal form:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

General form:

$$Ax + By + Cz = D$$

where $D = Ax_0 + By_0 + Cz_0$



A plane in 3D space defined by a point P_0 and a normal vector $\mathbf{n}$ extending from P_0 in the perpendicular direction.

Example 2: Plane from Point and Normal

Find the equation of the plane through $(2, -1, 3)$ with normal $\mathbf{n} = \langle 4, 5, -2 \rangle$.

1. Point-normal form:

$$4(x - 2) + 5(y + 1) - 2(z - 3) = 0$$

2. Expand:

$$4x - 8 + 5y + 5 - 2z + 6 = 0$$

3. Result: $4x + 5y - 2z = -3$

Example 3: Plane from Three Points

Find the equation of the plane through $P_1 = (1, 2, 3)$, $P_2 = (4, -1, 2)$, $P_3 = (-2, 3, 5)$.

1. Two vectors in the plane:

$$\overrightarrow{P_1P_2} = \langle 3, -3, -1 \rangle, \quad \overrightarrow{P_1P_3} = \langle -3, 1, 2 \rangle$$

2. Normal via cross product:

$$\mathbf{n} = \langle 3, -3, -1 \rangle \times \langle -3, 1, 2 \rangle = \langle -6 + 1, 3 - 6, 3 - 9 \rangle = \langle -5, -3, -6 \rangle$$

3. Use $P_1 = (1, 2, 3)$:

$$-5(x - 1) - 3(y - 2) - 6(z - 3) = 0 \implies 5x + 3y + 6z = 29$$

3. Intersections and Angles

Line–Plane Intersection

– Strategy: substitute parametric equations of the line into the plane equation, solve for t

Example 4: Line–Plane Intersection

Find where $\mathbf{r}(t) = \langle 1 + 2t, 2 - t, 3 + t \rangle$ intersects $3x - y + 2z = 5$.

1. Substitute:

$$3(1 + 2t) - (2 - t) + 2(3 + t) = 5$$

2. Solve:

$$3 + 6t - 2 + t + 6 + 2t = 5 \implies 9t + 7 = 5 \implies t = -\frac{2}{9}$$

3. Point:

$$\left(1 - \frac{4}{9}, 2 + \frac{2}{9}, 3 - \frac{2}{9}\right) = \left(\frac{5}{9}, \frac{20}{9}, \frac{25}{9}\right)$$

Angle Between Two Planes

Angle Between Planes

If planes have normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|}$$

– Parallel: $\mathbf{n}_1 \parallel \mathbf{n}_2$ – Perpendicular: $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$

Line of Intersection of Two Planes

Strategy

Given planes Π_1 and Π_2 with normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

1. **Direction:** $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$
2. **Point:** Set one variable (e.g. $z = 0$) and solve the 2×2 system

Example 5: Intersection of Two Planes

Find the line of intersection of $2x - y + z = 3$ and $x + y - 2z = -4$.

1. Direction: $\mathbf{n}_1 = \langle 2, -1, 1 \rangle$, $\mathbf{n}_2 = \langle 1, 1, -2 \rangle$

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \langle (-1)(-2) - (1)(1), (1)(1) - (2)(-2), (2)(1) - (-1)(1) \rangle = \langle 1, 5, 3 \rangle$$

2. Point (set $z = 0$): $2x - y = 3$ and $x + y = -4$.
3. Adding: $3x = -1 \implies x = -1/3$, $y = -11/3$.
4. Result: $\mathbf{r}(t) = \langle -1/3 + t, -11/3 + 5t, 3t \rangle$

4. Distances

Theorem 1: Distance from a Point to a Plane

Distance from $P_1 = (x_1, y_1, z_1)$ to $Ax + By + Cz = D$:

$$d = \frac{|Ax_1 + By_1 + Cz_1 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

Example 6: Point-to-Plane Distance

Find the distance from $Q = (1, -1, -3)$ to the plane $x + 2y + 3z = 18$.

1. Apply formula:

$$d = \frac{|1 + 2(-1) + 3(-3) - 18|}{\sqrt{1 + 4 + 9}} = \frac{|1 - 2 - 9 - 18|}{\sqrt{14}} = \frac{28}{\sqrt{14}}$$

2. Simplify: $d = \frac{28\sqrt{14}}{14} = \boxed{2\sqrt{14}}$

5. Vector Projection onto a Plane

Projection onto a Plane

To project $\mathbf{a}$ onto a plane with normal $\mathbf{n}$:

$$\mathbf{a}_{\text{plane}} = \mathbf{a} - \text{proj}_{\mathbf{n}} \mathbf{a} = \mathbf{a} - \frac{\mathbf{a} \cdot \mathbf{n}}{\|\mathbf{n}\|^2} \mathbf{n}$$

– Subtract the normal component from the vector

Example 7: Projection onto a Plane

Project $\mathbf{a} = \langle 3, 4, 5 \rangle$ onto the plane with normal $\mathbf{n} = \langle 1, 1, 1 \rangle$.

1. Normal component:

$$\text{proj}_{\mathbf{n}} \mathbf{a} = \frac{3 + 4 + 5}{3} \langle 1, 1, 1 \rangle = 4 \langle 1, 1, 1 \rangle = \langle 4, 4, 4 \rangle$$

2. Plane projection:

$$\mathbf{a}_{\text{plane}} = \langle 3, 4, 5 \rangle - \langle 4, 4, 4 \rangle = \langle -1, 0, 1 \rangle$$

3. Check: $\langle -1, 0, 1 \rangle \cdot \langle 1, 1, 1 \rangle = -1 + 0 + 1 = 0$ ✓ (perpendicular to $\mathbf{n}$)

Show all work for full credit.

Practice Problems

Problem 1. Write the parametric equation of the line through $P = (2, 1, -3)$ in direction $\mathbf{d} = \langle 1, -2, 4 \rangle$.

Answer: $\mathbf{r}(t) = \langle 2 + t, 1 - 2t, -3 + 4t \rangle$

Problem 2. Find the equation of the plane through $(3, 0, -1)$ with normal $\mathbf{n} = \langle 2, 1, -5 \rangle$.

Answer: $2x + y - 5z = 11$

Problem 3. Find where the line $\mathbf{r}(t) = \langle -t, 2t, t + 3 \rangle$ intersects the plane $-x + y = 6$.

Answer: $t = 2$, point = $(-2, 4, 5)$

Problem 4. Find the distance from $(2, 3, -1)$ to the plane $x - 2y + 2z = 4$.

Answer: $d = 10/3$

Problem 5. Given $\Pi_1 : x + 2y - z = 3$ and $\Pi_2 : 2x - y + 3z = 1$: (a) find the angle between the planes, (b) find the line of intersection.

Answer (a): $\theta = \arccos\left(\frac{3}{2\sqrt{21}}\right) \approx 70.9^\circ$

Answer (b): $\mathbf{r}(t) = \langle 1 + t, 1 - t, -t \rangle$

Lecture 4: Vector-Valued Functions

Background & Motivation

We now shift from static geometry to motion. A vector-valued function $\mathbf{r}(t)$ traces a curve through space as the parameter t varies—think of it as the position of a particle at time t . This chapter connects the vector algebra from Chapter 1 to calculus: we'll differentiate and integrate these functions component by component, leading to tangent vectors, arc length, curvature, and projectile motion.

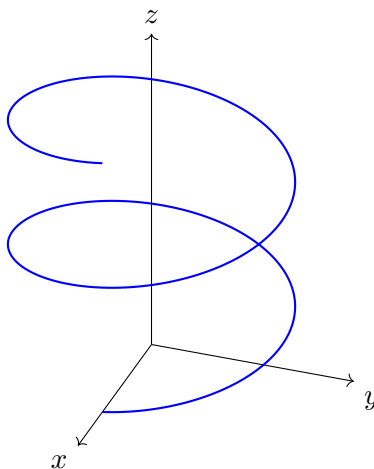
1. Parametric Curves in $\mathbb{R}^3$

Definition 1: Vector-Valued Function (VVF)

A VVF maps a scalar t to a vector:

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$$

- f, g, h are the **component functions**
- As t varies, the tip of $\mathbf{r}(t)$ traces a **space curve**



Helix $\mathbf{r}(t) = \langle R \cos t, R \sin t, ct \rangle$ spiraling around the z -axis. As t increases, $\mathbf{r}(t)$ rotates in the xy -plane while rising linearly in z .

Common Parametric Curves

- **Line:** $\mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{d}$
- **Circle** (radius R , xy -plane): $\mathbf{r}(t) = \langle R \cos t, R \sin t, 0 \rangle$, $t \in [0, 2\pi]$
- **Helix** (radius R , rising rate c): $\mathbf{r}(t) = \langle R \cos t, R \sin t, ct \rangle$
- **Ellipse:** $\mathbf{r}(t) = \langle a \cos t, b \sin t, 0 \rangle$

Example 1: Identifying a Helix

Describe the curve $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 2t \rangle$ for $t \geq 0$.

1. Check: $x^2 + y^2 = 9 \cos^2 t + 9 \sin^2 t = 9$ – lies on the cylinder $x^2 + y^2 = 9$.
2. The z -component $z = 2t$ increases linearly.
3. Result: a **helix** of radius 3, rising at rate 2 per radian, winding around the z -axis.

2. Derivatives of VVFs

Definition 2: Derivative of a VVF

The derivative $\mathbf{r}'(t)$ is the instantaneous rate of change of the position vector, obtained by differentiating each component separately. Geometrically it is the **velocity vector**: it points in the direction the curve is moving at that instant, and its magnitude is the speed at which the particle is traversing the curve.

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

- $\mathbf{r}'(t)$ is a **tangent vector** to the curve at $\mathbf{r}(t)$
- $\|\mathbf{r}'(t)\|$ is the **speed**

Derivative Rules for VVFs

- Scalar product rule: $\frac{d}{dt}[f(t)\mathbf{r}(t)] = f'(t)\mathbf{r}(t) + f(t)\mathbf{r}'(t)$
- Dot product rule: $\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{s}(t)] = \mathbf{r}'(t) \cdot \mathbf{s}(t) + \mathbf{r}(t) \cdot \mathbf{s}'(t)$
- Cross product rule: $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{s}(t)] = \mathbf{r}'(t) \times \mathbf{s}(t) + \mathbf{r}(t) \times \mathbf{s}'(t)$
- Chain rule: $\frac{d}{dt}[\mathbf{r}(g(t))] = \mathbf{r}'(g(t)) \cdot g'(t)$

Example 2: Derivative and Analysis at a Point

Let $\mathbf{r}(t) = \langle t^3, \sin t, e^t \rangle$. Find $\mathbf{r}'(t)$ and analyze at $t = \pi$.

1. Derivative: $\mathbf{r}'(t) = \langle 3t^2, \cos t, e^t \rangle$
2. At $t = \pi$:
 - Position: $\mathbf{r}(\pi) = \langle \pi^3, 0, e^\pi \rangle$
 - Tangent vector: $\mathbf{r}'(\pi) = \langle 3\pi^2, -1, e^\pi \rangle$
3. Component analysis: x -comp > 0 (increasing), y -comp < 0 (decreasing), z -comp > 0 (increasing)
4. Speed: $\|\mathbf{r}'(\pi)\| = \sqrt{9\pi^4 + 1 + e^{2\pi}}$

3. Integrals of VVFs

Definition 3: Integral of a VVF

Also **component-wise**:

$$\int \mathbf{r}(t) dt = \left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle + \mathbf{C}$$

– $\mathbf{C} = \langle C_1, C_2, C_3 \rangle$ is a **vector constant** of integration

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle$$

Example 3: Definite Integral of a VVF

Evaluate $\int_0^1 \langle 3t^2, 2 \sin t, e^{2t} \rangle dt$.

1. x -component: $\int_0^1 3t^2 dt = [t^3]_0^1 = 1$
2. y -component: $\int_0^1 2 \sin t dt = [-2 \cos t]_0^1 = 2 - 2 \cos 1$
3. z -component: $\int_0^1 e^{2t} dt = \left[\frac{e^{2t}}{2} \right]_0^1 = \frac{e^2 - 1}{2}$
4. Result: $\left\langle 1, 2 - 2 \cos 1, \frac{e^2 - 1}{2} \right\rangle$

4. Tangent Lines and Curves on Surfaces

Tangent Line at $t = t_0$

$$\mathbf{L}(s) = \mathbf{r}(t_0) + s \mathbf{r}'(t_0)$$

– Point of tangency: $\mathbf{r}(t_0)$ – Direction: $\mathbf{r}'(t_0)$

– To show a curve lies on a surface: substitute the components into the surface equation and verify

Example 4: Tangent Line + Curve on a Surface

The curve $\mathbf{r}(t) = \langle e^t, e^{-t}, t \rangle$. Show it lies on $xy = 1$ and find the tangent line at $t = 0$.

1. On the surface: $x \cdot y = e^t \cdot e^{-t} = e^0 = 1$ ✓ for all t .
2. At $t = 0$: $\mathbf{r}(0) = \langle 1, 1, 0 \rangle$
3. Tangent vector: $\mathbf{r}'(t) = \langle e^t, -e^{-t}, 1 \rangle \implies \mathbf{r}'(0) = \langle 1, -1, 1 \rangle$
4. Tangent line: $\mathbf{L}(s) = \langle 1 + s, 1 - s, s \rangle$

Show all work for full credit.

Practice Problems

Problem 1. If $\mathbf{r}(t) = \langle t^2, \ln(t+1), \cos t \rangle$, find $\mathbf{r}'(t)$ and $\mathbf{r}'(0)$.

Answer: $\mathbf{r}'(t) = \langle 2t, \frac{1}{t+1}, -\sin t \rangle$, $\mathbf{r}'(0) = \langle 0, 1, 0 \rangle$

Problem 2. Evaluate $\int_0^\pi \langle \cos t, \sin t, 1 \rangle dt$.

Answer: $\langle 0, 2, \pi \rangle$

Problem 3. A particle has velocity $\mathbf{v}(t) = \langle 2t, 3, -1 \rangle$ and $\mathbf{r}(0) = \langle 1, 0, 5 \rangle$. Find $\mathbf{r}(t)$.

Answer: $\mathbf{r}(t) = \langle t^2 + 1, 3t, 5 - t \rangle$

Problem 4. Show that $\mathbf{r}(t) = \langle 2 \cos t, 3 \sin t, t \rangle$ lies on the elliptic cylinder $\frac{x^2}{4} + \frac{y^2}{9} = 1$. Find the tangent line at $t = \pi/2$.

Answer: tangent line = $\langle -2s, 3, \pi/2 + s \rangle$

Problem 5. Prove: if $\|\mathbf{r}(t)\| = c$ (constant), then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$ for all t .

Hint: differentiate $\mathbf{r}(t) \cdot \mathbf{r}(t) = c^2$.

$$\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] = \frac{d}{dt}[c^2] = 0$$

By dot product rule: $\mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 0 \implies 2\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$. $\square$

Interpretation: a curve on a sphere has velocity always perpendicular to the position vector.

Lecture 5: Motion Along a Curve & Arc Length

Background & Motivation

Last time we defined vector-valued functions and their derivatives. Today we put them to work: the derivative $\mathbf{r}'(t)$ gives velocity, its magnitude gives speed, and integrating the speed gives arc length. We also derive the projectile motion equations you may have seen in physics—but now with the full vector calculus treatment.

1. Kinematics of Space Curves

Definition 1: Position, Velocity, Acceleration

For a particle moving along $\mathbf{r}(t)$:

Position:	$\mathbf{r}(t)$	where the particle is
Velocity:	$\mathbf{v}(t) = \mathbf{r}'(t)$	direction and rate of movement
Speed:	$\ \mathbf{v}(t)\ = \ \mathbf{r}'(t)\ $	how fast (scalar!)
Acceleration:	$\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$	rate of change of velocity

Recovering Position from Velocity/Acceleration

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt + \mathbf{C}_1 \quad \mathbf{r}(t) = \int \mathbf{v}(t) dt + \mathbf{C}_2$$

– Use initial conditions $\mathbf{v}(0)$ and $\mathbf{r}(0)$ to find the constants

- **Displacement** (net change): $\int_a^b \mathbf{v}(t) dt$ – a vector
- **Total distance traveled:** $\int_a^b \|\mathbf{v}(t)\| dt$ – a scalar (always $\geq$ displacement magnitude)

Example 1: Displacement vs. Distance

A particle has velocity $\mathbf{v}(t) = \langle 3t, 4, t^2 \rangle$ on $[0, 2]$.

- Find the displacement.
- Set up the integral for total distance traveled.

(a) Displacement:

- Integrate each component:

$$\int_0^2 \mathbf{v}(t) dt = \left\langle \left[\frac{3t^2}{2} \right]_0^2, [4t]_0^2, \left[\frac{t^3}{3} \right]_0^2 \right\rangle = \left\langle 6, 8, \frac{8}{3} \right\rangle$$

(b) Total distance:

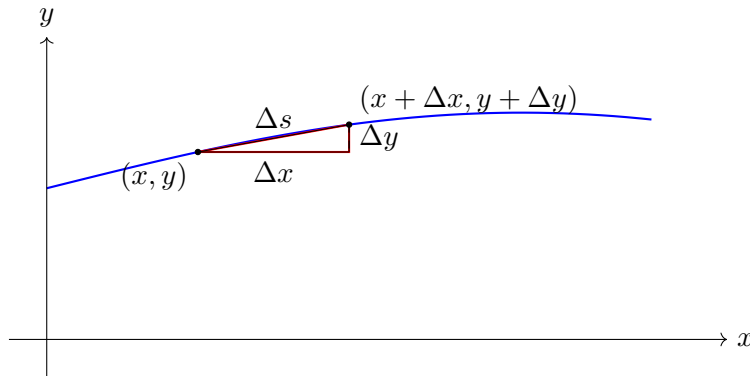
1. Speed: $\|\mathbf{v}(t)\| = \sqrt{9t^2 + 16 + t^4}$
2. Integral: $\int_0^2 \sqrt{9t^2 + 16 + t^4} dt$ – no nice closed form, use numerical methods.

2. Arc Length

Theorem 1: Arc Length of a Parametric Curve

Idea. Approximate the curve by a chain of tiny straight segments. Each segment between $\mathbf{r}(t)$ and $\mathbf{r}(t + dt)$ has length $\|\mathbf{r}'(t)\| dt$ (velocity times time). Adding these up in the limit gives the total arc length:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$



Approximating the arc length of a curve using a right triangle with Δs as the hypotenuse, Δx as the base, and Δy as the height. Summing Δs over infinitely many such triangles gives the integral formula for L .

Arc Length in Other Coordinates

Cartesian ($y = f(x)$):

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Polar ($r = f(\theta)$):

$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Example 2: Arc Length of a Helix

Find the arc length of $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 3t \rangle$ for $t \in [0, 2\pi]$.

1. Derivative: $\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t, 3 \rangle$

2. Speed: $\|\mathbf{r}'(t)\| = \sqrt{4 \sin^2 t + 4 \cos^2 t + 9} = \sqrt{4 + 9} = \sqrt{13}$

3. Arc length: $L = \int_0^{2\pi} \sqrt{13} dt = \boxed{2\pi\sqrt{13}}$

Note: the speed is constant – the helix is traversed at a uniform rate.

Example 3: Arc Length of a t^2/t^3 Curve

Find the arc length of $\mathbf{r}(t) = \langle t^2, t^3 \rangle$ for $t \in [0, 1]$.

1. Derivative: $\mathbf{r}'(t) = \langle 2t, 3t^2 \rangle$

2. Speed: $\|\mathbf{r}'(t)\| = \sqrt{4t^2 + 9t^4} = t\sqrt{4 + 9t^2}$

3. Arc length:

$$L = \int_0^1 t\sqrt{4 + 9t^2} dt$$

4. Substitution: $u = 4 + 9t^2$, $du = 18t dt$:

$$L = \frac{1}{18} \int_4^{13} \sqrt{u} du = \frac{1}{18} \cdot \frac{2}{3} \left[u^{3/2} \right]_4^{13} = \frac{1}{27} (13\sqrt{13} - 8) = \boxed{\frac{13\sqrt{13} - 8}{27}}$$

3. Projectile Motion

– Projectile launched from origin, initial speed v_0 , angle θ above horizontal, no air resistance

Projectile Motion Equations

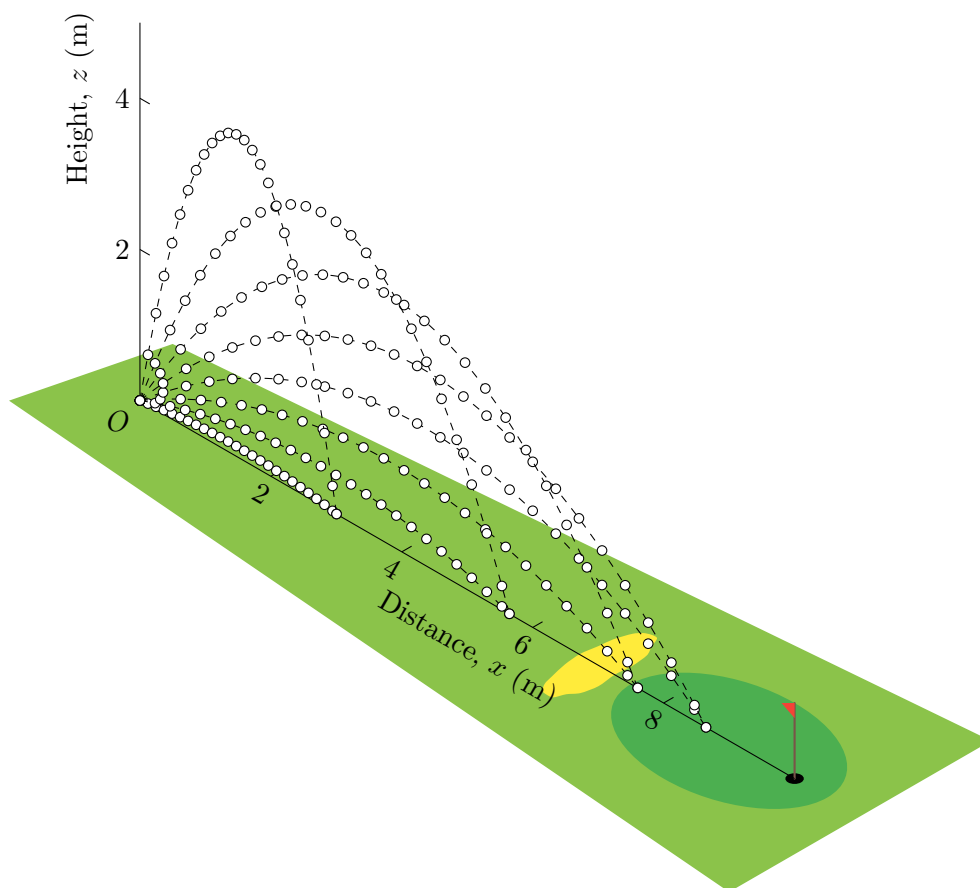
$$\mathbf{r}(t) = \langle v_0 \cos \theta \cdot t, v_0 \sin \theta \cdot t - \frac{1}{2}gt^2 \rangle$$

$$\text{Max height: } H = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$\text{Time of flight: } T = \frac{2v_0 \sin \theta}{g}$$

$$\text{Range: } R = \frac{v_0^2 \sin(2\theta)}{g}$$

– Max range at $\theta = 45^\circ$ – Max height at $\theta = 90^\circ$



Projectile trajectories at several launch angles, shown on a golf course. In the absence of air resistance each ball follows a parabolic path; the 45° launch gives the longest range.

Example 4: Projectile Problem

A ball is launched at $v_0 = 20$ m/s at $\theta = 45^\circ$. Find max height, time of flight, and range. ($g = 9.8$ m/s²)

1. Setup: $\sin 45^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}$
2. Max height: $H = \frac{20^2 \cdot (1/2)}{2(9.8)} = \frac{200}{19.6} \approx \boxed{10.2 \text{ m}}$
3. Time of flight: $T = \frac{2(20)(\sqrt{2}/2)}{9.8} = \frac{20\sqrt{2}}{9.8} \approx \boxed{2.89 \text{ s}}$
4. Range: $R = \frac{400 \sin 90^\circ}{9.8} = \frac{400}{9.8} \approx \boxed{40.8 \text{ m}}$

Note: $\theta = 45^\circ$ maximizes range since $\sin(2 \cdot 45^\circ) = 1$.

Show all work for full credit.

Practice Problems

Problem 1. Find the velocity, speed, and acceleration of $\mathbf{r}(t) = \langle \sqrt{t}, t^3, e^t \rangle$ at $t = 1$.

Answer: $\mathbf{v}(1) = \langle 1/2, 3, e \rangle$, speed = $\sqrt{37/4 + e^2}$, $\mathbf{a}(1) = \langle -1/4, 6, e \rangle$

Problem 2. Find the arc length of $\mathbf{r}(t) = \langle t, t^2/\sqrt{2}, t^3/3 \rangle$ for $t \in [0, 1]$.

Hint: simplify $\|\mathbf{r}'(t)\|$ – it factors.

Answer: $L = 4/3$

Problem 3. Find the arc length of the cardioid $r = 1 + \cos \theta$ for $\theta \in [0, 2\pi]$.

Hint: use $1 + \cos \theta = 2 \cos^2(\theta/2)$.

Answer: $L = 8$

Problem 4. A golfer hits a ball at $v_0 = 45$ m/s. What angle gives maximum range, and what is that range?

Answer: $\theta = 45^\circ$, $R = \frac{2025}{9.8} \approx 206.6$ m

Problem 5. A particle has $\mathbf{a}(t) = \langle 0, 0, -g \rangle$, $\mathbf{v}(0) = \langle 3, 4, 10 \rangle$ m/s, $\mathbf{r}(0) = \langle 0, 0, 100 \rangle$ m. Find $\mathbf{r}(t)$ and the time it hits the ground ($z = 0$).

Answer: $\mathbf{r}(t) = \langle 3t, 4t, -\frac{g}{2}t^2 + 10t + 100 \rangle$

Time: $t = \frac{10 + \sqrt{2060}}{9.8} \approx 5.65$ s

Lecture 6: Functions of Several Variables

Background & Motivation

This is where Calculus III truly begins. Until now we've worked with vectors and curves—extensions of familiar geometry. Starting today, we study functions $f(x, y)$ and $f(x, y, z)$ that take multiple inputs and return a single output. Temperature at a point in a room, pressure in the atmosphere, cost as a function of two inputs—these are all multivariate functions. We need new tools to visualize them (level curves, cross-sections) and new techniques to analyze them (partial derivatives, gradients, multiple integrals). Everything from here to the end of the course builds on this foundation.

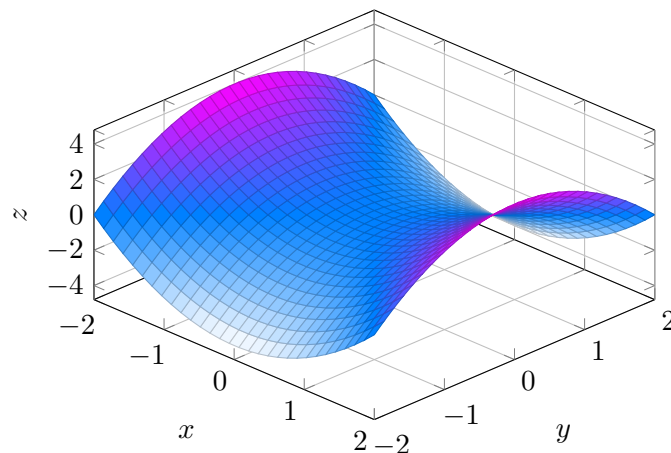
1. Functions of Two Variables

Definition 1: Function of Two Variables

A **function of two variables** $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ assigns to every point (x, y) in its domain a single real number $f(x, y)$. Whereas a single-variable function $y = f(x)$ draws a curve in the plane, a two-variable function draws a **surface** in 3D: for each input (x, y) you plot the point $(x, y, f(x, y))$, and the set of all such points is the **graph** $z = f(x, y)$. Think of the (x, y) -plane as the “floor” and the surface as a landscape whose height above that floor is determined by f .

– The graph $z = f(x, y)$ is a **surface in $\mathbb{R}^3$**

Surface Plot of $z = x^2 - y^2$



Graph of $z = f(x, y)$ as a 3D surface. Each point (x, y) in the plane is lifted to height $f(x, y)$; here $z = x^2 - y^2$ is the classic saddle.

Example 1: Real-World Multivariable Functions

Write each as $f(\text{inputs})$:

- (a) Ohm's Law: $V = IR$ (V depends on current I and resistance R)
- (b) Kinetic energy: $KE = \frac{1}{2}mv^2$ (depends on mass m and velocity v)
- (c) Ideal gas law: $P = \frac{nRT}{V}$ (at fixed n, R : depends on T and V)

2. Domains and Ranges

Domain Restrictions to Watch For

- **Denominators** – cannot equal 0
- **Square roots** – argument ≥ 0
- **Logarithms** – argument > 0
- **Inverse trig** – $|\text{argument}| \leq 1$ (for arcsin, arccos)

Example 2: Finding Domains

Find the domain of each function:

(a) $f(x, y) = \frac{1}{x + y}$

1. **Restriction:** $x + y \neq 0$

2. **Result:** $\text{dom}(f) = \{(x, y) : y \neq -x\}$ – all of $\mathbb{R}^2$ except the line $y = -x$

Range: $(-\infty, 0) \cup (0, \infty)$

(b) $g(x, y) = \frac{\ln(x - y)}{\sqrt{x^2 + y^2 - 4}}$

1. **Restrictions:** $x - y > 0$ (for $\ln$) AND $x^2 + y^2 - 4 > 0$ (for $\sqrt{\quad}$ in denominator)

2. **Result:** $\text{dom}(g) = \{(x, y) : x > y \text{ and } x^2 + y^2 > 4\}$

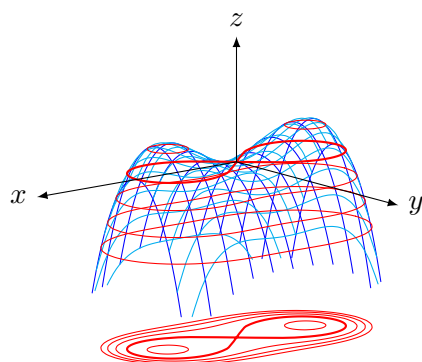
– Region below the line $y = x$ and outside the circle of radius 2

3. Level Curves

Definition 2: Level Curves

A **level curve** (or **contour**) of $f(x, y)$ at height k is the set of all points in the (x, y) -plane where f takes the value k — equivalently, the horizontal slice of the graph at height $z = k$, projected down onto the floor. Drawing many of these for different values of k produces a **topographic map** of the surface: regions where curves are packed close together correspond to *steep* parts of the surface, while widely-spaced curves correspond to *gentle* slopes.

$$\{(x, y) : f(x, y) = k\}$$



A 3D surface with its level curves projected straight down onto the xy -plane. The infinity-shaped level curve is a Bernoulli lemniscate. Together, the surface and its contour projection give complementary views of the same function.

Plotting Level Curves

1. Choose several values of k in the range of f
2. For each k , set $f(x, y) = k$ and simplify
3. Identify and sketch each resulting 2D curve
4. Label each curve with its k -value

Example 3: Level Curves of a Paraboloid

Sketch level curves of $f(x, y) = x^2 + y^2$ for $k = 1, 4, 9, 16$.

1. **Set up:** $x^2 + y^2 = k$ – circles of radius $\sqrt{k}$
2. **Identify curves:**
 - $k = 1$: circle of radius 1
 - $k = 4$: circle of radius 2
 - $k = 9$: circle of radius 3
 - $k = 16$: circle of radius 4
3. **Result:** Concentric circles – they get closer together as k increases (steeper slope at higher z).

Example 4: Level Curves of a Saddle

Sketch level curves of $f(x, y) = x^2 - y^2$ for $k = -4, -1, 0, 1, 4$.

1. **Set up:** $x^2 - y^2 = k$
2. **Identify curves:**
 - $k > 0$: hyperbolas opening along the x -axis
 - $k < 0$: hyperbolas opening along the y -axis
 - $k = 0$: $y = \pm x$ (two lines through origin)
3. **Result:** Topographic map of a hyperbolic paraboloid (saddle surface). Two families of hyperbolas cross at the saddle point.

4. Common 3D Equations

Surfaces You Should Recognize

Equation	Surface
$x^2 + y^2 + z^2 = r^2$	Sphere
$x^2 + y^2 = r^2$	Cylinder
$z = x^2 + y^2$	Paraboloid
$x^2 + y^2 = z^2$	Cone
$z = x^2 - y^2$	Saddle (hyperbolic paraboloid)
$Ax + By + Cz = D$	Plane

5. Functions of Three or More Variables

Definition 3: Level Surfaces

For $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, we cannot graph $w = f(x, y, z)$ (requires 4D). Instead we use **level surfaces**:

$$\{(x, y, z) : f(x, y, z) = k\}$$

– The 3D analog of level curves

Example 5: Level Surfaces of $x^2 + y^2 + z^2$

Describe the level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$.

1. Set up: $x^2 + y^2 + z^2 = k$

2. Identify:

- $k > 0$: spheres of radius $\sqrt{k}$ centered at the origin
- $k = 0$: single point (the origin)
- $k < 0$: empty set

3. Result: f measures squared distance from origin, so level surfaces are concentric spheres.

Practice Problems

Problem 1. Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

Answer: $x^2 + y^2 \leq 9$ – the closed disk of radius 3

Problem 2. Find the domain and range of $f(x, y) = e^{-(x^2+y^2)}$.

Domain: all of $\mathbb{R}^2$ (exponent always defined).

Range: $(0, 1]$. Max $f = 1$ at origin; $f \rightarrow 0$ as $\|(x, y)\| \rightarrow \infty$.

Problem 3. Sketch level curves of $f(x, y) = xy$ for $k = -2, -1, 0, 1, 2$.

$xy = k \implies y = k/x$: hyperbolas with asymptotes on the axes.

$k = 0$: the axes themselves ($x = 0$ or $y = 0$).

$k > 0$: curves in Q1 and Q3. $k < 0$: curves in Q2 and Q4.

Problem 4. Describe the level surfaces of $T(x, y, z) = 100 - x^2 - y^2 - z^2$.

$T = k \implies x^2 + y^2 + z^2 = 100 - k$. Spheres of radius $\sqrt{100 - k}$ centered at origin.

Valid for $k \leq 100$. Max temperature $T = 100$ at origin.

Problem 5. The wind chill index is $W(v, T) = 35.74 + 0.6215T - 35.75v^{0.16} + 0.4275Tv^{0.16}$. Find W at $T = 0^\circ\text{F}$ and $v = 25$ mph. What do the level curves represent physically?

$$v^{0.16} = 25^{0.16} \approx 1.673$$

$$W = 35.74 + 0 - 35.75(1.673) + 0 = 35.74 - 59.83 \approx -24.1^\circ\text{F}$$

Level curves $W = k$: combinations of wind speed and temperature that “feel” the same.

Lecture 7: Limits and Continuity in Several Variables

Background & Motivation

Before we can differentiate multivariate functions, we need limits. The key new difficulty: in one variable, you approach a point from the left or right. In two variables, you can approach from infinitely many directions. This makes proving limits exist much harder—and gives us a powerful tool (path analysis) for showing they don't.

1. Multivariable Limits

Definition 1: Limit of $f(x, y)$ as $(x, y) \rightarrow (a, b)$

We write $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ if $f(x, y)$ can be made arbitrarily close to L for all (x, y) sufficiently close to (a, b) .

Key difference from single-variable:

- In 1D – only **two directions** (left and right)
- In 2D – must approach from **infinitely many directions and paths**
- If *any two paths* give different limits $\implies$ limit **DNE**

Properties of Multivariable Limits

If $\lim_{(x,y) \rightarrow (a,b)} f = L$ and $\lim_{(x,y) \rightarrow (a,b)} g = M$ exist:

- $\lim(f \pm g) = L \pm M$
- $\lim(cf) = cL$
- $\lim(fg) = LM$
- $\lim(f/g) = L/M$ if $M \neq 0$

Polynomials and rational functions (where defined) – use **direct substitution**.

2. Techniques for Evaluating Limits

Technique 1: Direct Substitution

– Always try first; works when f is continuous at (a, b)

Technique 2: Algebraic Simplification / Rationalization

– Factor, multiply by conjugate, or simplify before substituting

Example 1: Rationalization

Evaluate $\lim_{(x,y) \rightarrow (1,1)} \frac{\sqrt{x} - \sqrt{y}}{x - y}$.

1. Rationalize:

$$\frac{\sqrt{x} - \sqrt{y}}{x - y} \cdot \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \frac{x - y}{(x - y)(\sqrt{x} + \sqrt{y})} = \frac{1}{\sqrt{x} + \sqrt{y}}$$

2. Substitute:

$$\lim_{(x,y) \rightarrow (1,1)} \frac{1}{\sqrt{x} + \sqrt{y}} = \frac{1}{1 + 1} = \boxed{\frac{1}{2}}$$

Technique 3: Polar / Spherical Coordinates

- For limits at the origin: $x = r \cos \theta$, $y = r \sin \theta$, $(x, y) \rightarrow (0, 0) \iff r \rightarrow 0^+$
- If the result is independent of θ and tends to L as $r \rightarrow 0$, the limit is L

Example 2: Polar Coordinates

Evaluate $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2}$.

1. Convert: In spherical coordinates, $x^2 + y^2 + z^2 = \rho^2$

2. Simplify: $\lim_{\rho \rightarrow 0^+} \frac{\sin(\rho^2)}{\rho^2}$

3. Evaluate: Let $u = \rho^2 \rightarrow 0$: $\lim_{u \rightarrow 0} \frac{\sin u}{u} = \boxed{1}$

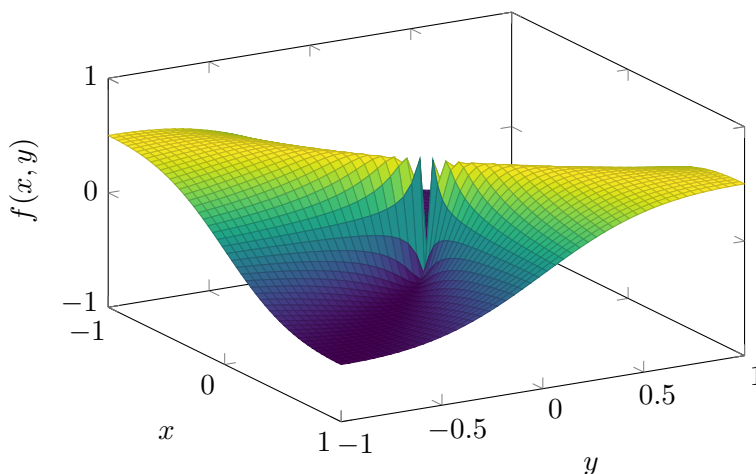
Technique 4: Path Analysis (for showing DNE)

- If two different paths give different limiting values $\implies$ limit DNE

Common Paths to Try

- Along axes: $y = 0$, then $x = 0$
- Along lines: $y = mx$ for various m
- Along curves: $y = x^2$, $y = x^3$, etc.

If all linear paths give the same value – try a curved path before concluding the limit exists!



Graph of $z = \frac{xy}{x^2 + y^2}$ near the origin. The surface takes different limiting values along different paths to $(0, 0)$: along $y = 0$ or $x = 0$ it equals 0, but along $y = x$ it equals $1/2$. Hence the limit does not exist.

Example 3: Limit DNE by Path Analysis

Show $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

1. **Path along $y = 0$:** $\frac{x^2 - 0}{x^2 + 0} = 1 \implies \text{limit} = 1$
2. **Path along $x = 0$:** $\frac{0 - y^2}{0 + y^2} = -1 \implies \text{limit} = -1$
3. **Result:** $1 \neq -1$, so the limit **DNE**.

Example 4: Tricky: All Lines Give 0, But Limit DNE

Does $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$ exist?

1. **Along $y = mx$:** $\frac{x \cdot m^2 x^2}{x^2 + m^4 x^4} = \frac{m^2 x}{1 + m^4 x^2} \rightarrow 0$ as $x \rightarrow 0$

Every straight line gives limit = 0.

2. **Along $x = y^2$:** $\frac{y^2 \cdot y^2}{y^4 + y^4} = \frac{y^4}{2y^4} = \frac{1}{2}$
3. **Result:** $0 \neq 1/2$, so the limit **DNE**.

Lesson: Checking only straight-line paths is not sufficient!

3. Continuity

Definition 2: Continuity of $f(x, y)$

f is **continuous at (a, b)** if all three hold:

1. $f(a, b)$ is defined
2. $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists
3. $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$

Polynomials, exponentials, trig functions, and their compositions are continuous on their domains.

Practice Problems

Problem 1. Evaluate $\lim_{(x,y) \rightarrow (2,3)} (x^2 y + xy^2 - 1)$.

Answer: $(4)(3) + (2)(9) - 1 = 12 + 18 - 1 = 29$

Problem 2. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$.

Polar: $\frac{r^2 \cos^2 \theta \cdot r \sin \theta}{r^2} = r \cos^2 \theta \sin \theta$

$|r \cos^2 \theta \sin \theta| \leq r \rightarrow 0$, so the limit is $\boxed{0}$ by Squeeze Theorem.

Problem 3. Show $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist.

Along $y = 0$: $0/x^2 = 0$, limit = 0.

Along $y = x$: $x^2/(2x^2) = 1/2$, limit = $1/2$.

$0 \neq 1/2 \implies$ DNE.

Problem 4. Evaluate $\lim_{(x,y) \rightarrow (1,1)} \frac{x^2 - y^2}{x - y}$.

Factor: $\frac{(x-y)(x+y)}{x-y} = x + y \rightarrow 1 + 1 = \boxed{2}$ (cancel since $x \neq y$ in the limit).

Problem 5. Show $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2}$ does not exist.

Along $y = 0$: $\frac{0}{0+x^2} = 0$, limit = 0.

Along $y = x$: $\frac{x^4}{x^4+0} = 1$, limit = 1.

$0 \neq 1 \implies$ DNE.

Problem 6. Prove using ϵ - δ that $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 y}{x^2 + y^2} = 0$.

Hint: Use $|y| \leq \sqrt{x^2 + y^2}$.

$$\left| \frac{3x^2 y}{x^2 + y^2} \right| = 3 \cdot \frac{x^2}{x^2 + y^2} \cdot |y| \leq 3 \cdot 1 \cdot |y| \leq 3\sqrt{x^2 + y^2}$$

Given $\epsilon > 0$, choose $\delta = \epsilon/3$. If $\sqrt{x^2 + y^2} < \delta$:

$$\left| \frac{3x^2 y}{x^2 + y^2} \right| \leq 3\sqrt{x^2 + y^2} < 3\delta = \epsilon. \quad \square$$

Lecture 8: Partial Derivatives

Background & Motivation

Now we differentiate. The idea is simple: hold all variables constant except one, and differentiate with respect to that one. This gives us partial derivatives—the multivariate analog of $f'(x)$. Today we define them, compute them, and see how they measure the slope of a surface in each coordinate direction.

1. Partial Derivatives

Definition 1: Partial Derivative (Limit Definition)

The **partial derivative** $f_x(a, b)$ is the instantaneous rate of change of $f(x, y)$ in the x -direction at the point (a, b) , with y held fixed at b . Geometrically, slice the graph of f with the vertical plane $y = b$; the resulting cross-section is a single-variable curve $z = f(x, b)$, and $f_x(a, b)$ is the **slope of the tangent line** to that curve at $x = a$. In other words, partial derivatives are ordinary single-variable slopes taken along axis-aligned slices of the surface.

$$f_x(x, y) = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

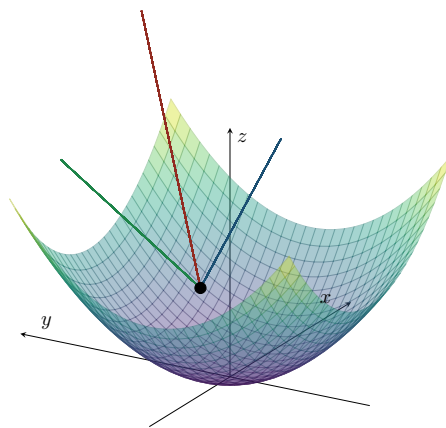
With respect to y :

$$f_y(x, y) = \frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Practical rule: To find f_x , treat y as a constant and differentiate with respect to x . For f_y , treat x as constant.

Geometric Interpretation

- $f_x(a, b)$ is the slope of the tangent line to $z = f(x, b)$ at $x = a$
- Measures rate of change of f in the x -direction while y is held fixed
- Similarly, $f_y(a, b)$ is the slope of $z = f(a, y)$ at $y = b$



Tangent line (red) at $(1, 1, 2)$ on $z = x^2 + y^2$. Blue dashed: xz -projection (slope $= f_x = 2$). Green dashed: yz -projection (slope $= f_y = 2$).

Example 1: Using the Limit Definition

Use the limit definition to find f_x for $f(x, y) = x^2y + 3xy^2$.

1. **Set up:** $f_x = \lim_{h \rightarrow 0} \frac{(x+h)^2y + 3(x+h)y^2 - x^2y - 3xy^2}{h}$

2. **Expand:** $= \lim_{h \rightarrow 0} \frac{x^2y + 2xhy + h^2y + 3xy^2 + 3hy^2 - x^2y - 3xy^2}{h}$
 $= \lim_{h \rightarrow 0} \frac{2xhy + h^2y + 3hy^2}{h}$

3. **Simplify and evaluate:** $= \lim_{h \rightarrow 0} (2xy + hy + 3y^2) = \boxed{2xy + 3y^2}$

Example 2: Computing Partial Derivatives

Find all first partial derivatives:

(a) $f(x, y) = x^3e^y + \sin(xy)$

$$f_x = 3x^2e^y + y \cos(xy)$$

$$f_y = x^3e^y + x \cos(xy)$$

(b) $g(x, y) = \frac{x^2}{x + y}$

$$g_x = \frac{2x(x+y) - x^2}{(x+y)^2} = \frac{x^2 + 2xy}{(x+y)^2} \quad (\text{quotient rule in } x)$$

$$g_y = \frac{0 - x^2}{(x+y)^2} = \frac{-x^2}{(x+y)^2}$$

(c) $h(x, y, z) = xye^z + z \ln(x + y)$

$$h_x = ye^z + \frac{z}{x+y}, \quad h_y = xe^z + \frac{z}{x+y}, \quad h_z = xye^z + \ln(x + y)$$

2. Derivative Rules for Partialials

Rules (Hold Other Variables Constant)

- **Linearity:** $\partial_x[af + bg] = af_x + bg_x$
- **Product:** $\partial_x[fg] = f_xg + fg_x$
- **Quotient:** $\partial_x[f/g] = (f_xg - fg_x)/g^2$
- **Chain:** $\partial_x[f(g(x, y))] = f'(g) \cdot g_x$
- **Zero partial:** if $f_x = 0$ everywhere $\implies f$ does not depend on x
- **Constants:** $\partial_x[c] = 0$; treat other variables as constants

3. Higher-Order Partial Derivatives

Definition 2: Second-Order Partial Derivatives

A **second-order partial derivative** is a partial derivative of a partial derivative. The two *pure* second partials f_{xx} and f_{yy} measure **concavity along coordinate directions** — how fast the slope f_x changes as you move further in x , and similarly for y — exactly analogous to the second derivative of a single-variable function. The two *mixed* partials f_{xy} and f_{yx} measure something new: how the slope in *one* direction changes as you move in the *other* direction, capturing the graph's twist or saddle-like behavior at a point.

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \quad (\text{pure, in } x)$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} \quad (\text{pure, in } y)$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \quad (\text{mixed: } x \text{ first, then } y)$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} \quad (\text{mixed: } y \text{ first, then } x)$$

Theorem 1: Clairaut's Theorem (Equality of Mixed Partial)

If f_{xy} and f_{yx} are both continuous on an open region, then:

$$f_{xy} = f_{yx}$$

The order of differentiation does not matter (under mild continuity conditions).

Why Clairaut's Theorem Works (Proof Outline)

- Consider the second-order difference quotient over a small rectangle $[x, x + h] \times [y, y + k]$
- Define $\Delta = f(x + h, y + k) - f(x + h, y) - f(x, y + k) + f(x, y)$
- Apply the Mean Value Theorem twice: first in x , then in $y \implies \Delta = f_{xy}(c_1, d_1) \cdot hk$
- Apply MVT in the other order: first in y , then in $x \implies \Delta = f_{yx}(c_2, d_2) \cdot hk$
- As $h, k \rightarrow 0$, continuity gives $f_{xy}(x, y) = f_{yx}(x, y)$

When Clairaut Fails – A Counterexample

- Consider $f(x, y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$
- One can show $f_{xy}(0, 0) = -1$ and $f_{yx}(0, 0) = 1$ – the mixed partials are *not equal* at the origin because they are not continuous there

Example 3: Clairaut Verification

Let $f(x, y) = x^3y^2 + e^{xy}$. Find all four second-order partials and verify $f_{xy} = f_{yx}$.

1. First partials:

$$f_x = 3x^2y^2 + ye^{xy}, \quad f_y = 2x^3y + xe^{xy}$$

2. Second partials:

$$f_{xx} = 6xy^2 + y^2e^{xy}$$

$$f_{yy} = 2x^3 + x^2e^{xy}$$

$$f_{xy} = \frac{\partial}{\partial y}[3x^2y^2 + ye^{xy}] = 6x^2y + e^{xy} + xye^{xy}$$

$$f_{yx} = \frac{\partial}{\partial x}[2x^3y + xe^{xy}] = 6x^2y + e^{xy} + xye^{xy}$$

3. Result: $f_{xy} = f_{yx}$ ✓

Example 4: Application: Container Design

A cylindrical container has volume $V(r, h) = \pi r^2h$ and surface area $S(r, h) = 2\pi rh + 2\pi r^2$. If $r = 5$ cm and $h = 10$ cm, how does V change if we increase r by 0.1 cm? If we increase h by 0.1 cm?

1. Compute partials of V :

$$V_r = 2\pi rh = 2\pi(5)(10) = 100\pi \approx 314 \text{ cm}^3/\text{cm}$$

$$V_h = \pi r^2 = 25\pi \approx 78.5 \text{ cm}^3/\text{cm}$$

2. Estimate changes:

$$\Delta r = 0.1: \Delta V \approx V_r \cdot 0.1 = 10\pi \approx 31.4 \text{ cm}^3$$

$$\Delta h = 0.1: \Delta V \approx V_h \cdot 0.1 = 2.5\pi \approx 7.85 \text{ cm}^3$$

3. Result: Increasing radius has $\sim 4\times$ more effect on volume than increasing height.

Practice Problems

Problem 1. Find f_x and f_y for $f(x, y) = x^2 \cos y + y \ln x$.

Answer: $f_x = 2x \cos y + y/x$, $f_y = -x^2 \sin y + \ln x$

Problem 2. Find all first partials of $f(x, y, z) = xyz + z^2 \sin(x)$.

$$f_x = yz + z^2 \cos x, \quad f_y = xz, \quad f_z = xy + 2z \sin x$$

Problem 3. For $f(x, y) = \arctan(y/x)$, show that $f_{xx} + f_{yy} = 0$ (i.e., f is **harmonic**).

$$f_x = \frac{-y/x^2}{1+y^2/x^2} = \frac{-y}{x^2+y^2}, \quad f_y = \frac{1/x}{1+y^2/x^2} = \frac{x}{x^2+y^2}$$

$$f_{xx} = \frac{2xy}{(x^2+y^2)^2}, \quad f_{yy} = \frac{-2xy}{(x^2+y^2)^2}$$

$$f_{xx} + f_{yy} = 0 \quad \checkmark$$

Problem 4. Use the limit definition to compute $f_x(0,0)$ for $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = \boxed{0}$$

Problem 5. Find f_{xxy} for $f(x,y) = e^x \sin(xy)$.

$$f_x = e^x \sin(xy) + ye^x \cos(xy)$$

$$f_{xx} = e^x \sin(xy) + ye^x \cos(xy) + ye^x \cos(xy) - y^2 e^x \sin(xy) = e^x [(1 - y^2) \sin(xy) + 2y \cos(xy)]$$

$$f_{xxy} = e^x [-2y \sin(xy) + x(1 - y^2) \cos(xy) + 2 \cos(xy) - 2xy \sin(xy)]$$

Lecture 9: The Chain Rule for Multivariable Functions

Background & Motivation

In Calc I, the chain rule handled compositions like $f(g(x))$. Now our functions have multiple intermediate variables, each depending on one or more parameters. The multivariate chain rule keeps track of all the paths from input to output—and in matrix form, it becomes the Jacobian.

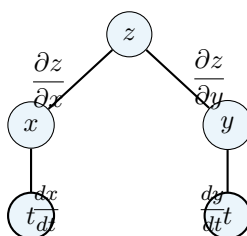
1. Chain Rule: One Parameter (Case 1)

Theorem 1: Chain Rule – Case 1: $z = f(x, y)$, $x = x(t)$, $y = y(t)$

If $z = f(x, y)$ where x and y are both functions of a single parameter t :

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

– Each intermediate variable contributes one term: (partial w.r.t. that variable) $\times$ (derivative of that variable w.r.t. t)



Tree diagram for Case 1. z depends on x and y ; each of x, y depends on t . To find $\frac{dz}{dt}$, walk every path from z down to t , multiply the derivatives along each path, and sum over all paths.

Example 1: $z = x^2y$ with $x = t^2$, $y = \sin t$

Find $\frac{dz}{dt}$.

1. Compute partials and derivatives:

$$\frac{\partial z}{\partial x} = 2xy = 2t^2 \sin t, \quad \frac{dx}{dt} = 2t$$

$$\frac{\partial z}{\partial y} = x^2 = t^4, \quad \frac{dy}{dt} = \cos t$$

2. Apply chain rule:

$$\frac{dz}{dt} = (2t^2 \sin t)(2t) + (t^4)(\cos t)$$

3. Result: $\frac{dz}{dt} = 4t^3 \sin t + t^4 \cos t$

2. Chain Rule: Multiple Parameters (Case 2)

Theorem 2: Chain Rule – Case 2: $z = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}$$

Matrix Form of the Chain Rule

$$\begin{pmatrix} \frac{\partial z}{\partial u} \\ \frac{\partial z}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{pmatrix}$$

– The chain rule in matrix form is the Jacobian matrix

Example 2: $z = \ln(xy)$ with $x = e^u$, $y = \cos v$

Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

1. Partial derivatives of z :

$$\frac{\partial z}{\partial x} = \frac{1}{x}, \quad \frac{\partial z}{\partial y} = \frac{1}{y}$$

2. Partial derivatives of x, y :

$$\frac{\partial x}{\partial u} = e^u, \quad \frac{\partial y}{\partial u} = 0, \quad \frac{\partial x}{\partial v} = 0, \quad \frac{\partial y}{\partial v} = -\sin v$$

3. Apply chain rule:

$$\frac{\partial z}{\partial u} = \frac{1}{x} \cdot e^u + \frac{1}{y} \cdot 0 = \frac{e^u}{e^u} = \boxed{1}$$

$$\frac{\partial z}{\partial v} = \frac{1}{x} \cdot 0 + \frac{1}{y} \cdot (-\sin v) = \frac{-\sin v}{\cos v} = \boxed{-\tan v}$$

3. Implicit Differentiation

Implicit Differentiation via Partial Derivatives

If $F(x, y) = 0$ defines y implicitly as a function of x :

$$\frac{dy}{dx} = -\frac{F_x}{F_y} \quad (\text{provided } F_y \neq 0)$$

If $F(x, y, z) = 0$ defines z implicitly as a function of x and y :

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

– *Why it works:* Differentiate $F(x, y(x)) = 0$ with respect to x using the chain rule: $F_x + F_y \frac{dy}{dx} = 0$, then solve

Example 3: Implicit Differentiation: $x^3 + y^3 = 6xy$

Find $\frac{dy}{dx}$.

1. **Set up:** Let $F(x, y) = x^3 + y^3 - 6xy$

2. **Compute partials:**

$$F_x = 3x^2 - 6y, \quad F_y = 3y^2 - 6x$$

3. **Apply formula:**

$$\frac{dy}{dx} = -\frac{3x^2 - 6y}{3y^2 - 6x} = \boxed{\frac{2y - x^2}{y^2 - 2x}}$$

4. Total Differentials

Definition 1: Total Differential

The **total differential** of $f(x, y)$:

$$df = f_x dx + f_y dy$$

– This is the best linear approximation: $\Delta f \approx df$ for small $dx = \Delta x$, $dy = \Delta y$

For $f(x, y, z)$: $df = f_x dx + f_y dy + f_z dz$

Example 4: Temperature Estimation with Differentials

The temperature in a room is modeled by $T(x, y) = 100 - x^2 - 2y^2$ (degrees). A sensor moves from $(1, 2)$ to $(1.05, 2.1)$. Estimate the change in temperature.

1. **Compute partials at $(1, 2)$:**

$$T_x = -2x = -2, \quad T_y = -4y = -8$$

2. **Apply differential:**

$$dT = T_x dx + T_y dy = (-2)(0.05) + (-8)(0.1) = -0.1 - 0.8$$

3. **Result:** $dT = \boxed{-0.9 \text{ degrees}}$ (temperature drops by $\approx 0.9^\circ$)

Practice Problems

Problem 1. If $z = x^2 + y^2$, $x = t$, $y = e^t$, find $\frac{dz}{dt}$.

Answer: $\frac{dz}{dt} = 2x \cdot 1 + 2y \cdot e^t = 2t + 2e^{2t}$

Problem 2. Let $w = xy + yz + xz$, $x = u + v$, $y = u - v$, $z = uv$. Find $\frac{\partial w}{\partial u}$.

$$w_x = y + z, w_y = x + z, w_z = x + y. \quad x_u = 1, y_u = 1, z_u = v.$$

$$\frac{\partial w}{\partial u} = (y + z)(1) + (x + z)(1) + (x + y)(v)$$

$$= (u - v + uv) + (u + v + uv) + v(2u) = 2u + 2uv + 2uv = \boxed{2u(1 + 2v)}$$

Problem 3. Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$ using implicit differentiation with partials.

$$F(x, y) = x^2 + y^2 - 25. \quad F_x = 2x, F_y = 2y.$$

$$\frac{dy}{dx} = -\frac{2x}{2y} = \boxed{-\frac{x}{y}}$$

Problem 4. Use differentials to approximate $\sqrt{(3.02)^2 + (3.97)^2}$.

$$f(x, y) = \sqrt{x^2 + y^2}. \text{ At } (3, 4): f = 5, f_x = x/f = 3/5, f_y = y/f = 4/5.$$

$$df = \frac{3}{5}(0.02) + \frac{4}{5}(-0.03) = 0.012 - 0.024 = -0.012$$

$$\sqrt{(3.02)^2 + (3.97)^2} \approx 5 - 0.012 = \boxed{4.988}$$

Problem 5. If $z = f(x - y, xy)$ where f is differentiable, express $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$ in terms of f_1 and f_2 (partials of f w.r.t. its first and second arguments).

$$\text{Let } u = x - y, v = xy. \text{ Then } z_x = f_1 \cdot 1 + f_2 \cdot y \text{ and } z_y = f_1 \cdot (-1) + f_2 \cdot x.$$

$$xz_x + yz_y = x(f_1 + yf_2) + y(-f_1 + xf_2) = (x - y)f_1 + 2xyf_2 = \boxed{uf_1 + 2vf_2}$$

Lecture 10: Directional Derivatives & the Gradient

Background & Motivation

Partial derivatives measure change along the coordinate axes. But what if we want the rate of change in an arbitrary direction? The directional derivative answers this, and the gradient vector ∇f tells us the direction of fastest increase. The gradient is one of the most important objects in all of applied mathematics—it drives optimization, machine learning, physics, and engineering.

1. Notation Review

- We write vectors in bold: $\mathbf{x} = \langle x_1, x_2 \rangle$ or $\mathbf{x} = \langle x_1, x_2, x_3 \rangle$
- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ takes a vector input and returns a scalar
- Partial derivatives: f_x , f_y , or $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$

2. The Directional Derivative

Definition 1: Directional Derivative (Limit Definition)

The **directional derivative** of f at $\mathbf{p}$ in the direction of unit vector $\hat{\mathbf{u}}$:

$$D_{\hat{\mathbf{u}}}f(\mathbf{p}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{p} + h\hat{\mathbf{u}}) - f(\mathbf{p})}{h}$$

- Measures the rate of change of f in the direction $\hat{\mathbf{u}}$
- Special cases: $D_{\hat{\mathbf{i}}}f = f_x$ and $D_{\hat{\mathbf{j}}}f = f_y$
- $\hat{\mathbf{u}}$ must be a unit vector

3. The Gradient Vector

Definition 2: Gradient

The **gradient** of $f(x, y)$ – the column of all partial derivatives:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

For $f(x, y, z)$:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Theorem 1: Directional Derivative via the Gradient

If f is differentiable, then:

$$D_{\hat{\mathbf{u}}}f(\mathbf{p}) = \nabla f(\mathbf{p}) \cdot \hat{\mathbf{u}}$$

Example 1: Computing the Gradient at a Point

Let $f(x, y, z) = x^2y + yz^3 - \sin(xz)$. Find ∇f at $(1, 2, 0)$ and the directional derivative in the direction $\langle 1, -1, 2 \rangle$.

1. **Compute partials:**

$$f_x = 2xy - z \cos(xz), \quad f_y = x^2 + z^3, \quad f_z = 3yz^2 - x \cos(xz)$$

2. **Evaluate at $(1, 2, 0)$:**

$$\nabla f(1, 2, 0) = \langle 4, 1, -1 \rangle$$

3. **Get unit vector:** $\hat{\mathbf{u}} = \frac{1}{\sqrt{6}} \langle 1, -1, 2 \rangle$

4. **Compute:** $D_{\hat{\mathbf{u}}}f = \langle 4, 1, -1 \rangle \cdot \frac{1}{\sqrt{6}} \langle 1, -1, 2 \rangle = \frac{4 - 1 - 2}{\sqrt{6}} = \frac{1}{\sqrt{6}}$

Example 2: Verification: Gradient Formula vs. Limit Definition

For $f(x, y) = x^2 + 3xy + y^2$ at $(1, 2)$ in direction $\hat{\mathbf{u}} = \frac{1}{\sqrt{2}} \langle 1, 1 \rangle$, compute $D_{\hat{\mathbf{u}}}f$ both ways.

Method 1 – Gradient formula:

1. **Gradient:** $\nabla f = \langle 2x + 3y, 3x + 2y \rangle$

2. **Evaluate:** $\nabla f(1, 2) = \langle 8, 7 \rangle$

3. **Dot product:** $D_{\hat{\mathbf{u}}}f = \langle 8, 7 \rangle \cdot \frac{1}{\sqrt{2}} \langle 1, 1 \rangle = \frac{15}{\sqrt{2}}$

Method 2 – Limit definition:

1. **Set up:** $D_{\hat{\mathbf{u}}}f = \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) - f(1, 2)}{h}$

2. **Expand:** $f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) = \left(1 + \frac{h}{\sqrt{2}}\right)^2 + 3\left(1 + \frac{h}{\sqrt{2}}\right)\left(2 + \frac{h}{\sqrt{2}}\right) + \left(2 + \frac{h}{\sqrt{2}}\right)^2$

After expanding and subtracting $f(1, 2) = 12$: numerator = $\frac{15h}{\sqrt{2}} + O(h^2)$

3. **Result:** $D_{\hat{\mathbf{u}}}f = \lim_{h \rightarrow 0} \frac{15h/\sqrt{2}}{h} = \frac{15}{\sqrt{2}} \quad \checkmark \quad \text{Same answer!}$

4. Key Properties of the Gradient

Theorem 2: Gradient Properties

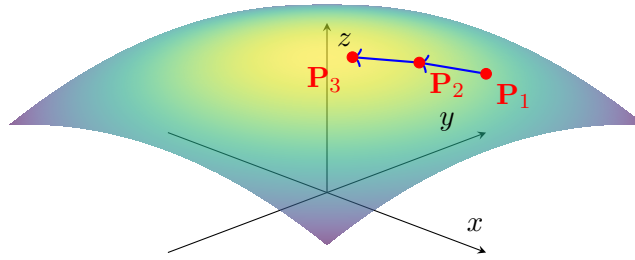
Let f be differentiable at $\mathbf{p}$ with $\nabla f(\mathbf{p}) \neq \mathbf{0}$. Then:

1. ∇f points in the direction of **steepest increase (ascent)**
2. $-\nabla f$ points in the direction of **steepest decrease (descent)**

3. $\|\nabla f\|$ equals the **maximum rate of change of f**
4. ∇f is **perpendicular to level curves/surfaces** of f
5. $D_{\hat{\mathbf{u}}}f = \|\nabla f\| \cos \theta$ where θ is the angle between ∇f and $\hat{\mathbf{u}}$

– **Why perpendicular to level curves?** If $\mathbf{r}(t)$ lies on $f = k$, then $\frac{d}{dt}[f(\mathbf{r}(t))] = \nabla f \cdot \mathbf{r}'(t) = 0$. Since $\mathbf{r}'(t)$ is tangent to the level curve, $\nabla f \perp$ level curve.

Gradient Ascent: Local Surface and Gradient Vectors



Gradient ascent on $f(x, y) = 4 - x^2 - y^2$. Starting at $\mathbf{P}_1$, each step moves in the direction of $+\nabla f$, climbing toward the maximum at the origin. Gradient descent (to minimize) uses $-\nabla f$ instead.

5. Gradient Ascent and Descent

Gradient Descent Algorithm

To minimize $f(\mathbf{x})$:

1. **Compute** $\nabla f(\mathbf{p}_n)$
2. **Unit direction:** $\hat{\mathbf{d}} = -\frac{\nabla f}{\|\nabla f\|}$ (steepest decrease)
3. **Update:** $\mathbf{p}_{n+1} = \mathbf{p}_n + \eta \hat{\mathbf{d}} = \mathbf{p}_n - \eta \frac{\nabla f(\mathbf{p}_n)}{\|\nabla f(\mathbf{p}_n)\|}$
4. **Iterate** until convergence

For **gradient ascent** (maximize), use $+\nabla f$ instead.

Example 3: Gradient Descent on $f = x^2 + 4y^2$

Minimize $f(x, y) = x^2 + 4y^2$ starting from $\mathbf{p}_0 = (4, 2)$ with step size $\eta = 0.5$. Do 3 iterations.

$$\nabla f = \langle 2x, 8y \rangle$$

$$\text{Iteration 1: } \nabla f(4, 2) = \langle 8, 16 \rangle, \quad \|\nabla f\| = \sqrt{64 + 256} = 8\sqrt{5}$$

$$\mathbf{p}_1 = (4, 2) - \frac{0.5}{8\sqrt{5}} \langle 8, 16 \rangle = (4, 2) - \langle 0.224, 0.447 \rangle \approx (3.776, 1.553)$$

$$f(\mathbf{p}_1) \approx 14.26 + 9.65 = 23.9 \quad (\text{down from } f(\mathbf{p}_0) = 32)$$

$$\text{Iteration 2: } \nabla f(3.776, 1.553) \approx \langle 7.55, 12.42 \rangle, \quad \|\nabla f\| \approx 14.53$$

$$\mathbf{p}_2 \approx (3.776, 1.553) - \frac{0.5}{14.53} \langle 7.55, 12.42 \rangle \approx (3.516, 1.126)$$

$$f(\mathbf{p}_2) \approx 12.36 + 5.07 = 17.4$$

$$\text{Iteration 3: } \nabla f(3.516, 1.126) \approx \langle 7.03, 9.01 \rangle, \quad \|\nabla f\| \approx 11.43$$

$$\mathbf{p}_3 \approx (3.516, 1.126) - \frac{0.5}{11.43} \langle 7.03, 9.01 \rangle \approx (3.209, 0.732)$$

$$f(\mathbf{p}_3) \approx 10.30 + 2.14 = 12.4$$

Each iteration decreases f : $32 \rightarrow 23.9 \rightarrow 17.4 \rightarrow 12.4$, converging toward the minimum at $(0, 0)$.

Example 4: Gradient Ascent on $f = -x^2 - y^2 + 4x + 4y$

Maximize $f(x, y) = -x^2 - y^2 + 4x + 4y$ starting from $\mathbf{p}_0 = (0, 0)$ with $\eta = 0.5$. Do 3 iterations.

$$\nabla f = \langle -2x + 4, -2y + 4 \rangle$$

Iteration 1: $\nabla f(0, 0) = \langle 4, 4 \rangle$, $\|\nabla f\| = 4\sqrt{2}$

$$\mathbf{p}_1 = (0, 0) + \frac{0.5}{4\sqrt{2}} \langle 4, 4 \rangle = \left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right) \approx (0.354, 0.354)$$

$$f(\mathbf{p}_1) \approx -0.125 - 0.125 + 1.414 + 1.414 = 2.58 \quad (\text{up from } f(\mathbf{p}_0) = 0)$$

Iteration 2: $\nabla f(0.354, 0.354) = \langle 3.29, 3.29 \rangle$, $\|\nabla f\| \approx 4.65$

$$\mathbf{p}_2 \approx (0.354, 0.354) + \frac{0.5}{4.65} \langle 3.29, 3.29 \rangle \approx (0.708, 0.708)$$

$$f(\mathbf{p}_2) \approx -0.501 - 0.501 + 2.832 + 2.832 = 4.66$$

Iteration 3: $\nabla f(0.708, 0.708) = \langle 2.58, 2.58 \rangle$, $\|\nabla f\| \approx 3.65$

$$\mathbf{p}_3 \approx (0.708, 0.708) + \frac{0.5}{3.65} \langle 2.58, 2.58 \rangle \approx (1.061, 1.061)$$

$$f(\mathbf{p}_3) \approx 5.88$$

Converging toward the maximum at $(2, 2)$ where $f(2, 2) = 8$: $0 \rightarrow 2.58 \rightarrow 4.66 \rightarrow 5.88$.

Connection to Machine Learning

- In ML, a model has parameters $\mathbf{w}$ and a **loss function** $L(\mathbf{w})$ measuring error
- Training is gradient descent on L : $\mathbf{w}_{n+1} = \mathbf{w}_n - \eta \nabla L(\mathbf{w}_n)$
- The step size η is called the **learning rate**
- This is how neural networks, logistic regression, and most modern AI models are trained

Practice Problems

Problem 1. Find ∇f for $f(x, y) = xe^y + \sin(xy)$ at the point $(0, \pi)$.

$$\nabla f = \langle e^y + y \cos(xy), xe^y + x \cos(xy) \rangle. \text{ At } (0, \pi): \nabla f = \langle e^\pi + \pi, 0 \rangle.$$

Answer: $\nabla f(0, \pi) = \langle e^\pi + \pi, 0 \rangle$

Problem 2. Find the maximum rate of change of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at $(3, 4, 12)$ and the direction in which it occurs.

$$\nabla f = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \langle x, y, z \rangle. \text{ At } (3, 4, 12): f = 13.$$

$$\nabla f = \frac{1}{13} \langle 3, 4, 12 \rangle. \text{ Max rate} = \|\nabla f\| = 1. \text{ Direction: } \frac{1}{13} \langle 3, 4, 12 \rangle.$$

Problem 3. Find the directional derivative of $f(x, y) = x^2 - 3y^2$ at $(2, -1)$ in the direction toward the point $(5, 3)$.

Direction: $\langle 5 - 2, 3 - (-1) \rangle = \langle 3, 4 \rangle$, $\hat{\mathbf{u}} = \langle 3/5, 4/5 \rangle$.

$\nabla f = \langle 2x, -6y \rangle$. At $(2, -1)$: $\nabla f = \langle 4, 6 \rangle$.

$D_{\hat{\mathbf{u}}}f = \langle 4, 6 \rangle \cdot \langle 3/5, 4/5 \rangle = 12/5 + 24/5 = 36/5$.

Problem 4. Perform 2 iterations of gradient descent on $f(x, y) = (x - 3)^2 + (y + 1)^2$ from $\mathbf{p}_0 = (0, 0)$ with $\eta = 1$.

$\nabla f = \langle 2(x - 3), 2(y + 1) \rangle$

Iteration 1: $\nabla f(0, 0) = \langle -6, 2 \rangle$, $\|\nabla f\| = \sqrt{40} = 2\sqrt{10}$.

$\mathbf{p}_1 = (0, 0) - \frac{1}{2\sqrt{10}} \langle -6, 2 \rangle = \left(\frac{3}{\sqrt{10}}, -\frac{1}{\sqrt{10}} \right) \approx (0.949, -0.316)$

$f(\mathbf{p}_1) \approx (0.949 - 3)^2 + (-0.316 + 1)^2 = 4.21 + 0.47 = 4.68$ (down from 10)

Iteration 2: $\nabla f(0.949, -0.316) = \langle -4.10, 1.37 \rangle$, $\|\nabla f\| \approx 4.32$

$\mathbf{p}_2 \approx (0.949, -0.316) - \frac{1}{4.32} \langle -4.10, 1.37 \rangle \approx (1.898, -0.633)$

$f(\mathbf{p}_2) \approx 1.21 + 0.13 = 1.35$ (converging toward min at $(3, -1)$)

Answer: $\mathbf{p}_2 \approx (1.90, -0.63)$

Problem 5. Show that ∇f is perpendicular to level curves. *Hint:* If $\mathbf{r}(t)$ is a curve on the level surface $f = k$, differentiate $f(\mathbf{r}(t)) = k$.

$$\frac{d}{dt}[f(\mathbf{r}(t))] = \frac{d}{dt}[k] = 0$$

By the chain rule: $\nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0$

Since $\mathbf{r}'(t)$ is tangent to the level curve, this says $\nabla f \perp$ the level curve. $\square$

Lecture 11: Tangent Planes & Linear Approximations

Background & Motivation

In Calc I, the tangent line at a point gives the best linear approximation: $f(x) \approx f(a) + f'(a)(x - a)$. Today we extend this to surfaces: the tangent plane is the best flat approximation to $z = f(x, y)$ near a point. This is the foundation of linearization, differentials, and error estimation in multiple dimensions.

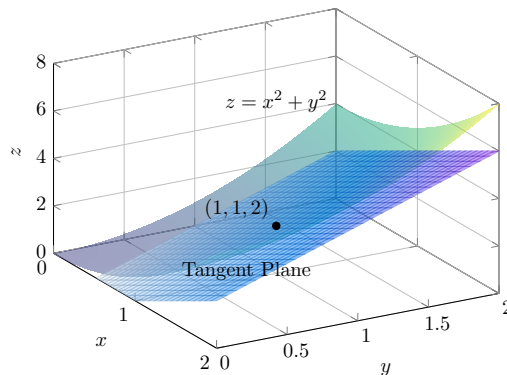
1. From Tangent Lines to Tangent Planes

- In Calc I: tangent line at $x = a$ gives $f(x) \approx f(a) + f'(a)(x - a)$
- In Calc III: the tangent *plane* at (a, b) is the best linear approximation to a surface

Theorem 1: Equation of the Tangent Plane

The tangent plane to $z = f(x, y)$ at the point $(a, b, f(a, b))$:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$



Example 1: Tangent Plane to a Paraboloid

Find the tangent plane to $z = x^2 + y^2$ at $(1, 1, 2)$.

1. **Compute partials:**

$$f_x = 2x, \quad f_y = 2y$$

2. **Evaluate at $(1, 1)$:**

$$f_x(1, 1) = 2, \quad f_y(1, 1) = 2, \quad f(1, 1) = 2$$

3. **Tangent plane:**

$$z = 2 + 2(x - 1) + 2(y - 1) = 2 + 2x - 2 + 2y - 2 = 2x + 2y - 2$$

Example 2: Tangent Plane to $z = y^4 + xy - 2y$ at $(1, 1)$

Find the equation of the tangent plane.

1. **Function value:** $f(1, 1) = 1 + 1 - 2 = 0$

2. **Partials:**

$$f_x = y \implies f_x(1, 1) = 1$$

$$f_y = 4y^3 + x - 2 \implies f_y(1, 1) = 4 + 1 - 2 = 3$$

3. **Result:** $z = 0 + 1(x - 1) + 3(y - 1) = x + 3y - 4$

2. Differentiability

Definition 1: Differentiability in Several Variables

$f(x, y)$ is **differentiable** at (a, b) if:

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x \Delta x + f_y \Delta y + E(\Delta x, \Delta y)$$

where the error E satisfies $\frac{E}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

- If f_x and f_y are **continuous** near (a, b) , then f is differentiable there
- Differentiable $\implies$ continuous (but not the other way around)

3. Linear Approximation

Linear Approximation Formula

$$f(x, y) \approx L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

- The tangent plane *is* the linear approximation
- Accuracy improves as (x, y) gets closer to (a, b)

Example 3: Practical Linear Approximation

Use linear approximation to estimate $f(2.1, 0.9)$ where $f(x, y) = x^2y - xy^3$.

1. **Base point:** $(a, b) = (2, 1)$, $f(2, 1) = 4 - 2 = 2$

2. **Partials at base point:**

$$f_x = 2xy - y^3 \implies f_x(2, 1) = 4 - 1 = 3$$

$$f_y = x^2 - 3xy^2 \implies f_y(2, 1) = 4 - 6 = -2$$

3. **Approximate:**

$$L(2.1, 0.9) = 2 + 3(0.1) + (-2)(-0.1) = 2 + 0.3 + 0.2 = 2.5$$

Check: $f(2.1, 0.9) = (4.41)(0.9) - (2.1)(0.729) = 3.969 - 1.531 = 2.438$. Error ≈ 0.06 .

4. Tangent Planes to Implicit Surfaces

– For a surface given implicitly as $F(x, y, z) = k$, the gradient ∇F is normal to the surface

Tangent Plane to $F(x, y, z) = k$ at (x_0, y_0, z_0)

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Equivalently: $\nabla F \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$

Example 4: Tangent Plane to a Sphere

Find the tangent plane to $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.

1. **Set up:** $F(x, y, z) = x^2 + y^2 + z^2$

2. **Gradient:** $\nabla F = \langle 2x, 2y, 2z \rangle$. At $(1, 2, 3)$: $\nabla F = \langle 2, 4, 6 \rangle$

3. **Tangent plane:**

$$2(x - 1) + 4(y - 2) + 6(z - 3) = 0$$

$$2x + 4y + 6z = 28 \implies \boxed{x + 2y + 3z = 14}$$

Practice Problems

Problem 1. Find the tangent plane to $z = xe^y$ at $(2, 0, 2)$.

$$f_x = e^y \implies f_x(2, 0) = 1. \quad f_y = xe^y \implies f_y(2, 0) = 2.$$

$$z = 2 + 1(x - 2) + 2(y - 0) = x + 2y.$$

Answer: $z = x + 2y$

Problem 2. Use linear approximation to estimate $\sqrt{(3.02)^2 + (3.97)^2}$.

Hint: Let $f(x, y) = \sqrt{x^2 + y^2}$ with base point $(3, 4)$.

$$f(3, 4) = 5. \quad f_x = x/\sqrt{x^2 + y^2} = 3/5, \quad f_y = y/\sqrt{x^2 + y^2} = 4/5.$$

$$L(3.02, 3.97) = 5 + \frac{3}{5}(0.02) + \frac{4}{5}(-0.03) = 5 + 0.012 - 0.024 = 4.988.$$

Answer: 4.988

Problem 3. Find the tangent plane to $x^2 + 2y^2 + 3z^2 = 36$ at $(1, 2, 3)$.

$F(x, y, z) = x^2 + 2y^2 + 3z^2$. $\nabla F = \langle 2x, 4y, 6z \rangle$. At $(1, 2, 3)$: $\nabla F = \langle 2, 8, 18 \rangle$.

$$2(x - 1) + 8(y - 2) + 18(z - 3) = 0 \implies x + 4y + 9z = 36.$$

Answer: $x + 4y + 9z = 36$

Problem 4. Find the tangent plane to $z = \sin(xy)$ at $(\pi/2, 1, 1)$ and use it to approximate $f(1.5, 1.1)$.

$$f_x = y \cos(xy) \implies f_x(\pi/2, 1) = \cos(\pi/2) = 0$$

$$f_y = x \cos(xy) \implies f_y(\pi/2, 1) = \frac{\pi}{2} \cos(\pi/2) = 0$$

Tangent plane: $z = 1$. So $f(1.5, 1.1) \approx 1$.

Both partials vanish — the surface is flat at this peak.

Answer: tangent plane $z = 1$, $f(1.5, 1.1) \approx 1$

Problem 5. Two surfaces $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$ intersect along a curve. Find the tangent line to this curve at $(1, 1, 2)$.

Tangent plane to $z_1 = x^2 + y^2$: $z = 2 + 2(x - 1) + 2(y - 1)$, i.e., $2x + 2y - z = 2$.

Tangent plane to $z_2 = 4 - x^2 - y^2$: $z = 2 - 2(x - 1) - 2(y - 1)$, i.e., $2x + 2y + z = 6$.

Tangent line direction: $\langle 2, 2, -1 \rangle \times \langle 2, 2, 1 \rangle = \langle 4, -4, 0 \rangle \propto \langle 1, -1, 0 \rangle$.

Tangent line: $\mathbf{r}(t) = \langle 1 + t, 1 - t, 2 \rangle$.

Answer: $\mathbf{r}(t) = \langle 1 + t, 1 - t, 2 \rangle$

Lecture 12: Extrema and Critical Points

Background & Motivation

In Calc I, you found maxima and minima by setting $f'(x) = 0$. The same idea works here: set $\nabla f = \mathbf{0}$. But in multiple dimensions, something new appears—saddle points, where the surface goes up in one direction and down in another. Today we develop the second derivative test for classifying critical points, then tackle global extrema on closed regions by analyzing boundaries.

1. Critical Points in Several Variables

Definition 1: Critical Point

A point (a, b) is a **critical point** of $f(x, y)$ if:

$$\nabla f(a, b) = \mathbf{0} \quad \text{i.e.,} \quad f_x(a, b) = 0 \text{ and } f_y(a, b) = 0$$

or if ∇f is undefined at (a, b) .

- All local extrema occur at critical points (necessary condition)
- But not every critical point is an extremum (could be a saddle)

2. The Second Derivative Test

Theorem 1: Second Derivative Test for $f(x, y)$

At a critical point (a, b) , compute the **discriminant**:

$$D = f_{xx}(a, b) f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

Classification:

- $D > 0$ and $f_{xx} > 0$: **local minimum**
- $D > 0$ and $f_{xx} < 0$: **local maximum**
- $D < 0$: **saddle point**
- $D = 0$: **inconclusive**

The Hessian Matrix

The discriminant D is the determinant of the **Hessian**:

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}, \quad D = \det(H) = f_{xx}f_{yy} - (f_{xy})^2$$

- $D > 0$ means both eigenvalues of H have the same sign (bowl shape)
- $D < 0$ means eigenvalues have opposite signs (saddle shape)

Example 1: $f(x, y) = x^3 + y^3 - 3xy$

Find and classify all critical points.

1. **Find critical points:**

$$f_x = 3x^2 - 3y = 0 \implies y = x^2$$

$$f_y = 3y^2 - 3x = 0 \implies x = y^2$$

$$\text{Substituting: } x = (x^2)^2 = x^4 \implies x^4 - x = 0 \implies x(x^3 - 1) = 0$$

$$x = 0 \implies y = 0; \quad x = 1 \implies y = 1$$

Critical points: $(0, 0)$ and $(1, 1)$.

2. **Second partials:** $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$

3. **Classify:**

At $(0, 0)$: $D = (0)(0) - 9 = -9 < 0 \implies$ **saddle point**

At $(1, 1)$: $D = (6)(6) - 9 = 27 > 0$, $f_{xx} = 6 > 0 \implies$ **local minimum**, $f(1, 1) = -1$

Example 2: $f(x, y) = 2x^2 + y^2 - 4x - 2y + 6$

Find and classify all critical points.

1. **Critical points:**

$$f_x = 4x - 4 = 0 \implies x = 1$$

$$f_y = 2y - 2 = 0 \implies y = 1$$

One critical point: $(1, 1)$.

2. **Second partials:** $f_{xx} = 4$, $f_{yy} = 2$, $f_{xy} = 0$

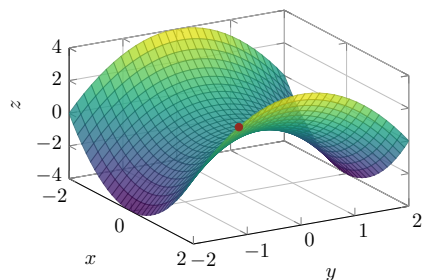
3. **Classify:** $D = (4)(2) - 0 = 8 > 0$, $f_{xx} = 4 > 0$

$\implies$ **Local minimum** at $(1, 1)$, $f(1, 1) = 2 + 1 - 4 - 2 + 6 = 3$

Also the global minimum since $f \rightarrow \infty$ as $\|(x, y)\| \rightarrow \infty$.

3. Saddle Points

- A **saddle point** is a critical point that is neither a max nor a min
- The surface curves up in one direction and down in another



A saddle point at $(0, 0)$ for $z = x^2 - y^2$. Along the x -axis it looks like a minimum; along the y -axis it looks like a maximum.

Example 3: The Classic Saddle: $z = x^2 - y^2$

$\nabla f = \langle 2x, -2y \rangle = \mathbf{0}$ at $(0, 0)$.

$f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$. $D = (2)(-2) - 0 = -4 < 0$. **Saddle point.**

- Along $y = 0$: $z = x^2$ (opens upward — looks like a minimum)
- Along $x = 0$: $z = -y^2$ (opens downward — looks like a maximum)
- The point $(0, 0)$ is simultaneously a min in the x -direction and a max in the y -direction!

4. Global Extrema on Closed Bounded Regions

Theorem 2: Extreme Value Theorem (Multivariate)

If f is continuous on a **closed, bounded** region D , then f attains both a global maximum and a global minimum on D .

Strategy for Finding Global Extrema

1. Find all critical points **inside** D (set $\nabla f = \mathbf{0}$)
2. Parameterize each piece of the **boundary**, reduce to 1D optimization on each edge
3. Evaluate f at all critical points (interior and boundary) and at all **corners**
4. Compare all values: largest = global max, smallest = global min

Example 4: Global Extrema on a Rectangle

Find the global max and min of $f(x, y) = x^2 + y^2 - 4x - 6y$ on $R = [0, 4] \times [0, 5]$.

1. Interior critical points:

$$f_x = 2x - 4 = 0 \implies x = 2, \quad f_y = 2y - 6 = 0 \implies y = 3$$

Interior critical point: $(2, 3) \in R$. $f(2, 3) = 4 + 9 - 8 - 18 = -13$.

2. Boundary analysis (4 edges):

Bottom ($y = 0, 0 \leq x \leq 4$): $f = x^2 - 4x$. $f' = 2x - 4 = 0 \implies x = 2$. $f(2, 0) = -4$.

Top ($y = 5, 0 \leq x \leq 4$): $f = x^2 - 4x - 5$. $f' = 2x - 4 = 0 \implies x = 2$. $f(2, 5) = -9$.

Left ($x = 0, 0 \leq y \leq 5$): $f = y^2 - 6y$. $f' = 2y - 6 = 0 \implies y = 3$. $f(0, 3) = -9$.

Right ($x = 4, 0 \leq y \leq 5$): $f = y^2 - 6y$. $f' = 2y - 6 = 0 \implies y = 3$. $f(4, 3) = -9$.

3. Corners:

$$f(0, 0) = 0, \quad f(4, 0) = 0, \quad f(0, 5) = -5, \quad f(4, 5) = -5$$

4. Compare all values: $\{-13, -4, -9, -9, -9, 0, 0, -5, -5\}$

Global max = 0 at $(0, 0)$ and $(4, 0)$. Global min = -13 at $(2, 3)$.

Practice Problems

Problem 1. Find and classify all critical points of $f(x, y) = x^2 + y^2 + 2x - 4y + 5$.

$$f_x = 2x + 2 = 0 \implies x = -1, \quad f_y = 2y - 4 = 0 \implies y = 2. \text{ Point: } (-1, 2).$$

$$D = (2)(2) - 0 = 4 > 0, \quad f_{xx} = 2 > 0. \text{ Local (and global) min. } f(-1, 2) = 1 + 4 - 2 - 8 + 5 = 0.$$

Answer: **local min at** $(-1, 2)$, $f = 0$

Problem 2. Find and classify all critical points of $f(x, y) = x^2y - x^2 - 2y^2$.

$$f_x = 2xy - 2x = 2x(y - 1) = 0 \implies x = 0 \text{ or } y = 1.$$

$$f_y = x^2 - 4y = 0.$$

$$\text{If } x = 0: -4y = 0 \implies y = 0. \text{ Point } (0, 0).$$

$$\text{If } y = 1: x^2 = 4 \implies x = \pm 2. \text{ Points } (\pm 2, 1).$$

$$f_{xx} = 2y - 2, \quad f_{yy} = -4, \quad f_{xy} = 2x.$$

$$(0, 0): D = (-2)(-4) - 0 = 8 > 0, \quad f_{xx} = -2 < 0. \text{ **Local max.** } f = 0.$$

$$(2, 1): D = (0)(-4) - 16 = -16 < 0. \text{ **Saddle.**}$$

$$(-2, 1): D = (0)(-4) - 16 = -16 < 0. \text{ **Saddle.**}$$

Answer: **local max at** $(0, 0)$; **saddles at** $(\pm 2, 1)$

Problem 3. Show that $f(x, y) = x^2 - y^2$ has a saddle point at the origin. Explain geometrically why it is neither a max nor a min.

$$f_x = 2x = 0, \quad f_y = -2y = 0 \implies \text{critical point } (0, 0).$$

$$D = (2)(-2) - 0 = -4 < 0. \text{ Saddle point.}$$

$$\text{Along } x\text{-axis } (y = 0): f = x^2 \geq 0, \text{ so } f(0, 0) = 0 \text{ is a minimum.}$$

$$\text{Along } y\text{-axis } (x = 0): f = -y^2 \leq 0, \text{ so } f(0, 0) = 0 \text{ is a maximum.}$$

Since f increases in some directions and decreases in others, it cannot be an extremum.

Problem 4. Find the global extrema of $f(x, y) = x^3 - 3x + y^2$ on $R = [-2, 2] \times [-1, 1]$.

Interior: $f_x = 3x^2 - 3 = 0 \implies x = \pm 1$, $f_y = 2y = 0 \implies y = 0$.

$$f(1, 0) = 1 - 3 = -2, \quad f(-1, 0) = -1 + 3 = 2.$$

$D = 6x \cdot 2 = 0$. At $(1, 0)$: $D = 12 > 0$, $f_{xx} = 6 > 0$, local min. At $(-1, 0)$: $D = -12 < 0$, saddle.

Boundaries: $f(2, y) = 2 + y^2$: min 2, max 3. $f(-2, y) = -2 + 6 + y^2 = 4 + y^2$: min 4, max 5.

$$f(x, \pm 1) = x^3 - 3x + 1: \text{crit at } x = \pm 1, \quad f(1, \pm 1) = -1, \quad f(-1, \pm 1) = 3.$$

Corners: $f(2, \pm 1) = 3$, $f(-2, \pm 1) = 5$.

Global max = 5 at $(-2, \pm 1)$. Global min = -2 at $(1, 0)$.

Answer: global max = 5 at $(-2, \pm 1)$; global min = -2 at $(1, 0)$

Problem 5. Find and classify all critical points of $f(x, y) = xy(1 - x - y)$.

$$f = xy - x^2y - xy^2$$

$$f_x = y - 2xy - y^2 = y(1 - 2x - y) = 0$$

$$f_y = x - x^2 - 2xy = x(1 - x - 2y) = 0$$

Case $y = 0$: $f_y = x(1 - x) = 0 \implies x = 0$ or $x = 1$. Points $(0, 0)$, $(1, 0)$.

Case $x = 0$: $f_x = y(1 - y) = 0 \implies y = 0$ or $y = 1$. Points $(0, 0)$, $(0, 1)$.

Case $1 - 2x - y = 0$ and $1 - x - 2y = 0$: subtract to get $x = y$. Then $1 - 3x = 0 \implies x = 1/3$. Point $(1/3, 1/3)$.

$$f_{xx} = -2y, \quad f_{yy} = -2x, \quad f_{xy} = 1 - 2x - 2y.$$

$(0, 0)$: $D = 0 - 1 = -1 < 0$. Saddle. $(1, 0)$: $D = 0 - (-1)^2 = -1 < 0$. Saddle. $(0, 1)$: $D = 0 - (-1)^2 = -1 < 0$. Saddle.

$(1/3, 1/3)$: $D = (-2/3)(-2/3) - (-1/3)^2 = 4/9 - 1/9 = 1/3 > 0$, $f_{xx} = -2/3 < 0$. **Local max.** $f = 1/27$.

Answer: **saddles at $(0, 0)$, $(1, 0)$, $(0, 1)$; local max at $(1/3, 1/3)$, $f = 1/27$**

Lecture 13: Double Integrals over Rectangular Regions

Background & Motivation

We now enter the second major part of the course: integration in multiple dimensions. Just as the single integral $\int_a^b f(x) dx$ computes area under a curve, the double integral $\iint_R f(x, y) dA$ computes volume under a surface. Today we start with rectangular regions and Fubini's Theorem, which lets us evaluate double integrals as iterated single integrals. This is the gateway to computing volumes, masses, centers of mass, and much more over two- and three-dimensional regions.

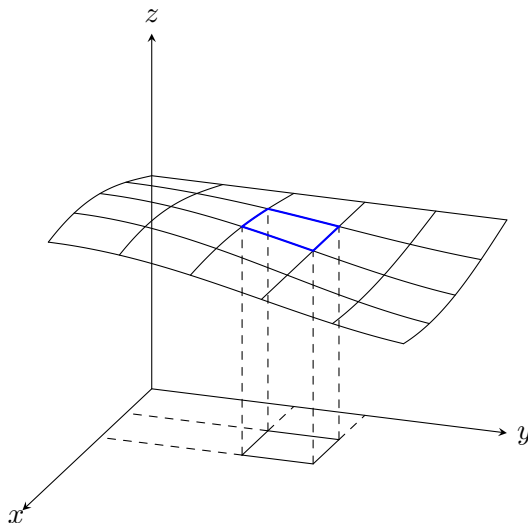
1. Double Integrals as Volume via Riemann Sums

Definition 1: Double Integral

The **double integral** of $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$:

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

- Partition R into $m \times n$ subrectangles, each with area $\Delta A = \Delta x \cdot \Delta y$
- Pick a sample point (x_{ij}^*, y_{ij}^*) in each subrectangle
- If $f \geq 0$: the double integral = **volume under** $z = f(x, y)$ **above** R



The double integral extends the idea of area under a curve to volume under a surface. Each sub-rectangle in the xy -plane gets a little prism of height $f(x, y)$; adding them up in the limit gives $\iint_R f dA$.

Example 1: Riemann Sum Approximation

Approximate $\iint_R (x^2 + y^2) dA$ over $R = [0, 2] \times [0, 1]$ using a 2×2 grid with upper-right sample points.

1. Set up grid:

$$\Delta x = 1, \quad \Delta y = 0.5, \quad \Delta A = 0.5$$

Upper-right sample points: $(1, 0.5)$, $(2, 0.5)$, $(1, 1)$, $(2, 1)$

2. Evaluate f at each sample point:

$$f(1, 0.5) = 1 + 0.25 = 1.25$$

$$f(2, 0.5) = 4 + 0.25 = 4.25$$

$$f(1, 1) = 1 + 1 = 2$$

$$f(2, 1) = 4 + 1 = 5$$

3. Compute sum:

$$\sum f \cdot \Delta A = (1.25 + 4.25 + 2 + 5)(0.5) = 12.5 \times 0.5 = 6.25$$

Exact answer (computed below): $10/3 \approx 3.33$. The Riemann sum overestimates because f is increasing and we used upper-right points.

2. Fubini's Theorem and Iterated Integrals

Theorem 1: Fubini's Theorem

If f is continuous on $R = [a, b] \times [c, d]$, then:

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

- The order of integration does not matter for continuous functions on rectangles
- Evaluate from the **inside out**: inner integral first, then outer

Example 2: Iterated Integral of xy

Evaluate $\iint_R xy dA$ where $R = [0, 3] \times [0, 2]$.

1. Set up (integrate y first):

$$\int_0^3 \int_0^2 xy dy dx$$

2. Inner integral (treat x as constant):

$$\int_0^2 xy dy = \left[\frac{xy^2}{2} \right]_0^2 = 2x$$

3. Outer integral:

$$\int_0^3 2x dx = [x^2]_0^3 = 9$$

Example 3: Symmetry Shortcut

Evaluate $\iint_R \frac{y}{x^2 + y^2} dA$ where $R = [1, 2] \times [-1, 1]$.

Key insight: $f(x, y) = \frac{y}{x^2 + y^2}$ is an **odd function of y** : $f(x, -y) = -f(x, y)$.

The region R is symmetric about the x -axis (y ranges from -1 to 1).

By symmetry: $\iint_R \frac{y}{x^2 + y^2} dA = \boxed{0}$

No computation needed!

Symmetry Principle

If f is **odd** in y (i.e., $f(x, -y) = -f(x, y)$) and R is symmetric about the x -axis:

$$\iint_R f(x, y) dA = 0$$

Same idea for odd in x on a region symmetric about the y -axis.

3. Strategic Order of Integration

- Sometimes one order is much easier than the other
- Look for which order avoids difficult antiderivatives

Example 4: Computational Iterated Integral

Evaluate $\int_0^1 \int_0^2 (x^3 \sqrt{y} + \cos x) dy dx$.

1. **Inner integral** (integrate with respect to y):

$$\int_0^2 (x^3 y^{1/2} + \cos x) dy = \left[\frac{2x^3}{3} y^{3/2} + y \cos x \right]_0^2 = \frac{2x^3}{3} \cdot 2\sqrt{2} + 2 \cos x = \frac{4\sqrt{2}}{3} x^3 + 2 \cos x$$

2. **Outer integral:**

$$\int_0^1 \left(\frac{4\sqrt{2}}{3} x^3 + 2 \cos x \right) dx = \frac{4\sqrt{2}}{3} \cdot \frac{x^4}{4} \Big|_0^1 + 2 \sin x \Big|_0^1 = \frac{\sqrt{2}}{3} + 2 \sin 1$$

3. **Result:** $\boxed{\frac{\sqrt{2}}{3} + 2 \sin 1}$

Practice Problems

Problem 1. Evaluate $\int_0^1 \int_0^2 (x + 2y) dx dy$.

$$\int_0^1 \left[\frac{x^2}{2} + 2xy \right]_0^2 dy = \int_0^1 (2 + 4y) dy = [2y + 2y^2]_0^1 = 4.$$

Answer: 4

Problem 2. Evaluate $\iint_R e^{x+y} dA$ where $R = [0, 1] \times [0, 1]$.

$$\int_0^1 \int_0^1 e^x e^y dy dx = \int_0^1 e^x [e^y]_0^1 dx = (e - 1) \int_0^1 e^x dx = (e - 1)[e^x]_0^1 = (e - 1)^2.$$

Answer: $(e - 1)^2$

Problem 3. Verify Fubini's Theorem by evaluating $\iint_R 2xy dA$ over $R = [1, 3] \times [0, 2]$ in both orders.

Order 1: $\int_1^3 \int_0^2 2xy dy dx = \int_1^3 [xy^2]_0^2 dx = \int_1^3 4x dx = [2x^2]_1^3 = 18 - 2 = 16.$

Order 2: $\int_0^2 \int_1^3 2xy dx dy = \int_0^2 [x^2y]_1^3 dy = \int_0^2 8y dy = [4y^2]_0^2 = 16. \quad \checkmark$

Answer: 16 (both orders agree)

Problem 4. Without computing, explain why $\iint_{[-1,1] \times [-1,1]} x^3 y dA = 0.$

$f(x, y) = x^3 y$ is odd in x : $f(-x, y) = -x^3 y = -f(x, y).$

The region $[-1, 1] \times [-1, 1]$ is symmetric about the y -axis.

By symmetry, the contributions from $x > 0$ and $x < 0$ cancel exactly, giving 0.

Answer: 0 (by symmetry — f is odd in x on a symmetric region)

Problem 5. Evaluate $\iint_R (x^2 y - y^2) dA$ where $R = [0, 2] \times [-1, 1]$.

Split: $\iint_R x^2 y dA - \iint_R y^2 dA.$

First integral: $x^2 y$ is odd in y on $[-1, 1]$, so $\iint_R x^2 y dA = 0.$

Second integral: $\int_0^2 \int_{-1}^1 y^2 dy dx = \int_0^2 \left[\frac{y^3}{3} \right]_{-1}^1 dx = \int_0^2 \frac{2}{3} dx = \frac{4}{3}.$

Total: $0 - \frac{4}{3} = -\frac{4}{3}.$

Answer: $-4/3$

Lecture 14: Double Integrals over General Regions

Background & Motivation

Last time we integrated over rectangles. Most regions of interest—disks, triangles, regions between curves—aren't rectangular. Today we handle general regions using Type I and Type II setups, and introduce polar coordinates for circular symmetry. The Gaussian integral, which cannot be evaluated by single-variable methods, yields beautifully to the polar approach.

1. Warm-Up: Double Integral over a Rectangle

Evaluate $\int_0^2 \int_1^3 (3xy^2) dy dx$.

1. Inner integral (y): $\int_1^3 3xy^2 dy = 3x \left[\frac{y^3}{3} \right]_1^3 = x(27 - 1) = 26x$

2. Outer integral (x): $\int_0^2 26x dx = [13x^2]_0^2 = 52$

3. Result: 52

2. Type I Regions (Vertical Slicing)

Definition 1: Type I Region

$$R = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

- Integrate y first (inner), then x (outer)
- Think: vertical lines sweep left to right across the region

3. Type II Regions (Horizontal Slicing)

Definition 2: Type II Region

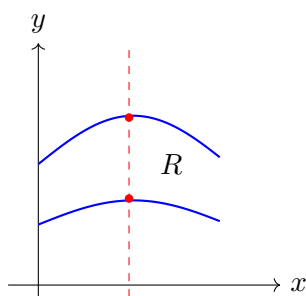
$$R = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

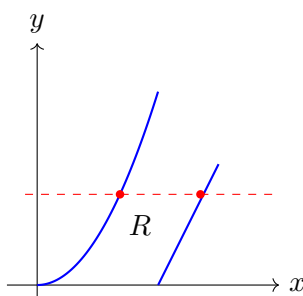
- Integrate x first (inner), then y (outer)
- Think: horizontal lines sweep bottom to top across the region

How to Classify a Region

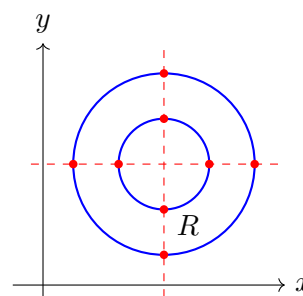
- Sketch the region and find all intersection points
- Type I – can every vertical line ($x = c$) enter through one curve and exit through one curve?
- Type II – can every horizontal line ($y = c$) enter through one curve and exit through one curve?
- Some regions work as both; pick whichever is simpler
- If neither works cleanly, **split** the region



Type I Region



Type II Region



Neither Type

A Type I region is bounded above and below by functions of x — every vertical line (red) hits the boundary exactly twice. Type II is analogous with horizontal lines. An annulus is *neither*: both horizontal and vertical slice lines can cross the boundary four times, so it must be split.

4. Examples

Example 1: Type I: Region Between Two Parabolas

Evaluate $\iint_R (x + 2y) dA$ where R is bounded by $y = x^2$ and $y = 2 - x^2$.

1. **Sketch and intersections:** $x^2 = 2 - x^2 \implies 2x^2 = 2 \implies x = \pm 1$.

Lower curve: $y = x^2$. Upper curve: $y = 2 - x^2$.

2. **Set up (Type I):**

$$\int_{-1}^1 \int_{x^2}^{2-x^2} (x + 2y) dy dx$$

3. **Inner integral (y):** $[xy + y^2]_{x^2}^{2-x^2} = x(2 - x^2) + (2 - x^2)^2 - x \cdot x^2 - x^4$
 $= 2x - x^3 + 4 - 4x^2 + x^4 - x^3 - x^4 = 2x - 2x^3 - 4x^2 + 4$

4. **Outer integral (x):** Note $2x - 2x^3$ is odd on $[-1, 1]$ so integrates to 0.

$$\int_{-1}^1 (-4x^2 + 4) dx = 2 \int_0^1 (-4x^2 + 4) dx = 2 \left[-\frac{4x^3}{3} + 4x \right]_0^1 = 2 \left(\frac{8}{3} \right)$$

5. **Result:** $\boxed{\frac{16}{3}}$

Example 2: Polar: Integral over a Disk

Evaluate $\iint_D (x^2 + y^2) dA$ where D is the disk $x^2 + y^2 \leq 4$.

1. **Convert to polar:** $x^2 + y^2 = r^2$, $dA = r dr d\theta$.

Disk: $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$.

2. **Set up:** $\int_0^{2\pi} \int_0^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} d\theta \cdot \int_0^2 r^3 dr$

3. **Evaluate:** $= 2\pi \cdot \left[\frac{r^4}{4}\right]_0^2 = 2\pi \cdot 4$

4. **Result:** $\boxed{8\pi}$

5. Polar Coordinates for Double Integrals

Double Integrals in Polar

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$$

– The extra factor of r comes from the area element: $dA = r dr d\theta$

– Use polar when the region or integrand involves $x^2 + y^2$, circles, or sectors

Example 3: The Gaussian Integral (I^2 Trick)

Evaluate $I = \int_{-\infty}^{\infty} e^{-x^2} dx$.

1. **Square the integral:** $I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy\right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA$

2. **Convert to polar:** $x^2 + y^2 = r^2$, region is all of $\mathbb{R}^2$.

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} \cdot r dr d\theta$$

3. **Inner integral:** Let $u = -r^2$, $du = -2r dr$: $\int_0^{\infty} r e^{-r^2} dr = \left[-\frac{1}{2}e^{-r^2}\right]_0^{\infty} = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$

4. **Outer integral:** $I^2 = 2\pi \cdot \frac{1}{2} = \pi$

5. **Result:** $I = \boxed{\sqrt{\pi}}$

This integral cannot be evaluated by single-variable techniques — the I^2 trick with polar coordinates is essential.

6. Complex Regions (Splitting)

When a region is neither purely Type I nor Type II – **split** it:

$$\iint_R f \, dA = \iint_{R_1} f \, dA + \iint_{R_2} f \, dA$$

You can even use Cartesian for one piece and polar for another.

Example 4: Split Region: Cartesian + Polar

Evaluate $\iint_R (1 + x^2 + y^2) \, dA$ where R is the region inside the square $[-1, 1] \times [-1, 1]$ but outside the unit disk $x^2 + y^2 < 1$, combined with the upper half-disk $x^2 + y^2 \leq 1, y \geq 0$.

1. **Split:** Let R_1 = upper half-disk ($x^2 + y^2 \leq 1, y \geq 0$) and R_2 = square minus full disk (corners).
2. R_1 **in polar:** $0 \leq r \leq 1, 0 \leq \theta \leq \pi$.

$$\iint_{R_1} (1 + r^2) \cdot r \, dr \, d\theta = \int_0^\pi \int_0^1 (r + r^3) \, dr \, d\theta = \pi \left[\frac{r^2}{2} + \frac{r^4}{4} \right]_0^1 = \frac{3\pi}{4}$$

3. R_2 : The corners of the square outside the disk can be set up as a Cartesian integral:

$$\iint_{\text{square}} (1 + x^2 + y^2) \, dA - \iint_{\text{disk}} (1 + x^2 + y^2) \, dA$$

Square integral: $\int_{-1}^1 \int_{-1}^1 (1 + x^2 + y^2) \, dy \, dx = \frac{20}{3}$.

Full disk integral (polar): $\int_0^{2\pi} \int_0^1 (1 + r^2)r \, dr \, d\theta = 2\pi \cdot \frac{3}{4} = \frac{3\pi}{2}$.

Corners: $\frac{20}{3} - \frac{3\pi}{2}$.

4. **Combine:** Total = $\frac{3\pi}{4} + \frac{20}{3} - \frac{3\pi}{2} = \frac{20}{3} - \frac{3\pi}{4}$

5. **Result:** $\frac{20}{3} - \frac{3\pi}{4}$

Practice Problems

Problem 1. Evaluate $\int_0^1 \int_0^x xy \, dy \, dx$.

Answer: $1/8$

Problem 2. Evaluate $\iint_R xy \, dA$ where R is bounded by $y = 0$, $y = \sqrt{x}$, and $x = 4$. (Type I)

Answer: $32/3$

Problem 3. Evaluate $\iint_D e^{-(x^2+y^2)} dA$ where D is the annulus $1 \leq x^2 + y^2 \leq 4$. (Use polar.)

Answer: $\pi(e^{-1} - e^{-4})$

Problem 4. Reverse the order of integration and evaluate: $\int_0^1 \int_0^{\sqrt{y}} f(x, y) dx dy$ where $f(x, y) = e^{y/x}$.

Answer: $\int_0^1 \int_{x^2}^1 e^{y/x} dy dx$

Problem 5. Evaluate $\iint_R \sqrt{x^2 + y^2} dA$ where R is the region in the first quadrant between $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$.

Answer: $\frac{\pi}{2} \cdot \frac{26}{3} = \frac{13\pi}{3}$

Lecture 15: Applications of Multiple Integrals

Background & Motivation

Now that we can integrate over general regions, we put multiple integrals to work. Mass, center of mass, moments of inertia, and average value all reduce to integrals with the right integrand — the same recipe in 2D and in 3D. We also derive the surface area formula, extending arc length to two dimensions.

1. Warm-Up: Double Integral Review

Evaluate $\int_0^1 \int_0^{1-x} (x+y) dy dx$.

1. Inner integral (y): $\int_0^{1-x} (x+y) dy = \left[xy + \frac{y^2}{2} \right]_0^{1-x} = x(1-x) + \frac{(1-x)^2}{2}$

2. Simplify: $x - x^2 + \frac{1-2x+x^2}{2} = \frac{1}{2} - \frac{x^2}{2}$

3. Outer integral: $\int_0^1 \left(\frac{1}{2} - \frac{x^2}{2} \right) dx = \left[\frac{x}{2} - \frac{x^3}{6} \right]_0^1 = \frac{1}{2} - \frac{1}{6} = \boxed{\frac{1}{3}}$

2. Mass and Center of Mass

Mass, Moments, and Center of Mass

For a lamina with density $\rho(x, y)$ over region R :

$$\text{Mass: } M = \iint_R \rho(x, y) dA$$

$$\text{Moments: } M_x = \iint_R y \rho(x, y) dA \quad (\text{about } x\text{-axis})$$

$$M_y = \iint_R x \rho(x, y) dA \quad (\text{about } y\text{-axis})$$

$$\text{Center of mass: } \bar{x} = \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M}$$

- M_x uses y (distance to x -axis); M_y uses x (distance to y -axis)
- If $\rho = \text{const}$, center of mass = centroid



Center of mass of a system of point masses.

Example 1: Center of Mass of a Triangular Lamina

Find the mass and center of mass of a triangular lamina with vertices $(0, 0)$, $(2, 0)$, $(0, 1)$ and density $\rho(x, y) = 1 + x$.

1. **Region (Type I):** $0 \leq x \leq 2$, $0 \leq y \leq 1 - x/2$.

$$2. \text{ Mass: } M = \int_0^2 \int_0^{1-x/2} (1+x) dy dx = \int_0^2 (1+x)(1-x/2) dx \\ = \int_0^2 \left(1 + \frac{x}{2} - \frac{x^2}{2}\right) dx = \left[x + \frac{x^2}{4} - \frac{x^3}{6}\right]_0^2 = 2 + 1 - \frac{4}{3} = \frac{5}{3}$$

$$3. \text{ Moment } M_y: M_y = \int_0^2 \int_0^{1-x/2} x(1+x) dy dx = \int_0^2 x(1+x)(1-x/2) dx \\ = \int_0^2 \left(x + \frac{x^2}{2} - \frac{x^3}{2}\right) dx = \left[\frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{8}\right]_0^2 = 2 + \frac{4}{3} - 2 = \frac{4}{3}$$

$$4. \text{ Moment } M_x: M_x = \int_0^2 \int_0^{1-x/2} y(1+x) dy dx = \int_0^2 (1+x) \frac{(1-x/2)^2}{2} dx = \frac{5}{12}$$

$$5. \text{ Center of mass: } \bar{x} = \frac{M_y}{M} = \frac{4/3}{5/3} = \frac{4}{5}, \quad \bar{y} = \frac{M_x}{M} = \frac{5/12}{5/3} = \frac{1}{4}$$

$$6. \text{ Result: } (\bar{x}, \bar{y}) = \left(\frac{4}{5}, \frac{1}{4}\right)$$

The center of mass shifts right because the density $1+x$ is heavier on the right.

3. Moments of Inertia

Moments of Inertia

$$I_x = \iint_R y^2 \rho dA \quad (\text{about the } x\text{-axis})$$

$$I_y = \iint_R x^2 \rho dA \quad (\text{about the } y\text{-axis})$$

$$I_z = I_0 = \iint_R (x^2 + y^2) \rho dA \quad (\text{polar moment / about the origin})$$

Note: $I_z = I_x + I_y$.

Physics connections:

- Rotational kinetic energy: $KE = \frac{1}{2}I\omega^2$
- Newton's second law for rotation: $\tau = I\alpha$

Example 2: DJ Turntable Platter

A circular platter (annular disk, inner radius $R_1 = 1$, outer radius $R_2 = 2$) has density $\rho(r) = kr^2$ (denser at the rim). If total mass is $M = 1$ kg, find k and the polar moment of inertia I_z .

1. **Find k from mass:** $M = \int_0^{2\pi} \int_1^2 kr^2 \cdot r \, dr \, d\theta = 2\pi k \int_1^2 r^3 \, dr = 2\pi k \left[\frac{r^4}{4} \right]_1^2 = 2\pi k \cdot \frac{15}{4} = \frac{15\pi k}{2}$

$$\frac{15\pi k}{2} = 1 \implies k = \frac{2}{15\pi}$$

2. **Polar moment of inertia:** $I_z = \iint (x^2 + y^2) \rho \, dA = \iint r^2 \cdot kr^2 \cdot r \, dr \, d\theta$

$$I_z = 2\pi k \int_1^2 r^5 \, dr = 2\pi k \left[\frac{r^6}{6} \right]_1^2 = 2\pi k \cdot \frac{63}{6} = 21\pi k$$

3. **Substitute k :** $I_z = 21\pi \cdot \frac{2}{15\pi} = \frac{42}{15} = \frac{14}{5}$

4. **Result:** $I_z = \frac{14}{5} \approx 2.8 \text{ kg} \cdot \text{m}^2$

Compare: a uniform annular disk of the same mass has $I_z = \frac{1}{2}M(R_1^2 + R_2^2) = 2.5$. The rim-heavy platter has a larger moment of inertia, as expected.

4. Surface Area

Theorem 1: Surface Area Formula

The surface area of $z = f(x, y)$ over a region R in the xy -plane:

$$A = \iint_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} \, dA$$

– This generalizes the arc length formula $\int \sqrt{1 + (f')^2} \, dx$ to surfaces

Example 3: Surface Area of a Paraboloid

Find the surface area of $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 4$.

1. **Partial derivatives:** $f_x = 2x$, $f_y = 2y$.

2. **Integrand:** $\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2}$ in polar.

3. **Set up (polar):** $A = \int_0^{2\pi} \int_0^2 \sqrt{1 + 4r^2} \cdot r \, dr \, d\theta = 2\pi \int_0^2 r\sqrt{1 + 4r^2} \, dr$

4. **Substitution:** $u = 1 + 4r^2$, $du = 8r \, dr$, so $r \, dr = du/8$.

When $r = 0$: $u = 1$. When $r = 2$: $u = 17$.

$$A = 2\pi \cdot \frac{1}{8} \int_1^{17} \sqrt{u} \, du = \frac{\pi}{4} \cdot \left[\frac{2}{3} u^{3/2} \right]_1^{17} = \frac{\pi}{6} (17\sqrt{17} - 1)$$

5. **Result:** $A = \frac{\pi}{6} (17\sqrt{17} - 1) \approx 36.2$

5. Applications via Triple Integrals

Mass, Center of Mass, and Moments of Inertia in 3D

For a solid E with density $\rho(x, y, z)$:

$$\text{Mass: } M = \iiint_E \rho \, dV$$

$$\text{Center of mass: } \bar{x} = \frac{1}{M} \iiint_E x \rho \, dV, \quad \bar{y} = \frac{1}{M} \iiint_E y \rho \, dV, \quad \bar{z} = \frac{1}{M} \iiint_E z \rho \, dV$$

$$\text{Moments of inertia: } I_x = \iiint_E (y^2 + z^2) \rho \, dV$$

$$I_y = \iiint_E (x^2 + z^2) \rho \, dV$$

$$I_z = \iiint_E (x^2 + y^2) \rho \, dV$$

- I_x uses $y^2 + z^2$ — the squared distance from the x -axis. Same pattern for I_y and I_z .
- Rotational kinetic energy about the z -axis: $KE = \frac{1}{2} I_z \omega^2$.

Example 4: Center of Mass of a Solid Tetrahedron

A solid tetrahedron E has vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ and uniform density $\rho = 1$. Find its center of mass.

1. **Region:** $0 \leq z \leq 1 - x - y$, $0 \leq y \leq 1 - x$, $0 \leq x \leq 1$.

2. **Mass:** $M = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1 \, dz \, dy \, dx = \int_0^1 \int_0^{1-x} (1 - x - y) \, dy \, dx = \int_0^1 \frac{(1-x)^2}{2} \, dx = \frac{1}{6}$.

3. **Moment about the yz -plane:** $\iiint_E x \, dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x \, dz \, dy \, dx = \int_0^1 x \cdot \frac{(1-x)^2}{2} \, dx = \frac{1}{24}$.

So $\bar{x} = \frac{1/24}{1/6} = \frac{1}{4}$.

4. By symmetry of the tetrahedron and density, $\bar{y} = \bar{z} = \frac{1}{4}$.

5. **Result:** $(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ — the centroid sits one-quarter of the way along each axis.

Example 5: Moment of Inertia of a Solid Cylinder

A solid cylinder E of radius R and height h sits along the z -axis with uniform density ρ_0 . Find I_z in terms of the total mass $M = \pi R^2 h \rho_0$.

1. **Cylindrical coordinates:** $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq h$, $dV = r \, dr \, d\theta \, dz$.

2. **Set up:** $I_z = \iiint_E (x^2 + y^2) \rho_0 \, dV = \rho_0 \int_0^h \int_0^{2\pi} \int_0^R r^2 \cdot r \, dr \, d\theta \, dz = \rho_0 \cdot h \cdot 2\pi \cdot \frac{R^4}{4}$.

3. **Simplify:** $I_z = \frac{\pi R^4 h \rho_0}{2} = \frac{1}{2} M R^2$.

4. **Result:** $I_z = \frac{1}{2} M R^2$ — the standard result for a uniform solid cylinder.

6. Average Value

Average Value of a Function

The average value of f over a region is its integral divided by the size of the region:

$$\text{2D: } \bar{f} = \frac{1}{\text{Area}(R)} \iint_R f \, dA$$

$$\text{3D: } \bar{f} = \frac{1}{\text{Vol}(E)} \iiint_E f \, dV$$

– Same idea as $\bar{f} = \frac{1}{b-a} \int_a^b f(x) \, dx$ from Calc I.

Example 6: Average Temperature in a Cube

The unit cube $E = [0, 1]^3$ has temperature $T(x, y, z) = x + y + z$. Find the average temperature.

1. **Volume:** $\text{Vol}(E) = 1$.

2. **Integrate:** $\iiint_E (x + y + z) \, dV = \int_0^1 \int_0^1 \int_0^1 (x + y + z) \, dz \, dy \, dx$.

By symmetry, $\iiint_E x \, dV = \iiint_E y \, dV = \iiint_E z \, dV = \frac{1}{2}$, so the sum is $\frac{3}{2}$.

3. **Result:** $\bar{T} = \frac{3/2}{1} = \boxed{\frac{3}{2}}$ — exactly the value of T at the center $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, which makes sense because T is linear.

Practice Problems

Problem 1. Find the mass of a lamina over $R = [0, 1] \times [0, 2]$ with density $\rho(x, y) = xy$.

Answer: $M = 1$

Problem 2. Find I_z for a uniform disk of radius R with constant density ρ_0 . Express in terms of the total mass $M = \pi R^2 \rho_0$.

Answer: $I_z = \frac{1}{2}MR^2$

Problem 3. A disk of radius 2 has density $\rho = 4 - x^2 - y^2$. A torque of $\tau = 12$ N·m is applied. Find the angular acceleration $\alpha = \tau/I_z$.

Answer: $I_z = 32\pi/3$, so $\alpha = 9/(8\pi) \approx 0.358$ rad/s²

Problem 4. Find the center of mass of the region bounded by $y = x^2$ and $y = 4$ with uniform density $\rho = 1$.

Answer: $(\bar{x}, \bar{y}) = (0, 12/5)$

Problem 5. Set up (do not evaluate) the surface area integral for $z = e^x \sin y$ over the rectangle $[0, 1] \times [0, \pi]$.

Answer: $A = \int_0^1 \int_0^\pi \sqrt{1 + e^{2x} \sin^2 y + e^{2x} \cos^2 y} \, dy \, dx = \int_0^1 \int_0^\pi \sqrt{1 + e^{2x}} \, dy \, dx$

Problem 6. Find the mass of the solid box $E = [0, 1] \times [0, 2] \times [0, 3]$ with density $\rho(x, y, z) = xyz$.

Answer: $M = \int_0^1 \int_0^2 \int_0^3 xyz \, dz \, dy \, dx = \frac{1}{2} \cdot 2 \cdot \frac{9}{2} = \frac{9}{2}$

Problem 7. A solid cone of radius R and height h has uniform density ρ_0 and total mass $M = \frac{1}{3}\pi R^2 h \rho_0$. Find its moment of inertia I_z about the central axis. *Hint: cylindrical coordinates with $0 \leq r \leq Rz/h$.*
Answer: $I_z = \frac{3}{10}MR^2$

Problem 8. Find the average value of $f(x, y, z) = x^2 + y^2 + z^2$ over the unit ball $x^2 + y^2 + z^2 \leq 1$. *Hint: spherical coordinates; $\text{Vol} = \frac{4\pi}{3}$.*

Answer: $\iiint_E \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{4\pi}{5}$, so $\bar{f} = \frac{4\pi/5}{4\pi/3} = \frac{3}{5}$

Lecture 16: Triple Integrals

Background & Motivation

Everything we did with double integrals extends naturally to three dimensions. Triple integrals compute volume, mass, and moments over solid regions. Cylindrical and spherical coordinates play the same role that polar coordinates did for double integrals—they simplify regions with the right symmetry.

1. Warm-Up: Double Integral Review

(a) Evaluate $\int_0^2 \int_1^3 xy \, dy \, dx$.

1. Inner (y): $\left[\frac{xy^2}{2}\right]_1^3 = \frac{9x}{2} - \frac{x}{2} = 4x$

2. Outer (x): $\int_0^2 4x \, dx = [2x^2]_0^2 = 8$

3. Result: $\boxed{8}$

(b) Evaluate $\iint_R y \, dA$ where R is bounded by $y = 0$, $x = 0$, and $x + y = 1$.

1. Region (Type I): $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$.

2. Inner: $\int_0^{1-x} y \, dy = \frac{(1-x)^2}{2}$

3. Outer: $\int_0^1 \frac{(1-x)^2}{2} \, dx = \left[-\frac{(1-x)^3}{6}\right]_0^1 = \frac{1}{6}$

(c) Evaluate $\int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\theta$.

1. Inner (r): $\int_0^1 r^3 \, dr = 1/4$

2. Outer (θ): $2\pi \cdot \frac{1}{4} = \boxed{\frac{\pi}{2}}$

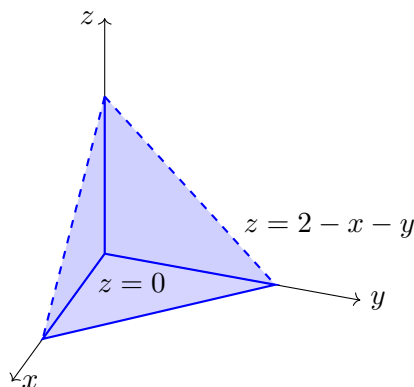
2. Triple Integrals and Fubini's Theorem

Definition 1: Triple Integral

$$\iiint_E f(x, y, z) \, dV = \int \int \int f(x, y, z) \, dz \, dy \, dx$$

– When $f = 1$: the triple integral gives the **volume** of E

– Fubini’s theorem: for continuous f on a box $[a, b] \times [c, d] \times [p, q]$, integrate in any order



A typical triple-integral Type I region: the solid tetrahedron in the first octant bounded below by $z = 0$ and above by the plane $z = 2 - x - y$. Integrate z first from the bottom to top surface, then project onto the xy -plane for the remaining double integral.

Types of Regions

- **Type I:** z bounded by functions of (x, y) – project onto the xy -plane
- **Type II:** x bounded by functions of (y, z) – project onto the yz -plane
- **Type III:** y bounded by functions of (x, z) – project onto the xz -plane

Choose the type that makes the bounds simplest.

3. Examples in Rectangular Coordinates

Example 1: Rectangular Box

Evaluate $\int_0^1 \int_0^2 \int_0^3 (x + y + z) \, dz \, dy \, dx$.

1. **Inner (z):** $\int_0^3 (x + y + z) \, dz = \left[(x + y)z + \frac{z^2}{2} \right]_0^3 = 3(x + y) + \frac{9}{2}$
2. **Middle (y):** $\int_0^2 \left(3x + 3y + \frac{9}{2} \right) \, dy = \left[3xy + \frac{3y^2}{2} + \frac{9y}{2} \right]_0^2 = 6x + 6 + 9 = 6x + 15$
3. **Outer (x):** $\int_0^1 (6x + 15) \, dx = [3x^2 + 15x]_0^1 = 18$
4. **Result:** 18

Example 2: Type I: First-Octant Tetrahedron

Find the volume of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

1. **Identify bounds (Type I):**

- z : from 0 to $1 - x - y$
- y : from 0 to $1 - x$
- x : from 0 to 1

2. **Inner (z):** $\int_0^{1-x-y} dz = 1 - x - y$

3. **Middle (y):** $\int_0^{1-x} (1 - x - y) \, dy = \left[(1 - x)y - \frac{y^2}{2} \right]_0^{1-x} = \frac{(1-x)^2}{2}$

4. **Outer (x):** $\int_0^1 \frac{(1-x)^2}{2} dx = \frac{1}{2} \left[-\frac{(1-x)^3}{3} \right]_0^1 = \frac{1}{6}$

5. **Result:** $V = \frac{1}{6}$

Check: $V = \frac{1}{6} |a_1(b_2c_3)| = \frac{1}{3} \cdot \frac{1}{2} \cdot 1 \cdot 1 \cdot 1 = 1/6 \checkmark$

4. Cylindrical Coordinates

Cylindrical Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$dV = r \, dz \, dr \, d\theta$$

- Use when the region has circular symmetry in x, y (cylinders, cones, paraboloids)
- Same as polar coordinates in xy , with z added

Example 3: Cone Volume via Cylindrical

Find the volume of the solid bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by $z = 3$.

1. **Convert:** In cylindrical, the cone is $z = r$. Region: $0 \leq r \leq 3, r \leq z \leq 3, 0 \leq \theta \leq 2\pi$.

2. **Set up:** $V = \int_0^{2\pi} \int_0^3 \int_r^3 r \, dz \, dr \, d\theta$

3. **Inner (z):** $\int_r^3 r \, dz = r(3 - r)$

4. **Middle (r):** $\int_0^3 r(3 - r) \, dr = \int_0^3 (3r - r^2) \, dr = \left[\frac{3r^2}{2} - \frac{r^3}{3} \right]_0^3 = \frac{27}{2} - 9 = \frac{9}{2}$

5. **Outer (θ):** $2\pi \cdot \frac{9}{2} = 9\pi$

6. **Result:** $V = 9\pi$

Check: $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (9)(3) = 9\pi \checkmark$

5. Spherical Coordinates

Spherical Coordinates

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

- ρ : distance from origin ϕ : angle from positive z -axis ($0 \leq \phi \leq \pi$) θ : angle in xy -plane
- Use when the region is a sphere/hemisphere or integrand involves $x^2 + y^2 + z^2$
- Key relation: $x^2 + y^2 + z^2 = \rho^2$

Example 4: Integral over a Sphere

Evaluate $\iiint_E (x^2 + y^2 + z^2) \, dV$ where E is the ball $x^2 + y^2 + z^2 \leq 4$.

1. **Convert:** $x^2 + y^2 + z^2 = \rho^2$. Bounds: $0 \leq \rho \leq 2$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$.
2. **Set up:** $\int_0^{2\pi} \int_0^\pi \int_0^2 \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$
3. **Separate (all bounds are constants):** $= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^\pi \sin \phi \, d\phi \right) \left(\int_0^2 \rho^4 \, d\rho \right)$
4. **Evaluate each:** $(2\pi)(2) \left(\frac{32}{5} \right) = \frac{128\pi}{5}$
5. **Result:** $\boxed{\frac{128\pi}{5}}$

Practice Problems

Problem 1. Evaluate $\int_0^1 \int_0^1 \int_0^1 (x + y + z) \, dz \, dy \, dx$.

Answer: $3/2$

Problem 2. Evaluate $\iiint_E z \, dV$ where E is bounded below by $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$. (Use cylindrical.)

Answer: $32\pi/3$

Problem 3. Find the volume of the upper hemisphere $x^2 + y^2 + z^2 \leq 9$, $z \geq 0$ using spherical coordinates.

Answer: 18π

Problem 4. Evaluate $\iiint_E \sqrt{x^2 + y^2 + z^2} e^{-(x^2+y^2+z^2)} dV$ over all of $\mathbb{R}^3$. (Use spherical.)

Answer: 2π

Lecture 17: Change of Variables & the Jacobian

Background & Motivation

We've used polar, cylindrical, and spherical coordinates with extra factors of r or $\rho^2 \sin \phi$ in the integrand. Today we explain where those factors come from: the Jacobian determinant, which measures how a coordinate transformation stretches or shrinks area and volume. This also lets us design custom substitutions for specific regions.

1. Warm-Up: Where Does $r \, dr \, d\theta$ Come From?

In polar coordinates, $dA = r \, dr \, d\theta$. We have used this factor of r many times – today we explain *why* it appears.

Evaluate $\int_0^{2\pi} \int_0^3 r \cdot r \, dr \, d\theta$ to find the area of a disk of radius 3.

1. Inner (r): $\int_0^3 r \, dr = 9/2$ (using $f = 1$, so integrand is just r from dA)
2. Outer (θ): $2\pi \cdot 9/2 = 9\pi$
3. Check: $\pi(3)^2 = 9\pi \checkmark$

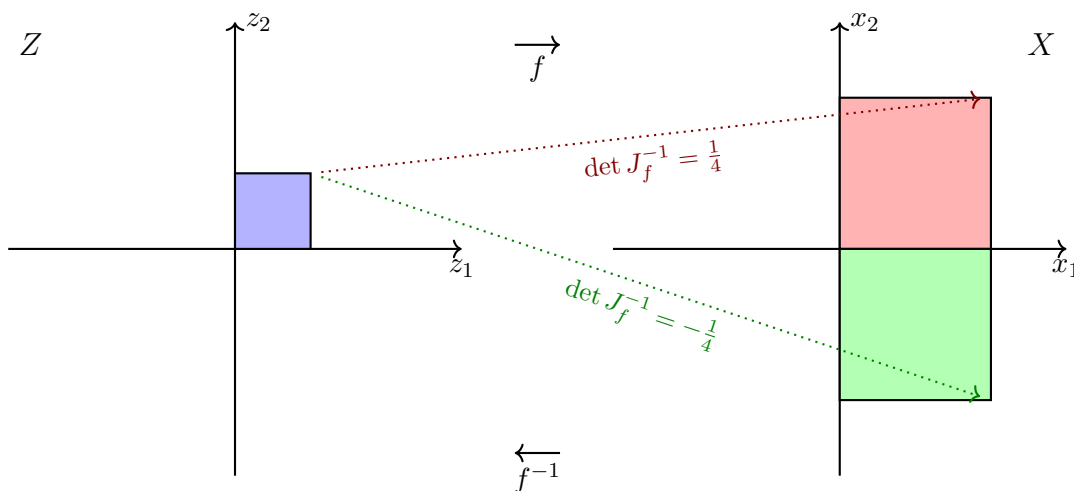
2. The Jacobian Determinant

Definition 1: Jacobian (2D)

For a transformation $T: (u, v) \mapsto (x, y)$, the **Jacobian** is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

- $|J|$ measures how much the transformation stretches or shrinks area at each point
- $dA_{xy} = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$



How the Jacobian determinant measures area scaling under a transformation. Different parts of the source region Z get scaled by different factors when mapped into X — the absolute value of $|J|$ gives the local scaling factor, and a negative determinant indicates a reflection.

Definition 2: Jacobian (3D)

For $(u, v, w) \mapsto (x, y, z)$:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

– Same idea: $dV_{xyz} = |J| du dv dw$

3. Change of Variables Formula

Theorem 1: Change of Variables (Double Integral)

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

where S is the region in the uv -plane corresponding to R .

Triple integral version:

$$\iiint_R f dV = \iiint_S f(x(u, v, w), \dots) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

4. Examples

Example 1: Polar Jacobian Verification

Verify that the Jacobian of $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ is r .

1. **Compute partials:** $\frac{\partial x}{\partial r} = \cos \theta$, $\frac{\partial x}{\partial \theta} = -r \sin \theta$

$\frac{\partial y}{\partial r} = \sin \theta$, $\frac{\partial y}{\partial \theta} = r \cos \theta$

2. **Determinant:** $J = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$

3. **Result:** $|J| = r$, confirming $dA = r dr d\theta$. $\boxed{J = r}$

Example 2: Custom Substitution for a Parallelogram

Evaluate $\iint_R (x - y) dA$ where R is the parallelogram with vertices $(0, 0)$, $(2, 1)$, $(3, 3)$, $(1, 2)$.

1. **Identify edges:**

- From $(0, 0)$ to $(2, 1)$: slope $1/2$, so $y = x/2$ i.e. $2y - x = 0$
- From $(1, 2)$ to $(3, 3)$: slope $1/2$, so $2y - x = 3$
- From $(0, 0)$ to $(1, 2)$: slope 2 , so $y = 2x$ i.e. $y - 2x = 0$
- From $(2, 1)$ to $(3, 3)$: slope 2 , so $y - 2x = -3$

2. **Choose transformation:** Let $u = 2y - x$ and $v = y - 2x$.

Bounds become: $0 \leq u \leq 3$, $-3 \leq v \leq 0$.

3. **Invert:** From $u = 2y - x$ and $v = y - 2x$:

$$u - 2v = 2y - x - 2y + 4x = 3x \implies x = \frac{u - 2v}{3}$$

$$2u - v = 4y - 2x - y + 2x = 3y \implies y = \frac{2u - v}{3}$$

4. **Jacobian:** $J = \begin{vmatrix} 1/3 & -2/3 \\ 2/3 & -1/3 \end{vmatrix} = \frac{-1}{9} + \frac{4}{9} = \frac{1}{3}$, $|J| = \frac{1}{3}$

5. **Integrand:** $x - y = \frac{u - 2v}{3} - \frac{2u - v}{3} = \frac{-u - v}{3}$

6. **Evaluate:** $\int_0^3 \int_{-3}^0 \frac{-u - v}{3} \cdot \frac{1}{3} dv du = \frac{1}{9} \int_0^3 \int_{-3}^0 (-u - v) dv du$

$$= \frac{1}{9} \int_0^3 \left[-uv - \frac{v^2}{2} \right]_{-3}^0 du = \frac{1}{9} \int_0^3 \left(-3u + \frac{9}{2} \right) du$$

$$= \frac{1}{9} \left[-\frac{3u^2}{2} + \frac{9u}{2} \right]_0^3 = \frac{1}{9} \left(-\frac{27}{2} + \frac{27}{2} \right) = 0$$

7. **Result:** $\boxed{0}$

(By symmetry of the parallelogram and the integrand, this makes sense.)

Example 3: Spherical Jacobian Verification

Show that the Jacobian for spherical coordinates is $\rho^2 \sin \phi$.

1. **Transformation:** $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$.

2. 3×3 **Jacobian matrix:**

$$J = \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix}$$

3. **Expand along Row 3:**

$$\begin{aligned} &= \cos \phi (-\rho^2 \sin \phi \cos \phi \sin^2 \theta - \rho^2 \sin \phi \cos \phi \cos^2 \theta) \\ &\quad + (-\rho \sin \phi)(\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta) \\ &= -\rho^2 \sin \phi \cos^2 \phi - \rho^2 \sin^3 \phi = -\rho^2 \sin \phi (\cos^2 \phi + \sin^2 \phi) \end{aligned}$$

4. **Result:** $J = -\rho^2 \sin \phi$, so $|J| = \boxed{\rho^2 \sin \phi}$

Example 4: Ellipse Area via Substitution

Use a change of variables to show that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab .

1. **Substitution:** Let $x = au$, $y = bv$.

2. **New region:** The ellipse becomes $u^2 + v^2 = 1$ (the unit disk).

3. **Jacobian:** $J = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab$, $|J| = ab$

4. **Area:** $A = \iint_R dA = \iint_{\text{unit disk}} ab \, du \, dv = ab \cdot (\text{area of unit disk}) = ab \cdot \pi$

5. **Result:** $A = \pi ab$

Practice Problems

Problem 1. Compute the Jacobian $|J|$ of the transformation $x = u + v$, $y = u - v$.

Answer: $|J| = 2$

Problem 2. Verify the cylindrical coordinate Jacobian: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$. Show $|J| = r$.

Answer: $|J| = r$

Problem 3. Use $u = x + y$, $v = x - y$ to evaluate $\iint_R (x^2 - y^2) \, dA$ where R is the square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, $(0, -1)$.

Hint: $x^2 - y^2 = (x + y)(x - y) = uv$ and the region in uv -space is the square $|u| \leq 1$, $|v| \leq 1$.

Answer: 0

Problem 4. Use $x = 2u$, $y = 3v$ to evaluate $\iint_D xy \, dA$ where D is the region inside the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$, first quadrant only.

Answer: $\int_0^{\pi/2} \int_0^1 (2r \cos \theta)(3r \sin \theta) \cdot 6r \, dr \, d\theta = 36 \int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \int_0^1 r^3 \, dr = 36 \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{9}{2}$

Problem 5. Let $T: (u, v) \mapsto (u^2 - v^2, 2uv)$. Compute the Jacobian and describe geometrically what this transformation does. *Hint:* think complex numbers, $z = u + iv \mapsto z^2$.

Answer: $J = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2 = 4(u^2 + v^2)$

Lecture 18: Vector Fields

Background & Motivation

We now begin the final chapter of the course: vector calculus. Until now, our functions have been scalar-valued— $f(x, y)$ returns a number. A vector field $\mathbf{F}(x, y)$ returns a vector at each point: think wind velocity, magnetic force, or fluid flow. Over the next six lectures, we'll integrate vector fields along curves (line integrals), across surfaces (flux integrals), and connect everything through the great theorems of Green, Stokes, and Gauss. These theorems unify all of calculus into a single principle: the integral of a derivative over a region equals the integral of the function over the boundary.

1. Warm-Up: Gradient Review

Find ∇f for $f(x, y, z) = x^2y + yz^3$.

1. $f_x = 2xy$
2. $f_y = x^2 + z^3$
3. $f_z = 3yz^2$
4. $\nabla f = \langle 2xy, x^2 + z^3, 3yz^2 \rangle$

– The gradient assigns a *vector* to each point – this is an example of a **vector field**.

2. Vector Field Definition

Definition 1: Vector Field

A **vector field** on $\mathbb{R}^2$ is a function $\mathbf{F}(x, y) = P(x, y)\hat{\mathbf{i}} + Q(x, y)\hat{\mathbf{j}}$.

A **vector field** on $\mathbb{R}^3$ is $\mathbf{F}(x, y, z) = P\hat{\mathbf{i}} + Q\hat{\mathbf{j}} + R\hat{\mathbf{k}}$.

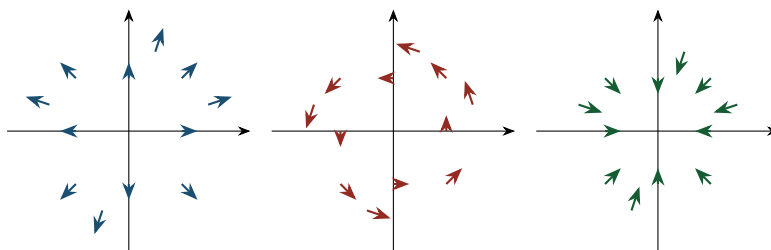
- At each point, $\mathbf{F}$ assigns a vector (direction and magnitude)
- Examples: wind velocity, gravitational force, electric field, fluid flow

3. How to Draw a Vector Field

Procedure for Sketching $\mathbf{F}(x, y)$

1. Pick a grid of sample points (e.g., $(\pm 1, 0)$, $(0, \pm 1)$, $(\pm 1, \pm 1)$, etc.)
 2. At each point (a, b) : evaluate $\mathbf{F}(a, b) = \langle P(a, b), Q(a, b) \rangle$
 3. Draw an arrow **starting at** (a, b) , pointing in the direction $\mathbf{F}(a, b)$
 4. Arrow length $\propto \|\mathbf{F}(a, b)\|$ (longer arrow = stronger field)
- **Key:** the arrow's *tail* is at the sample point, not its midpoint
 - **Normalization:** $\hat{\mathbf{F}} = \mathbf{F}/\|\mathbf{F}\|$ makes all arrows length 1 (shows direction only)

Common Patterns



Source: $\mathbf{F} = \langle x, y \rangle$ **Rotation:** $\mathbf{F} = \langle -y, x \rangle$ **Sink:** $\mathbf{F} = \langle -x, -y \rangle$

- **Source:** vectors point away from origin – fluid is being “created”
- **Sink:** vectors point toward origin – fluid is being “absorbed”
- **Rotation:** vectors swirl around a point – “whirlpool” pattern

4. Examples

Example 1: Evaluating and Sketching a Rotation Field

Let $\mathbf{F}(x, y) = \langle -y, x \rangle$. Evaluate at several points and describe the pattern.

1. **Evaluate at sample points:**

(a, b)	$\mathbf{F}(a, b) = \langle -b, a \rangle$	Direction
$(1, 0)$	$\langle 0, 1 \rangle$	up
$(0, 1)$	$\langle -1, 0 \rangle$	left
$(-1, 0)$	$\langle 0, -1 \rangle$	down
$(0, -1)$	$\langle 1, 0 \rangle$	right
$(1, 1)$	$\langle -1, 1 \rangle$	upper-left
$(2, 0)$	$\langle 0, 2 \rangle$	up (longer)

2. **Sketch:** At each point, draw arrow *starting at that point* in the direction computed.

3. **Pattern:** Vectors circulate **counterclockwise** around the origin.

Magnitude: $\|\mathbf{F}\| = \sqrt{x^2 + y^2} = r$ – arrows get longer farther from the origin.

This is a **rotation field** (also called a **vortex**).

Example 2: Evaluating a Source Field

Let $\mathbf{F}(x, y) = \langle x, y \rangle$. Evaluate at $(1, 0)$, $(0, 2)$, $(-1, 1)$.

1. $\mathbf{F}(1, 0) = \langle 1, 0 \rangle$ – points right from $(1, 0)$

2. $\mathbf{F}(0, 2) = \langle 0, 2 \rangle$ – points up from $(0, 2)$
3. $\mathbf{F}(-1, 1) = \langle -1, 1 \rangle$ – points upper-left from $(-1, 1)$
4. **Pattern:** Every vector points **radially outward** from the origin. This is a **source field**.
Magnitude $= \sqrt{x^2 + y^2} = r$; arrows grow with distance.

5. Conservative Vector Fields

Definition 2: Conservative Field

$\mathbf{F}$ is **conservative** if there exists a scalar function f (the **potential function**) such that:

$$\mathbf{F} = \nabla f$$

- Work done by a conservative field is **path-independent**
- In physics: conservative $\iff$ energy is conserved (e.g., gravity, electrostatics)

Theorem 1: Test for Conservative Fields (2D)

$\mathbf{F} = \langle P, Q \rangle$ is conservative on a simply connected domain if and only if:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

3D version: $\mathbf{F} = \langle P, Q, R \rangle$ is conservative iff $\nabla \times \mathbf{F} = \mathbf{0}$:

$$P_y = Q_x, \quad P_z = R_x, \quad Q_z = R_y$$

Caution: The domain must be **simply connected** (no holes) for the test to be sufficient.

Simply Connected Domains

- **Simply connected** – every closed curve can be shrunk to a point without leaving the domain
- $\mathbb{R}^2$ and $\mathbb{R}^3$: simply connected ✓
- $\mathbb{R}^2 \setminus \{(0, 0)\}$: **not** simply connected (has a “hole”)

6. Finding Potential Functions

Algorithm for Finding f (2D)

Given $\mathbf{F} = \langle P, Q \rangle$ that passes the conservative test:

1. **Integrate:** $f = \int P \, dx$ (treat y as constant) $\implies f = (\dots) + g(y)$
2. **Differentiate:** $f_y = (\dots) + g'(y)$. Set equal to Q and solve for $g'(y)$.
3. **Integrate** $g'(y)$ to find $g(y)$. Combine.

For 3D: same idea but with $g(y, z)$, then $h(z)$.

Example 3: Testing and Finding a Potential Function

Is $\mathbf{F} = \langle 2xy + y^2 \cos x, x^2 + 2y \sin x \rangle$ conservative? If so, find f .

1. **Test:** $P = 2xy + y^2 \cos x$, $Q = x^2 + 2y \sin x$.

$$P_y = 2x + 2y \cos x$$

$$Q_x = 2x + 2y \cos x$$

$$P_y = Q_x \quad \checkmark \quad \text{Conservative!}$$

2. **Integrate P with respect to x :**

$$f = \int (2xy + y^2 \cos x) dx = x^2 y + y^2 \sin x + g(y)$$

3. **Differentiate with respect to y and match to Q :**

$$f_y = x^2 + 2y \sin x + g'(y) = Q = x^2 + 2y \sin x$$

$$\implies g'(y) = 0 \implies g(y) = C$$

4. **Result:** $f(x, y) = x^2 y + y^2 \sin x + C$

Example 4: Non-Conservative Field: The Hole Counterexample

Show that $\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$ passes $P_y = Q_x$ but is NOT conservative.

1. **Check partials:**

$$P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$P_y = Q_x \quad \checkmark \quad \text{test satisfied!}$$

2. **But the domain is $\mathbb{R}^2 \setminus \{(0, 0)\}$** – not simply connected (hole at origin).

3. **Proof:** Integrate around the unit circle C : $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = 2\pi \neq 0$$

If $\mathbf{F}$ were conservative, every closed-loop integral would be 0. Contradiction!

4. **Lesson:** Simply connected domain is **essential** for the $P_y = Q_x$ test.

Practice Problems

Problem 1. Evaluate $\mathbf{F}(x, y) = \langle x^2, xy \rangle$ at $(1, 2)$, $(2, 1)$, $(-1, 3)$. For each, state where the arrow starts and what direction it points.

$\mathbf{F}(1, 2) = \langle 1, 2 \rangle$: arrow starts at $(1, 2)$, points in direction $\langle 1, 2 \rangle$ (up-right, steep).

$\mathbf{F}(2, 1) = \langle 4, 2 \rangle$: arrow starts at $(2, 1)$, points in direction $\langle 4, 2 \rangle$ (right-ish, shallow).

$\mathbf{F}(-1, 3) = \langle 1, -3 \rangle$: arrow starts at $(-1, 3)$, points in direction $\langle 1, -3 \rangle$ (down-right).

Problem 2. Determine if $\mathbf{F} = \langle ye^{xy}, xe^{xy} \rangle$ is conservative. If so, find f .

1. **Test:** $P_y = e^{xy} + xye^{xy}$, $Q_x = e^{xy} + xye^{xy}$. Equal $\checkmark$

2. **Integrate:** $f = \int ye^{xy} dx = e^{xy} + g(y)$

3. **Match:** $f_y = xe^{xy} + g'(y) = xe^{xy} \implies g'(y) = 0$

4. **Result:** $f = e^{xy} + C$

Problem 3. Show that $\mathbf{F} = \langle x^2 - y^2, 2xy \rangle$ is NOT conservative.

$$P_y = -2y, \quad Q_x = 2y.$$

Since $-2y \neq 2y$ (for $y \neq 0$), $P_y \neq Q_x$. NOT conservative.

Problem 4. Find the potential function for $\mathbf{F} = \langle 2xz + y, x + z^2, x^2 + 2yz \rangle$.

1. **Integrate P w.r.t. x :** $f = \int (2xz + y) dx = x^2z + xy + g(y, z)$

2. **Match $f_y = Q$:** $x + g_y = x + z^2 \implies g_y = z^2 \implies g = yz^2 + h(z)$

3. **Match $f_z = R$:** $x^2 + 2yz + h'(z) = x^2 + 2yz \implies h'(z) = 0$

4. **Result:** $f = x^2z + xy + yz^2 + C$

Problem 5. Let $\mathbf{F} = \langle y \cos(xy) + 1, x \cos(xy) + 2z, 2y + 3z^2 \rangle$.

(a) Verify $\mathbf{F}$ is conservative (check all three conditions).

(b) Find the potential function.

(a) Test:

$$P_y = \cos(xy) - xy \sin(xy), \quad Q_x = \cos(xy) - xy \sin(xy) \quad \checkmark$$

$$Q_z = 2, \quad R_y = 2 \quad \checkmark$$

$$P_z = 0, \quad R_x = 0 \quad \checkmark$$

All three match – conservative.

(b) Find f :

1. $f = \int (y \cos(xy) + 1) dx = \sin(xy) + x + g(y, z)$

2. $f_y = x \cos(xy) + g_y = x \cos(xy) + 2z \implies g_y = 2z \implies g = 2yz + h(z)$

3. $f_z = 2y + h'(z) = 2y + 3z^2 \implies h'(z) = 3z^2 \implies h = z^3$

4. **Result:** $f = \sin(xy) + x + 2yz + z^3 + C$

Lecture 19: Line Integrals

Background & Motivation

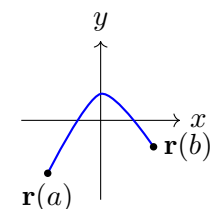
With vector fields defined, we now integrate them along curves. The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ computes the work done by a force along a path. For conservative fields, the Fundamental Theorem of Line Integrals gives us a shortcut: the integral depends only on the endpoints, not the path.

1. Contour Types

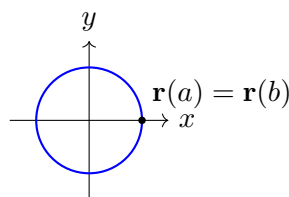
A **contour** (or **curve**) C is a path in $\mathbb{R}^2$ or $\mathbb{R}^3$ given by a parameterization $\mathbf{r}(t)$, $a \leq t \leq b$.

Definition 1: Types of Contours

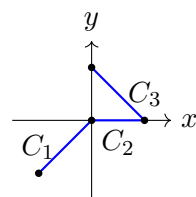
- **Open** – endpoints $\mathbf{r}(a) \neq \mathbf{r}(b)$
- **Closed** – $\mathbf{r}(a) = \mathbf{r}(b)$ (loop); use notation $\oint_C$
- **Simple** – does not cross itself
- **Piecewise smooth** – finitely many smooth pieces joined end-to-end



Open Contour



Closed Contour



Piecewise Contour

Common Parameterizations

- Line segment from A to B – $\mathbf{r}(t) = (1-t)A + tB$, $t \in [0, 1]$
- Circle $x^2 + y^2 = R^2$ – $\mathbf{r}(t) = \langle R \cos t, R \sin t \rangle$, $t \in [0, 2\pi]$
- Graph $y = f(x)$, $a \leq x \leq b$ – $\mathbf{r}(t) = \langle t, f(t) \rangle$, $t \in [a, b]$
- Helix – $\mathbf{r}(t) = \langle a \cos t, a \sin t, bt \rangle$

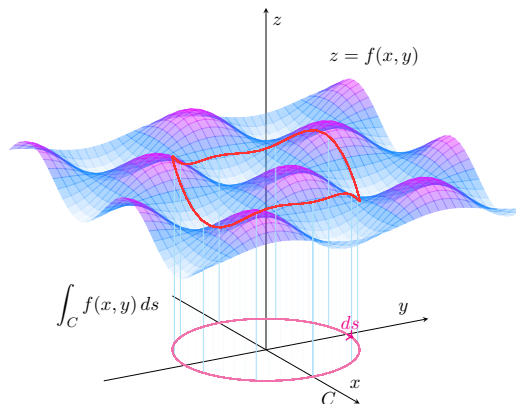
2. Scalar Line Integrals

Definition 2: Scalar Line Integral

For a scalar function f along a curve C parameterized by $\mathbf{r}(t)$:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt$$

- $ds = \|\mathbf{r}'(t)\| \, dt$ is the **arc length element**
- **Interpretation:** area of the “curtain” under f along C
- Direction does not matter: $\int_C f \, ds = \int_{-C} f \, ds$



Example 1: Scalar Line Integral Along a Line Segment

Evaluate $\int_C (x + 2y) \, ds$ where C is the segment from $(0, 0)$ to $(3, 4)$.

1. **Parameterize:** $\mathbf{r}(t) = \langle 3t, 4t \rangle$, $t \in [0, 1]$
2. **Speed:** $\mathbf{r}'(t) = \langle 3, 4 \rangle$, $\|\mathbf{r}'(t)\| = \sqrt{9 + 16} = 5$
3. **Substitute:** $f(\mathbf{r}(t)) = 3t + 2(4t) = 11t$
4. **Integrate:** $\int_0^1 11t \cdot 5 \, dt = 55 \int_0^1 t \, dt = 55 \cdot \frac{1}{2} = \boxed{\frac{55}{2}}$

3. Vector Field Line Integrals

Definition 3: Line Integral of a Vector Field

For a vector field $\mathbf{F}$ along a curve C :

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$$

- **Interpretation:** work done by the force $\mathbf{F}$ along path C
- Reversing direction **negates** the integral: $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$
- Equivalent notation: $\int_C P \, dx + Q \, dy + R \, dz$

Example 2: Work Around a Circle

Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y, x \rangle$ along the circle $x^2 + y^2 = 4$ (CCW).

1. **Parameterize:** $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t \rangle$, $\mathbf{r}' = \langle -2 \sin t, 2 \cos t \rangle$, $t \in [0, 2\pi]$
2. **Evaluate $\mathbf{F}$:** $\mathbf{F}(\mathbf{r}(t)) = \langle -2 \sin t, 2 \cos t \rangle$
3. **Dot product:** $\mathbf{F} \cdot \mathbf{r}' = (-2 \sin t)(-2 \sin t) + (2 \cos t)(2 \cos t) = 4 \sin^2 t + 4 \cos^2 t = 4$
4. **Integrate:** $\int_0^{2\pi} 4 \, dt = \boxed{8\pi}$

4. Fundamental Theorem of Line Integrals (FTOLI)

Theorem 1: Fundamental Theorem of Line Integrals

If $\mathbf{F} = \nabla f$ (conservative) and C goes from A to B :

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

Consequences:

- The integral depends only on **the endpoints** – **path independence**
- For any closed curve: $\oint_C \nabla f \cdot d\mathbf{r} = 0$
- If $\oint_C \mathbf{F} \cdot d\mathbf{r} \neq 0$ for some closed C , then $\mathbf{F}$ is **not** conservative

Example 3: Applying FTOLI

Evaluate $\int_C \langle ye^{xy}, xe^{xy} \rangle \cdot d\mathbf{r}$ from $(0,0)$ to $(1,2)$.

1. **Recognize:** From Worksheet 21, $\mathbf{F} = \nabla f$ with $f(x, y) = e^{xy}$
2. **Apply FTOLI:** $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = e^{(1)(2)} - e^0 = \boxed{e^2 - 1}$

No parameterization needed — any path from $(0,0)$ to $(1,2)$ gives the same answer.

Example 4: Piecewise 3D Line Integral

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2, z^2, x^2 \rangle$ along the path $(0,0,0) \rightarrow (1,0,0) \rightarrow (1,1,0) \rightarrow (1,1,1)$.

Segment 1: $(0,0,0) \rightarrow (1,0,0)$: $\mathbf{r} = \langle t, 0, 0 \rangle$, $\mathbf{r}' = \langle 1, 0, 0 \rangle$

$\mathbf{F} = \langle 0, 0, t^2 \rangle$, $\mathbf{F} \cdot \mathbf{r}' = 0$. Integral = 0.

Segment 2: $(1,0,0) \rightarrow (1,1,0)$: $\mathbf{r} = \langle 1, t, 0 \rangle$, $\mathbf{r}' = \langle 0, 1, 0 \rangle$

$\mathbf{F} = \langle t^2, 0, 1 \rangle$, $\mathbf{F} \cdot \mathbf{r}' = 0$. Integral = 0.

Segment 3: $(1,1,0) \rightarrow (1,1,1)$: $\mathbf{r} = \langle 1, 1, t \rangle$, $\mathbf{r}' = \langle 0, 0, 1 \rangle$

$\mathbf{F} = \langle 1, t^2, 1 \rangle$, $\mathbf{F} \cdot \mathbf{r}' = 1$. Integral = $\int_0^1 1 \, dt = 1$.

Total: $0 + 0 + 1 = \boxed{1}$

Practice Problems

Problem 1. Evaluate $\int_C xy \, ds$ where C is the line segment from $(0,0)$ to $(1,1)$.

Answer: $\frac{\sqrt{2}}{3}$

Problem 2. Evaluate $\int_C \langle y^2, 2xy \rangle \cdot d\mathbf{r}$ from $(0,0)$ to $(1,1)$ along $y = x^2$.

Answer: 1

Problem 3. Evaluate $\oint_C \langle x^2, y^2 \rangle \cdot d\mathbf{r}$ where C is the unit circle. (*Hint: is the field conservative?*)

Answer: 0

Problem 4. Evaluate $\int_C \langle 2xz + y, x + z^2, x^2 + 2yz \rangle \cdot d\mathbf{r}$ from $(0,0,0)$ to $(1,2,3)$ using FTOLI.

(*Hint: find the potential function first.*)

Answer: $f = x^2z + xy + yz^2$; $f(1,2,3) - f(0,0,0) = 3 + 2 + 18 = 23$

Problem 5. Evaluate $\int_C (x^2 + y^2) ds$ where C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$, $0 \leq t \leq 2\pi$.

Answer: $2\sqrt{2}\pi$

Lecture 20: Green's Theorem

Background & Motivation

Green's Theorem is the first of the three big theorems in vector calculus. It converts a line integral around a closed curve into a double integral over the enclosed region. This is powerful both computationally—sometimes one side is much easier than the other—and theoretically, as it connects circulation to curl.

1. Green's Theorem

- Converts a line integral around a closed curve into a double integral over the enclosed region (and vice versa)

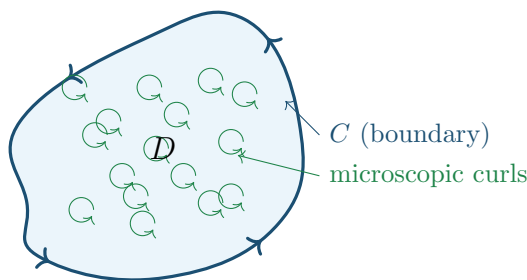
Theorem 1: Green's Theorem (Circulation Form)

Let C be a positively oriented (CCW), simple, closed curve bounding a region D . If P and Q have continuous partial derivatives on an open region containing D :

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Requirements:

- C must be **simple, closed, positively oriented**
- P, Q must be **continuously differentiable on all of D**



Macroscopic circulation around C = sum of all microscopic curls $(Q_x - P_y)$ inside D .

Theorem 2: Green's Theorem (Vector Form)

Recall that $\oint_C P dx + Q dy$ is really a line integral of $\mathbf{F} = \langle P, Q \rangle$ along C . Writing Green's Theorem

entirely in terms of $\mathbf{F}$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

The middle form is what we compute directly (parameterize C , dot, integrate). The right side is what Green's lets us swap to—a double integral over D .

Preview: The quantity $Q_x - P_y$ is what we will call the **curl** of $\mathbf{F}$ (next lecture). It measures local rotation. When we later learn **Stokes' Theorem**, we will see that Green's is just Stokes' with the surface flat in the xy -plane. Swap $Q_x - P_y$ for curl and you get Stokes'.

When to use Green's Theorem

- Hard line integral but easy double integral $\implies$ use Green's left-to-right
- Hard double integral but easy line integral $\implies$ use Green's right-to-left
- To compute areas via line integrals

Example 1: Verification on a Triangle (Both Sides)

Verify Green's Theorem for $\oint_C (x^2y dx + xy^2 dy)$ around the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.

Double integral side:

1. **Compute:** $P = x^2y$, $Q = xy^2$. $Q_x - P_y = y^2 - x^2$
2. **Set up:** $\iint_D (y^2 - x^2) dA = \int_0^1 \int_0^{1-x} (y^2 - x^2) dy dx$
3. **Inner integral:** $\left[\frac{y^3}{3} - x^2y \right]_0^{1-x} = \frac{(1-x)^3}{3} - x^2(1-x)$
4. **Evaluate:** $\int_0^1 \left[\frac{(1-x)^3}{3} - x^2(1-x) \right] dx = \frac{1}{12} - \frac{1}{12} = 0$

Line integral side:

C_1 : $(0,0) \rightarrow (1,0)$: $\mathbf{r} = \langle t, 0 \rangle$, integral $= \int_0^1 0 dt = 0$

C_2 : $(1,0) \rightarrow (0,1)$: $\mathbf{r} = \langle 1-t, t \rangle$, $dx = -dt$, $dy = dt$

$$\begin{aligned} \text{Integral} &= \int_0^1 [-(1-t)^2t + (1-t)t^2] dt = \int_0^1 t(1-t)(2t-1) dt \\ &= \int_0^1 (2t^3 - 3t^2 + t) dt = \frac{1}{2} - 1 + \frac{1}{2} = 0 \end{aligned}$$

C_3 : $(0,1) \rightarrow (0,0)$: $\mathbf{r} = \langle 0, 1-t \rangle$, integral $= 0$

Total: $0 + 0 + 0 = 0$ ✓

Example 2: Using Green's to Simplify a Circle Integral

Evaluate $\oint_C (e^x + y^3) dx + (x^3 - \sin y) dy$ where C is the unit circle (CCW).

1. **Identify:** $P = e^x + y^3$, $Q = x^3 - \sin y$
2. **Compute:** $Q_x - P_y = 3x^2 - 3y^2$
3. **Convert to polar:** $\iint_D (3x^2 - 3y^2) dA = \int_0^{2\pi} \int_0^1 3r^2(\cos^2 \theta - \sin^2 \theta) \cdot r dr d\theta$
4. **Evaluate:** $= 3 \int_0^1 r^3 dr \cdot \int_0^{2\pi} \cos 2\theta d\theta = 3 \cdot \frac{1}{4} \cdot 0 = \boxed{0}$

2. Area via Green's Theorem

Area Formulas via Green's Theorem

Setting $Q_x - P_y = 1$ in Green's Theorem gives area formulas. After parameterizing C with $\mathbf{r}(t) = \langle x(t), y(t) \rangle$:

$$\text{Area}(D) = \frac{1}{2} \int_a^b \left(x \frac{dy}{dt} - y \frac{dx}{dt} \right) dt$$

Also written in differential notation: $\frac{1}{2} \oint_C (x dy - y dx)$, or equivalently $\oint_C x dy$ or $-\oint_C y dx$.

Example 3: Area of an Ellipse

Find the area enclosed by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using Green's Theorem.

1. **Parameterize:** $x = a \cos t$, $y = b \sin t$, $t \in [0, 2\pi]$
2. **Compute differentials:** $dx = -a \sin t dt$, $dy = b \cos t dt$
3. **Apply area formula:**

$$A = \frac{1}{2} \oint_C (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} [a \cos t \cdot b \cos t - b \sin t \cdot (-a \sin t)] dt$$

4. **Simplify:** $= \frac{ab}{2} \int_0^{2\pi} (\cos^2 t + \sin^2 t) dt = \frac{ab}{2} \cdot 2\pi = \boxed{\pi ab}$

3. When Green's Theorem Fails

- Green's requires P, Q to have continuous partials on **all** of D
- If $\mathbf{F}$ is undefined inside D (e.g., at the origin), the theorem does **not** apply directly

Example 4: The $1/(x^2 + y^2)$ Counterexample

Compute $\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$ around the unit circle. Show Green's Theorem does not apply.

1. **Check partials:** $P = \frac{-y}{x^2 + y^2}$, $Q = \frac{x}{x^2 + y^2}$.

$$P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}. \quad \text{So } Q_x - P_y = 0!$$

If Green's applied, the answer would be $\iint_D 0 dA = 0$. But...

2. **Direct computation:** $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$, $\mathbf{r}' = \langle -\sin t, \cos t \rangle$

$$\mathbf{F} \cdot \mathbf{r}' = (-\sin t)(-\sin t) + (\cos t)(\cos t) = 1$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} 1 dt = \boxed{2\pi} \neq 0$$

3. **Why:** $\mathbf{F}$ is undefined at $(0, 0)$, which lies inside C . The domain has a “hole,” so Green's Theorem does not apply.

Practice Problems

Problem 1. Use Green's Theorem to evaluate $\oint_C (2y \, dx + 3x \, dy)$ around the unit square $[0, 1]^2$ (CCW).

Answer: 1

Problem 2. Evaluate $\oint_C (y^2 \, dx + x^2 \, dy)$ where C is the boundary of the region between $y = x$ and $y = x^2$, $0 \leq x \leq 1$ (CCW).

Answer: $\frac{1}{30}$

Problem 3. Use Green's Theorem to find the area enclosed by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq 2\pi$.

Answer: $\frac{3\pi}{8}$

Problem 4. Evaluate $\oint_C (x^2 y \, dx - x y^2 \, dy)$ where C is the circle $x^2 + y^2 = 4$ (CCW).

Answer: $Q_x - P_y = -y^2 - x^2 = -(x^2 + y^2)$; $\iint_D -(x^2 + y^2) \, dA = -\int_0^{2\pi} \int_0^2 r^3 \, dr \, d\theta = -8\pi$

Problem 5. True or false (explain briefly):

(a) If $Q_x - P_y = 0$ everywhere in D , then $\oint_C P dx + Q dy = 0$.

(b) Green's Theorem can be applied to $\mathbf{F} = \langle x/r^2, y/r^2 \rangle$ (where $r = \sqrt{x^2 + y^2}$) on any region not containing the origin.

Answer: (a) True — by Green's Theorem, $\oint 0 dA = 0$, provided P, Q are smooth on all of D .

(b) True — as long as D does not contain the origin, $\mathbf{F}$ is smooth on D and Green's applies.

Lecture 21: Curl and Divergence

Background & Motivation

Green's Theorem involved the quantity $Q_x - P_y$. Today we give it a name: the **curl**. We also introduce the **divergence**, which measures expansion or compression. These two differential operators are the vocabulary of the final theorems of the course (Stokes' and the Divergence Theorem).

1. Curl: Measuring Rotation

Definition 1: Curl of a Vector Field

For $\mathbf{F} = \langle P, Q, R \rangle$, the **curl** is (all the same thing, different notation):

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \nabla \times \mathbf{F} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$$

How to compute (component by component):

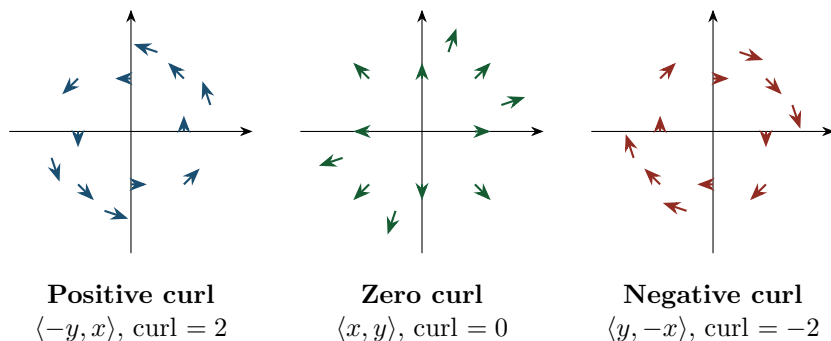
$$\bullet \hat{\mathbf{i}}: \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \quad \hat{\mathbf{j}}: \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \quad \hat{\mathbf{k}}: \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Memory trick: each component uses the two variables it doesn't point along, in cyclic order ($x \rightarrow y \rightarrow z \rightarrow x \rightarrow \dots$). Always subtract in "reverse" order. If you know determinants, this equals

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}, \text{ but that is not required.}$$

- **Interpretation:** measures the **local rotation** ("swirling") of the field at a point
- $\nabla \times \mathbf{F} = \mathbf{0} \implies$ **irrotational** (no spinning)
- Direction of $\nabla \times \mathbf{F}$ = axis of rotation (right-hand rule), magnitude = rate of rotation

Visualizing Curl



The Paddlewheel Test

Think of placing a tiny paddlewheel at a point in the flow. The fluid pushes on each blade. If the flow is faster or differently directed on one side than the other, the imbalance creates a torque and the wheel spins.

- **Positive curl** ($\nabla \times \mathbf{F} > 0$ in 2D): paddlewheel spins **CCW**. Right-hand rule: thumb out of page ($+\hat{\mathbf{k}}$).
- **Negative curl** ($\nabla \times \mathbf{F} < 0$ in 2D): paddlewheel spins **CW**. Right-hand rule: thumb into page ($-\hat{\mathbf{k}}$).
- **Zero curl**: paddlewheel doesn't spin, even if fluid is moving fast.

Key insight: curl measures *local spin*, not global shape. A field can have curved streamlines but zero curl, and *straight* streamlines but nonzero curl!

Example — shear flow: In $\mathbf{F} = \langle y, 0 \rangle$ every arrow is horizontal, yet the paddlewheel spins. The top blade ($y > 0$) is pushed right; the bottom blade ($y < 0$) is pushed left. This imbalance $\implies$ **CW** rotation, and indeed $Q_x - P_y = 0 - 1 = -1 < 0$. Negative curl despite no circular streamlines.

In 3D, $\nabla \times \mathbf{F}$ is a vector: its **direction** is the axis of fastest rotation (right-hand rule) and its **magnitude** is the rate of rotation.

Example 1: Determining Curl Sign from the Paddlewheel

For each field, predict the paddlewheel behavior at the origin, then verify by computing the 2D curl.

- (a) $\mathbf{F} = \langle 0, x \rangle$
- (b) $\mathbf{F} = \langle -y, 0 \rangle$
- (c) $\mathbf{F} = \langle y, x \rangle$

(a) Right of origin ($x > 0$): upward push. Left ($x < 0$): downward push. $\implies$ **CCW**.
 $Q_x - P_y = 1 - 0 = 1 > 0$ ✓ Positive curl (CCW).

(b) Above origin ($y > 0$): leftward push. Below ($y < 0$): rightward push. $\implies$ **CCW**.
 $Q_x - P_y = 0 - (-1) = 1 > 0$ ✓ Positive curl (CCW). Different-looking field, same curl as (a)!

(c) From $P = y$: top-right, bottom-left (CW tendency). From $Q = x$: right-up, left-down (CCW tendency). They **cancel**.
 $Q_x - P_y = 1 - 1 = 0$ ✓ Zero curl. In fact $\mathbf{F} = \nabla(xy)$ — conservative.

2D Reduction

For $\mathbf{F} = \langle P, Q \rangle$ (2D, so $R = 0$), only the $\hat{\mathbf{k}}$ component survives: $\nabla \times \mathbf{F} = (Q_x - P_y)\hat{\mathbf{k}}$. We call the scalar $Q_x - P_y$ the **2D curl** — exactly the integrand in Green's Theorem!

Example 2: 3D Curl Computation

Find $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle xz, xyz, -y^2 \rangle$.

$$P = xz, \quad Q = xyz, \quad R = -y^2.$$

$$\hat{\mathbf{i}}: R_y - Q_z = (-y^2)_y - (xyz)_z = -2y - xy$$

$$\hat{\mathbf{j}}: P_z - R_x = (xz)_z - (-y^2)_x = x - 0 = x$$

$$\hat{\mathbf{k}}: Q_x - P_y = (xyz)_x - (xz)_y = yz - 0 = yz$$

$$\nabla \times \mathbf{F} = \langle -2y - xy, \quad x, \quad yz \rangle$$

Example 3: 2D Curl: Rotation vs. Source

Compute the 2D curl of (a) $\mathbf{F} = \langle -y, x \rangle$ and (b) $\mathbf{F} = \langle x, y \rangle$.

(a) $Q_x - P_y = (x)_x - (-y)_y = 1 - (-1) = 2$. Positive $\implies$ CCW rotation everywhere.

(b) $Q_x - P_y = (y)_x - (x)_y = 0 - 0 = 0$. Irrotational! Despite pointing outward everywhere, there is no local spinning. (This field is conservative: $\mathbf{F} = \nabla\left(\frac{x^2+y^2}{2}\right)$.)

2. Divergence: Measuring Expansion

Definition 2: Divergence of a Vector Field

For $\mathbf{F} = \langle P, Q, R \rangle$, the **divergence** is (same thing, different notation):

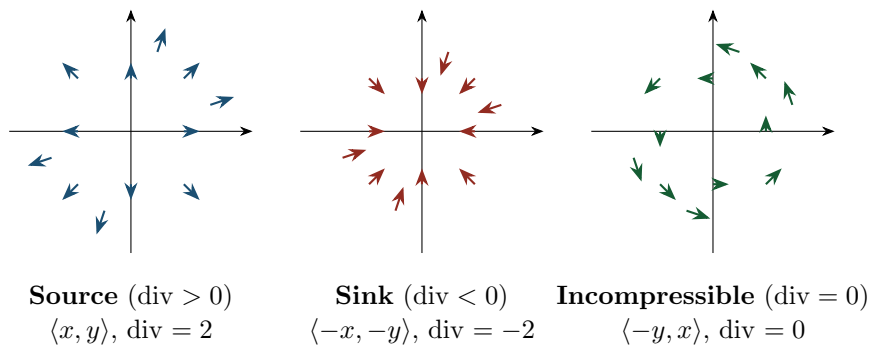
$$\operatorname{div} \mathbf{F} = \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Just add up how much each component changes along its own direction.

– **Interpretation:** measures the net “**outflow**” or **expansion at a point**

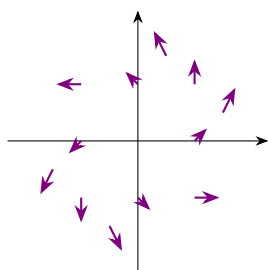
– $\operatorname{div} \mathbf{F} > 0$: **source** (fluid spreading out) $\operatorname{div} \mathbf{F} < 0$: **sink** (fluid compressing) $\operatorname{div} \mathbf{F} = 0$: **incompressible**

Visualizing Divergence

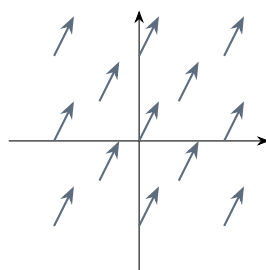


Both Properties / Neither

Curl and divergence are **independent**. A field can have any combination:



Both ($\text{curl} \neq 0, \text{div} \neq 0$)
 $\langle x - y, x + y \rangle$
 $\text{curl} = 2, \text{div} = 2$



Neither ($\text{curl} = 0, \text{div} = 0$)
 $\langle 1, 2 \rangle$
 Uniform flow

Example 4: Incompressible Pipe Flow

A velocity field in a pipe is $\mathbf{F} = \langle 2x, -y, -z \rangle$. Is the flow incompressible?

$$\text{div } \mathbf{F} = (2x)_x + (-y)_y + (-z)_z = 2 + (-1) + (-1) = 0$$

Yes — $\text{div } \mathbf{F} = 0$, so the flow is **incompressible**. The fluid speeds up along x (pipe narrows) but compensates by slowing in y and z . No fluid is created or destroyed.

Example 5: Classifying a Field

For $\mathbf{F} = \langle x, y, z \rangle$, find $\text{div } \mathbf{F}$ and $\nabla \times \mathbf{F}$. Interpret.

$$\text{div } \mathbf{F} = 1 + 1 + 1 = 3 \quad (\text{source everywhere — uniform expansion})$$

$$\nabla \times \mathbf{F}: \quad \hat{\mathbf{i}}: 0 - 0 = 0, \quad \hat{\mathbf{j}}: 0 - 0 = 0, \quad \hat{\mathbf{k}}: 0 - 0 = 0. \quad \text{So } \nabla \times \mathbf{F} = \mathbf{0} \text{ (irrotational).}$$

Like an explosion: everything moves outward, nothing rotates.

Example 6: Another Classification

For $\mathbf{F} = \langle -y, x, 0 \rangle$, find $\text{div } \mathbf{F}$ and $\nabla \times \mathbf{F}$. Interpret.

$$\text{div } \mathbf{F} = (-y)_x + (x)_y + (0)_z = 0 \quad (\text{incompressible})$$

$$\nabla \times \mathbf{F}: \quad \hat{\mathbf{i}}: 0 - 0 = 0, \quad \hat{\mathbf{j}}: 0 - 0 = 0, \quad \hat{\mathbf{k}}: 1 - (-1) = 2. \quad \text{So } \nabla \times \mathbf{F} = 2\hat{\mathbf{k}} \text{ (CCW about } z\text{).}$$

Like a merry-go-round: everything spins, nothing expands or compresses.

3. Key Identities

Vector Calculus Identities

1. $\nabla \times (\nabla f) = \mathbf{0}$ — conservative fields are always irrotational (by Clairaut's: $f_{xy} = f_{yx}$)

Contrapositive: if $\nabla \times \mathbf{F} \neq \mathbf{0}$, then $\mathbf{F}$ is **not** conservative.

2. $\text{div}(\nabla \times \mathbf{F}) = 0$ — curl fields are always divergence-free (Clairaut's again)

3. **Laplacian:** $\nabla^2 f = \text{div}(\nabla f) = f_{xx} + f_{yy} + f_{zz}$ — divergence of the gradient

4. **Product rules:**

$$\nabla(fg) = f\nabla g + g\nabla f \quad \text{div}(f\mathbf{F}) = f \text{div } \mathbf{F} + \mathbf{F} \cdot \nabla f \quad \nabla \times (f\mathbf{F}) = f \nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$$

Operator Relationships

$$\text{Scalar field } f \xrightarrow{\text{grad}} \text{Vector field } \mathbf{F} \xrightarrow{\text{curl}} \text{Vector field } \mathbf{G} \xrightarrow{\text{div}} \text{Scalar field } h$$

The two zero identities say: $\text{grad} \rightarrow \text{curl} = \mathbf{0}$, and $\text{curl} \rightarrow \text{div} = 0$. In other words, the “derivative of a derivative” is zero in this chain.

Example 7: Verifying $\nabla \times (\nabla f) = \mathbf{0}$

Let $f(x, y, z) = x^2yz + \sin(yz)$. Verify $\nabla \times (\nabla f) = \mathbf{0}$.

Gradient: $\nabla f = \langle 2xyz, x^2z + z \cos(yz), x^2y + y \cos(yz) \rangle$

Curl:

$\hat{\mathbf{i}}$: $(x^2y + y \cos(yz))_y - (x^2z + z \cos(yz))_z = [x^2 + \cos(yz) - yz \sin(yz)] - [x^2 + \cos(yz) - yz \sin(yz)] = 0$

✓

$\hat{\mathbf{j}}$: $(2xyz)_z - (x^2y + y \cos(yz))_x = 2xy - 2xy = 0$ ✓

$\hat{\mathbf{k}}$: $(x^2z + z \cos(yz))_x - (2xyz)_y = 2xz - 2xz = 0$ ✓

This always works by Clairaut’s theorem ($f_{xy} = f_{yx}$, etc.).

Example 8: Proving $\text{div}(\nabla \times \mathbf{F}) = 0$

Show that $\text{div}(\nabla \times \mathbf{F}) = 0$ for any smooth $\mathbf{F} = \langle P, Q, R \rangle$.

$\nabla \times \mathbf{F} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$

$\text{div}(\nabla \times \mathbf{F}) = (R_y - Q_z)_x + (P_z - R_x)_y + (Q_x - P_y)_z$

$= R_{yx} - Q_{zx} + P_{zy} - R_{xy} + Q_{xz} - P_{yz}$

By Clairaut’s: $R_{yx} = R_{xy}$, $Q_{zx} = Q_{xz}$, $P_{zy} = P_{yz}$. Every term cancels. $\square$

4. Green’s Theorem Reformulated

Green’s Theorem via Curl

$$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \iint_D (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} dA$$

The circulation of $\mathbf{F}$ around C equals the total curl over D . This is the 2D special case of **Stokes’ Theorem** (coming soon!).

Practice Problems

Problem 1. Compute $\text{div } \mathbf{F}$ and $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$.

Answer: $\text{div } \mathbf{F} = 2x + 2y + 2z$, $\nabla \times \mathbf{F} = \mathbf{0}$

Problem 2. Show that $\mathbf{F} = \langle yz, xz, xy \rangle$ is both irrotational and incompressible.

Answer: $\nabla \times \mathbf{F} = \mathbf{0}$, $\text{div } \mathbf{F} = 0$

Problem 3. Find $\nabla \times \mathbf{F}$ and $\operatorname{div} \mathbf{F}$ for $\mathbf{F} = \langle y^2z, x^2z, x^2y \rangle$.

Answer: $\nabla \times \mathbf{F} = \langle 0, y^2 - 2xy, 2z(x - y) \rangle$, $\operatorname{div} \mathbf{F} = 0$

Problem 4. Compute $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle e^x \cos y, -e^x \sin y, 0 \rangle$. What does the result tell you?

Answer: $\nabla \times \mathbf{F} = \mathbf{0} \implies$ conservative, potential $f = e^x \cos y$

Problem 5. Verify $\operatorname{div}(f\mathbf{F}) = f \operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f$ for $f = xy$ and $\mathbf{F} = \langle 1, 1, 0 \rangle$.

Answer: both sides $= x + y$ ✓

Problem 6. Prove: if $\mathbf{F} = \nabla f$ and f is harmonic ($\nabla^2 f = 0$), then $\mathbf{F}$ is both irrotational and incompressible.

Irrotational: $\nabla \times \mathbf{F} = \nabla \times (\nabla f) = \mathbf{0}$ (identity 1).

Incompressible: $\operatorname{div} \mathbf{F} = \operatorname{div}(\nabla f) = \nabla^2 f = 0$ (since f is harmonic). $\square$

Problem 7. Without computing, predict whether the 2D curl is positive, negative, or zero at the origin. Then verify.

(a) $\mathbf{F} = \langle -2y, 3x \rangle$ (b) $\mathbf{F} = \langle y, y \rangle$ (c) $\mathbf{F} = \langle x^2, -y^2 \rangle$

Answer: (a) $Q_x - P_y = 3 - (-2) = 5$ (positive, CCW). (b) $Q_x - P_y = 0 - 1 = -1$ (negative, CW).

(c) $Q_x - P_y = 0$ (irrotational).

Problem 8. The velocity field of a whirlpool is modeled by $\mathbf{F} = \langle -y, x, 2z \rangle$. Find $\nabla \times \mathbf{F}$ and $\operatorname{div} \mathbf{F}$. Does this flow rotate, expand, or both?

$\nabla \times \mathbf{F} = \langle 0 - 0, 0 - 0, 1 - (-1) \rangle = 2\hat{\mathbf{k}}$ (CCW rotation about z -axis)

$\operatorname{div} \mathbf{F} = 0 + 0 + 2 = 2$ (expanding vertically)

Both! Like a tornado: spinning while air is pulled upward.

Lecture 22: Surface Integrals

Background & Motivation

We've integrated functions along curves (line integrals) and over flat regions (double integrals). Today we integrate over curved surfaces in 3D. Surface integrals compute mass of a shell, and flux integrals measure how much a vector field flows through a surface. These are the ingredients we need for Stokes' and the Divergence Theorem.

1. Parametric Surfaces

Definition 1: Parametric Surface

A **parametric surface** S is the image of a continuous map $\mathbf{r} : D \rightarrow \mathbb{R}^3$, where $D \subseteq \mathbb{R}^2$. Each pair $(u, v) \in D$ is sent to a point on S — think of latitude and longitude on a globe.

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D.$$

- **Tangent vectors:** $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}$, $\mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$
- **Normal vector:** $\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v$ (perpendicular to S ; nonzero where S is smooth)
- **Surface area element:** $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv$

Common Parameterizations

Surface	Parameterization $\mathbf{r}(u, v)$	Domain	$dS = \ \mathbf{r}_u \times \mathbf{r}_v\ du dv$
Graph $z = g(x, y)$	$\langle x, y, g(x, y) \rangle$	$(x, y) \in D$	$\sqrt{1 + g_x^2 + g_y^2} dA$
Sphere $x^2 + y^2 + z^2 = a^2$	$\langle a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi \rangle$	$\theta \in [0, 2\pi], \phi \in [0, \pi]$	$a^2 \sin \phi d\phi d\theta$
Cylinder $x^2 + y^2 = R^2$	$\langle R \cos \theta, R \sin \theta, z \rangle$	$\theta \in [0, 2\pi], z \in [c, d]$	$R d\theta dz$
Cone $z = \sqrt{x^2 + y^2}$	$\langle r \cos \theta, r \sin \theta, r \rangle$	$\theta \in [0, 2\pi], r \in [0, h]$	$r\sqrt{2} dr d\theta$
Plane thru P_0 , dirs $\mathbf{a}, \mathbf{b}$	$P_0 + s\mathbf{a} + t\mathbf{b}$	$(s, t) \in D$	$\ \mathbf{a} \times \mathbf{b}\ ds dt$
Rev. of $y = f(x)$ about x -axis	$\langle x, f(x) \cos \theta, f(x) \sin \theta \rangle$	$x \in [a, b], \theta \in [0, 2\pi]$	$ f(x) \sqrt{1 + f'(x)^2} dx d\theta$

Special Case: $z = g(x, y)$

$$dS = \sqrt{1 + g_x^2 + g_y^2} dA$$

$$\mathbf{n} dS = \langle -g_x, -g_y, 1 \rangle dA \quad (\text{upward-pointing normal})$$

For downward normal, negate: $\langle g_x, g_y, -1 \rangle dA$.

2. Scalar Surface Integrals

Definition 2: Surface Integral of a Scalar Function

The **surface integral** $\iint_S f \, dS$ generalizes the double integral from a flat 2D region to a curved surface in $\mathbb{R}^3$. It sums values of f over S weighted by the true area element dS .

$$\iint_S f \, dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| \, du \, dv$$

For $z = g(x, y)$:

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dA$$

– **Applications:** mass ($f = \rho$), average value on a surface

Special Case: $f \equiv 1$ gives Surface Area

Setting $f = 1$ in the surface integral turns it into the **surface area** of S :

$$\text{Area}(S) = \iint_S 1 \, dS = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| \, du \, dv$$

For a graph $z = g(x, y)$ over a region D in the xy -plane:

$$\text{Area}(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} \, dA$$

Example 1: Mass of a Hemispherical Shell

Find the mass of the upper hemisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$, with density $\rho = z$.

1. **Parameterize:** $\mathbf{r}(\theta, \phi) = \langle 3 \sin \phi \cos \theta, 3 \sin \phi \sin \theta, 3 \cos \phi \rangle$

$$\theta \in [0, 2\pi], \phi \in [0, \pi/2]; \quad \|\mathbf{r}_\theta \times \mathbf{r}_\phi\| = 9 \sin \phi$$

2. **Set up:** $\rho = z = 3 \cos \phi$

$$M = \int_0^{2\pi} \int_0^{\pi/2} (3 \cos \phi)(9 \sin \phi) \, d\phi \, d\theta$$

3. **Evaluate:** $= 27 \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin \phi \cos \phi \, d\phi = 27 \cdot 2\pi \cdot \frac{1}{2} = \boxed{27\pi}$

3. Flux Integrals

Definition 3: Flux of a Vector Field Through a Surface

The **flux** of a vector field $\mathbf{F}$ across an oriented surface S measures the net rate at which $\mathbf{F}$ crosses S in the direction of the chosen unit normal $\hat{\mathbf{n}}$. If $\mathbf{F}$ is a fluid velocity field, the flux is the volume

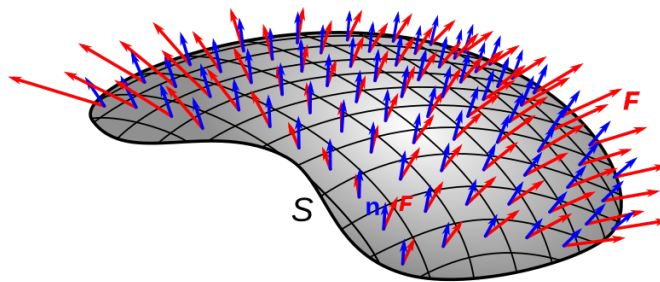
of fluid passing through S per unit time.

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv$$

For $z = g(x, y)$ with **upward** normal:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D (-P g_x - Q g_y + R) dA$$

- **Interpretation: net flow of $\mathbf{F}$ through the surface per unit time**
- Positive flux = net flow in direction of chosen normal
- **Orientation matters:** flipping normal negates the flux



Surface S with normal vectors $\mathbf{n}$ (blue) and vector field $\mathbf{F}$ (red). The flux integral measures how much $\mathbf{F}$ flows through S .

Example 2: Flux Through a Paraboloid

Find the flux of $\mathbf{F} = \langle y, x, z \rangle$ through the paraboloid $z = 1 - x^2 - y^2$, $z \geq 0$, with upward normal.

1. **Set up:** $g(x, y) = 1 - x^2 - y^2$, so $g_x = -2x$, $g_y = -2y$

$$P = y, Q = x, R = z = 1 - x^2 - y^2$$

Domain D : $x^2 + y^2 \leq 1$ (disk of radius 1)

2. **Apply flux formula:**

$$\begin{aligned} \text{Flux} &= \iint_D (-y(-2x) - x(-2y) + (1 - x^2 - y^2)) dA \\ &= \iint_D (2xy + 2xy + 1 - x^2 - y^2) dA \\ &= \iint_D (4xy + 1 - x^2 - y^2) dA \end{aligned}$$

3. **Polar coordinates:** $\int_0^{2\pi} \int_0^1 (4r^2 \cos \theta \sin \theta + 1 - r^2) r dr d\theta$

The $4r^2 \cos \theta \sin \theta$ term integrates to 0 over $[0, 2\pi]$.

$$= \int_0^{2\pi} \int_0^1 (r - r^3) dr d\theta = 2\pi \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 = 2\pi \cdot \frac{1}{4} = \boxed{\frac{\pi}{2}}$$

Example 3: Flux Through a Sphere

Find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ through the sphere $x^2 + y^2 + z^2 = a^2$.

1. **Outward unit normal:** On the sphere, $\hat{\mathbf{n}} = \frac{1}{a} \langle x, y, z \rangle$

2. **Dot product:** $\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{x^2 + y^2 + z^2}{a} = \frac{a^2}{a} = a$

3. **Integrate:** $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S a \, dS = a \cdot 4\pi a^2 = \boxed{4\pi a^3}$

Practice Problems

Problem 1. Find the surface area of the plane $z = 3 - x - y$ over the triangle $0 \leq x \leq 1, 0 \leq y \leq 1 - x$.

Answer: $\frac{\sqrt{3}}{2}$

Problem 2. Find the flux of $\mathbf{F} = \langle 0, 0, 1 \rangle$ upward through $z = 1 - x^2 - y^2$ above $z = 0$.

Answer: π

Problem 3. Compute $\iint_S z \, dS$ where S is the cone $z = \sqrt{x^2 + y^2}, 0 \leq z \leq 2$.

(Hint: $g_x = x/\sqrt{x^2 + y^2}, g_y = y/\sqrt{x^2 + y^2}$, so $\sqrt{1 + g_x^2 + g_y^2} = \sqrt{2}$.)

Answer: $\sqrt{2} \int_0^{2\pi} \int_0^2 r \cdot r \, dr \, d\theta = \sqrt{2} \cdot 2\pi \cdot \frac{8}{3} = \frac{16\sqrt{2}\pi}{3}$

Problem 4. Find the flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ outward through the unit sphere $x^2 + y^2 + z^2 = 1$.

(Hint: on the unit sphere the outward unit normal is $\hat{\mathbf{n}} = \langle x, y, z \rangle$, so $\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$. Use symmetry.)

Answer: 0 ($\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$; each term is odd in one variable on a symmetric sphere)

Problem 5. Find the flux of $\mathbf{F} = \langle z, x, y \rangle$ upward through the triangle cut from the plane $x + y + z = 1$ in the first octant.

Answer: $-Pg_x - Qg_y + R = -z(-1) - x(-1) + y = z + x + y = 1$; Flux = $\iint_D 1 \, dA = \frac{1}{2}$

Lecture 23: Stokes' Theorem & the Divergence Theorem

Background & Motivation

This is the culmination of the entire course. Stokes' Theorem and the Divergence Theorem are the 3D generalizations of Green's Theorem, which itself generalized the Fundamental Theorem of Calculus. Every major theorem we've seen—FTOC, FTOLI, Green's, Stokes', Divergence—is the same idea in different dimensions: the integral of a derivative over a region equals the integral over the boundary. These theorems are the foundation of electromagnetism, fluid dynamics, and modern physics.

1. Stokes' Theorem

Picture: tiny paddle wheels scattered across a surface S . Add up all their local spinning ($\nabla \times \mathbf{F}$) and you recover the net circulation of $\mathbf{F}$ around the boundary curve C .

Theorem 1: Stokes' Theorem

Let S be an oriented surface with unit normal $\hat{\mathbf{n}}$ and boundary curve C oriented by the right-hand rule (curl the fingers of your right hand along C ; thumb points along $\hat{\mathbf{n}}$). For $\mathbf{F} = \langle P, Q, R \rangle$ with continuous partial derivatives, the following are equivalent:

(1) **Unit-normal form:**

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

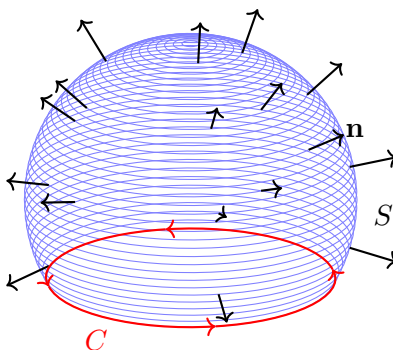
(2) **Vector differential form:**

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}, \quad d\mathbf{S} = \hat{\mathbf{n}} dS$$

(3) **Component form:**

$$\oint_C P dx + Q dy + R dz = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

- Circulation of $\mathbf{F}$ around C = flux of $\nabla \times \mathbf{F}$ through S
- When S is flat in the xy -plane ($\hat{\mathbf{n}} = \hat{\mathbf{k}}$), reduces to **Green's Theorem**



A surface S (here a hemisphere) with outward-pointing normals $\mathbf{n}$ and boundary curve $C = \partial S$ oriented CCW by the right-hand rule. Stokes' Theorem says the circulation around the red rim equals the total curl-flux through all the little normal spikes on the surface above.

Key observations

- The surface S can be *any* surface bounded by C — pick the easiest one!
- If $\nabla \times \mathbf{F} = \mathbf{0}$, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve
- If $\mathbf{F} = \nabla f$, then $\nabla \times \mathbf{F} = \mathbf{0}$, recovering path independence (FTOLI)

Example 1: Stokes' Verification on a Paraboloid

Verify Stokes' Theorem for $\mathbf{F} = \langle y^2, z^2, x^2 \rangle$ over the paraboloid $z = 1 - x^2 - y^2$, $z \geq 0$, with boundary C : the unit circle $x^2 + y^2 = 1$, $z = 0$.

Line integral side:

1. **Parameterize C :** $\mathbf{r}(t) = \langle \cos t, \sin t, 0 \rangle$, $\mathbf{r}' = \langle -\sin t, \cos t, 0 \rangle$
2. **Evaluate:** $\mathbf{F}(\mathbf{r}(t)) = \langle \sin^2 t, 0, \cos^2 t \rangle$
3. **Dot product:** $\mathbf{F} \cdot \mathbf{r}' = -\sin^3 t + 0 + 0 = -\sin^3 t$
4. **Integrate:** $\int_0^{2\pi} (-\sin^3 t) dt = 0$ (odd power of sin over full period)

Surface integral side:

1. **Curl:** $\nabla \times \mathbf{F} = \langle -2z, -2x, -2y \rangle$
2. **Surface:** $g = 1 - x^2 - y^2$, $g_x = -2x$, $g_y = -2y$
 $\nabla \times \mathbf{F} \cdot \langle -g_x, -g_y, 1 \rangle = (-2z)(2x) + (-2x)(2y) + (-2y)(1)$
 $= -4xz - 4xy - 2y$
3. **Substitute $z = 1 - x^2 - y^2$:**
 $= -4x(1 - x^2 - y^2) - 4xy - 2y$
4. **Integrate over D :** By symmetry, each term is an odd function of x or y over the disk $x^2 + y^2 \leq 1$,
so $\iint_D = \boxed{0} \checkmark$

2. The Divergence Theorem

Theorem 2: The Divergence Theorem (Gauss's Theorem)

Let E be a solid region bounded by the closed surface S with outward unit normal $\hat{\mathbf{n}}$. For $\mathbf{F} = \langle P, Q, R \rangle$ with continuous partial derivatives, the following are equivalent:

(1) **Unit-normal form:**

$$\oint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS = \iiint_E \operatorname{div} \mathbf{F} dV$$

(2) **Vector differential form** (computed piece by piece on a parameterization):

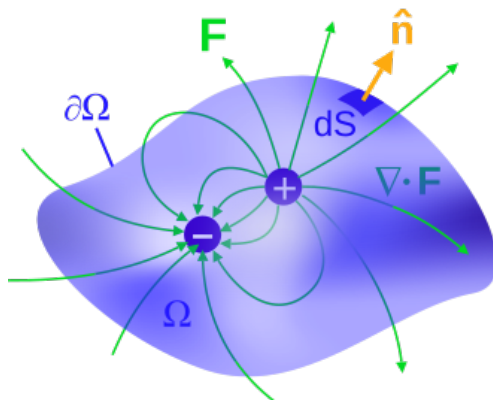
$$\oint_S \mathbf{F} \cdot d\mathbf{S} = \sum_i \iint_{D_i} \mathbf{F}(\mathbf{r}_i(u, v)) \cdot (\mathbf{r}_{i,u} \times \mathbf{r}_{i,v}) du dv = \iiint_E \operatorname{div} \mathbf{F} dV$$

where $S = \bigcup_i S_i$ is split into smooth pieces, each parameterized by $\mathbf{r}_i : D_i \rightarrow \mathbb{R}^3$ with outward orientation, and $d\mathbf{S} = \hat{\mathbf{n}} dS$.

(3) **Component form:**

$$\oint_S (P dy dz + Q dz dx + R dx dy) = \iiint_E \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV$$

- Total flux out of S = total “source strength” inside E
- **Connection to Green’s:** The flux form of Green’s Theorem says $\oint_C \mathbf{F} \cdot \hat{\mathbf{n}} ds = \iint_D \operatorname{div} \mathbf{F} dA$ in 2D. The Divergence Theorem is the same statement one dimension up — swap the closed curve for a closed surface and the 2D region for a 3D solid.



Field lines (green “spikes”) emerge from positive sources ($\operatorname{div} \mathbf{F} > 0$) and terminate on negative sinks ($\operatorname{div} \mathbf{F} < 0$). Draw any closed surface (the blue “bubble”) around the region: the net flux through the bubble equals the total source strength inside it. What flows out must have been created within.

When to use it

- S is a complicated closed surface but $\operatorname{div} \mathbf{F}$ is simple $\implies$ compute the volume integral
- S is not closed $\implies$ close it off with a “cap,” apply the theorem, then subtract the cap

Example 2: Divergence Theorem on the Unit Sphere

Verify the Divergence Theorem for $\mathbf{F} = \langle x, y, z \rangle$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

Volume integral side:

1. **Divergence:** $\operatorname{div} \mathbf{F} = 1 + 1 + 1 = 3$
2. **Integrate:** $\iiint_E 3 dV = 3 \cdot \frac{4}{3}\pi(1)^3 = 4\pi$

Surface integral side:

1. **Outward normal:** On the unit sphere, $\hat{\mathbf{n}} = \langle x, y, z \rangle$

2. **Dot product:** $\mathbf{F} \cdot \hat{\mathbf{n}} = x^2 + y^2 + z^2 = 1$

3. **Integrate:** $\iint_S 1 \, dS = 4\pi$

Both sides = 4π ✓

Example 3: Divergence Theorem on the Unit Cube

Find the outward flux of $\mathbf{F} = \langle xy, yz, xz \rangle$ through the unit cube $[0, 1]^3$.

1. **Divergence:** $\operatorname{div} \mathbf{F} = y + z + x$

2. **Integrate:** $\iiint_{[0,1]^3} (x + y + z) \, dV = \int_0^1 \int_0^1 \int_0^1 (x + y + z) \, dx \, dy \, dz$

3. **By symmetry:** $\int x \, dV = \int y \, dV = \int z \, dV = \frac{1}{2}$

4. **Result:** Flux = $3 \cdot \frac{1}{2} = \boxed{\frac{3}{2}}$

Practice Problems

Problem 1. Use the Divergence Theorem to compute $\iint_S \langle x, y, z \rangle \cdot d\mathbf{S}$ over the sphere of radius R .

Answer: $4\pi R^3$

Problem 2. Use Stokes' Theorem to evaluate $\oint_C \langle -y, x, z \rangle \cdot d\mathbf{r}$ where C is the circle $x^2 + y^2 = 4$, $z = 3$ (CCW when viewed from above).

$\nabla \times \mathbf{F} = \langle 0 - 1, 0 - 0, 1 - (-1) \rangle = \langle -1, 0, 2 \rangle$

Surface: disk $x^2 + y^2 \leq 4$, $z = 3$. Normal: $\hat{\mathbf{n}} = \hat{\mathbf{k}}$.

$\iint_S \langle -1, 0, 2 \rangle \cdot \langle 0, 0, 1 \rangle \, dA = \iint_S 2 \, dA = 2 \cdot 4\pi = \boxed{8\pi}$

Problem 3. Use the Divergence Theorem to compute $\iint_S \langle x^2, y^2, z^2 \rangle \cdot d\mathbf{S}$ over the unit cube $[0, 1]^3$.

Answer: 3

Problem 4. Verify Stokes' Theorem for $\mathbf{F} = \langle z, x, y \rangle$ on the triangular portion of the plane $x+y+z = 1$ in the first octant, with upward normal.

Surface integral side:

$$\nabla \times \mathbf{F} = \langle 1, 1, 1 \rangle. \quad \text{Surface: } g = 1 - x - y, \text{ so } \langle -g_x, -g_y, 1 \rangle = \langle 1, 1, 1 \rangle.$$

$$\nabla \times \mathbf{F} \cdot \langle 1, 1, 1 \rangle = 3. \quad \iint_D 3 \, dA = 3 \cdot \frac{1}{2} = \frac{3}{2}.$$

Line integral side: (CCW from above: $(1, 0, 0) \rightarrow (0, 1, 0) \rightarrow (0, 0, 1) \rightarrow (1, 0, 0)$)

$$C_1: \mathbf{r} = \langle 1 - t, t, 0 \rangle, \mathbf{F} = \langle 0, 1 - t, t \rangle, \mathbf{r}' = \langle -1, 1, 0 \rangle. \quad \text{Dot} = 1 - t. \quad \int = 1/2.$$

$$C_2: \mathbf{r} = \langle 0, 1 - t, t \rangle, \mathbf{F} = \langle t, 0, 1 - t \rangle, \mathbf{r}' = \langle 0, -1, 1 \rangle. \quad \text{Dot} = 1 - t. \quad \int = 1/2.$$

$$C_3: \mathbf{r} = \langle t, 0, 1 - t \rangle, \mathbf{F} = \langle 1 - t, t, 0 \rangle, \mathbf{r}' = \langle 1, 0, -1 \rangle. \quad \text{Dot} = 1 - t. \quad \int = 1/2.$$

$$\text{Total: } \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \boxed{\frac{3}{2}} \quad \checkmark$$

Problem 5. Use the Divergence Theorem to derive $\text{Vol}(E) = \frac{1}{3} \oiint_S \langle x, y, z \rangle \cdot d\mathbf{S}$.

$\text{div} \langle x, y, z \rangle = 1 + 1 + 1 = 3$. By the Divergence Theorem:

$$\oiint_S \langle x, y, z \rangle \cdot d\mathbf{S} = \iiint_E 3 \, dV = 3 \text{Vol}(E)$$

$$\therefore \text{Vol}(E) = \frac{1}{3} \oiint_S \langle x, y, z \rangle \cdot d\mathbf{S}. \quad \square$$